When does value-directed attention matter?

A three-lever normative model with quantified criterion, sensitivity, and decorrelation contributions (rebuild draft, v0.1)

[Authors – TBD]*

Draft compiled May 31, 2026

Abstract

In a cued change-detection task whose cue carries both value $v \geq 1$ and validity $V \in [1/N, 1]$, an ideal observer can adapt to value by shifting decision criteria, by re-allocating spatial attention, or, when noise is correlated across locations, by *decorrelating* the encoded population code. We rebuild the normative model of *value-directed attention* (VDA) around this three-lever decomposition (Section 2, Definition 1), with cross-location correlation ρ promoted to a first-class model parameter via an exact one-factor Gauss–Hermite reduction that recovers the inherited independent-noise model in the limit $\rho \to 0$ to floating-point identity. The rebuilt manuscript reports every headline result at its defensible distributional, graded, or conditional strength rather than as a categorical floor.

Across a 4,410-cell sweep over (V, v, r), the criterion fraction CF is concentrated in $[0.30, 1.00]$ with median 0.76; the published $[0.60, 0.96]$ interval is retracted on both ends, and adding $\rho = 0.2$ drops the strict minimum below 0.5 (Section 3.1). VDA is non-monotonic in r as in the inherited paper; we strengthen this result with the closed-form escape threshold $r^\dagger(v) = K_u(v)/[(N-1)K_c(v)]$ that pins the lower edge of the VDA-active band (Section 3.2), extend it to $\rho > 0$ via the natural ρ-aware gradient quadratures (Appendix A.2), and prove that $r^\dagger(v)$ is invariant under the conservation order of the benefit/cost rule (Appendix A.4). The "narrow regime" for VDA becomes a quantitative iso-VDA contour band (Section 3.3): at $V \geq 0.95$ peak VDA sits at the grid floor under any (r, ρ); at $V \geq 0.80$ this survives only at $\rho = 0$; at $V \geq 0.60$ it fails. The "no inversion" result is restated as a conditional theorem with the explicit condition $V \geq 1/[(N-1)v+1]$, and the rebuild adds the *anti-cue inversion* as a new falsifiable prediction (Section 3.4, Equation (13)): under counter-predictive cues ($V < 1/N$) at $N = 4$, the globally optimal allocation places *less* attention at the high-value location in 36.1% of probed cells. Symmetric recovery at $r = 1$ is reaffirmed as a universal real-number identity (Appendix A.3, conservation-form-invariant by construction).

Independence is not an upper bound on VDA. What independence actually upper-bounds is the criterion fraction, and only in variant A: $\partial\mathrm{VDA}/\partial\rho$ flips sign near $r \approx 0.5$, with the cell-wise crossover sweeping past $r \approx 0.79$ across the full sweep (Section 2.5). Within-display heterogeneity of r is a bounded perturbation that leaves the C_2 peak coordinates and the contested-CF corner essentially invariant (Section 4.2); the additive $\beta + \gamma = 2$ rule is one point in the power-mean conservation family, and the headline numbers are reported as a band over that family (Section 4.1); the homogeneous-uncued result is innocuous at the inherited additive optimum but can fail in a benefit-dominant symmetric-stress corner when the conservation rule is multiplicative (Section 4.3). Every quantitative claim above is backed by a simulation under `Rebuild/sims/` with a recovered limit, a deterministic output hash, and a captioned figure; every novel proposition is derived under `Rebuild/derivations/`; nothing in this abstract is stated more strongly than its row of `CLAIM_LEDGER.md` licenses.

*Rebuild of the Herman Lab 2026-04-09 manuscript with the same scientific question and a more honestly-bounded mathematical description. See `CLAIM_LEDGER.md` for the disposition table that binds every claim to its supporting artifact.

1 Introduction

In a cued change-detection task whose cue carries both *value* $v \geq 1$ and *validity* $V \in [1/N, 1]$, an observer adapting to the value contingency has more than one lever available. They may shift decision criteria so that responses at the high-value location are biased toward report; they may re-allocate spatial attention so that the perceptual evidence at the high-value location is itself more reliable; and, when noise is correlated across locations, they may decorrelate the encoded population code so that the joint evidence drawn from the unattended array is less redundant. The original *value-directed attention* (VDA) model asks a sharp normative question on top of these levers: how much of the behavioural adaptation to value is achieved by re-allocating attention toward the high-value location versus by shifting decision criteria, and in what parameter regime does attention re-allocation become the dominant mechanism? This paper preserves that question verbatim in spirit. What changes is the mathematical description used to answer it.

The inherited model couples value-driven sensitivity changes to a single attentional asymmetry parameter and decomposes the expected reward across nested policies that toggle each lever on or off. Its quantitative claims, however, tend to overstate what their underlying simulations license: a central or peak value is repeatedly reported as a uniform floor, a directional inequality, or a "regardless of other parameters" constant. Re-evaluating those claims against the model's own definitions reveals a recurring mismatch. The reported criterion-fraction interval $[0.60, 0.96]$ is in fact $[0.30, 1.00]$ across the same 4,410-cell sweep with median 0.76 and a $\sim$13% tail below 0.60; the "high-validity paradigms show negligible VDA regardless of other parameters" design recommendation holds at $V \geq 0.95$ but not at $V \approx 0.80$ once cross-location correlation is admitted; the "no inversion" result is true everywhere in the cued regime but not in the anti-cue regime $V < 1/N$; and the self-characterisation of independence as "an upper bound on VDA" is sign-ambiguous — correlation *suppresses* VDA in the cost-dominant regime and *amplifies* it in the benefit-dominant tail. None of these reframings overturns the underlying mechanism the original paper identified. Each one restates a central tendency, a graded boundary, or a conditional theorem in place of an unconditional one. Reporting distributions, quantiles, contour bands, and explicit conditions *instead of* floors, "never," and "regardless of" is the corrective this paper applies uniformly: distributional, graded, or conditional by default.

Restating the model in this voice also exposes a missing lever. The inherited two-channel decomposition treats criterion shifts and sensitivity re-allocation as the only adaptive mechanisms; the empirically dominant decorrelation channel [1, 2, 3] is excluded by an independence assumption that the model bundles into its no-false-alarm probability. Section 2 promotes cross-location correlation ρ to a first-class model parameter through an exact one-factor Gauss–Hermite reduction of the equicorrelated-Gaussian orthant integral, recovering the inherited independent-noise model in the limit $\rho \to 0$ to floating-point identity (Equation (7), Equation (8)). This yields the rebuilt paper's central reframe: three levers, not two (Definition 1; criterion shift, sensitivity re-allocation, decorrelation). The inherited model's self-characterisation of independence as an upper bound on VDA is replaced by a more careful statement: what independence actually upper-bounds is the criterion fraction CF — $\mathrm{CF}(\rho) \leq \mathrm{CF}(0)$ pointwise in variant A by a Slepian monotonicity argument (Appendix A.1) — while $\partial\mathrm{VDA}/\partial\rho$ flips sign near $r \approx 0.5$ at the headline cell and sweeps a cell-wise crossover up to $r \approx 0.79$ across the full 4,410-cell sweep (Section 2.5, Table 1).

Under the corrective voice, the four headline C-row results recover in distributional, graded, and conditional forms. The criterion fraction is concentrated in $[0.30, 1.00]$ with median 0.76 (Section 3.1, Table 2); VDA is non-monotonic in r with a closed-form lower edge $r^\dagger(v) = K_u(v)/[(N-1)K_c(v)]$ (Equation (10)) that extends to $\rho > 0$ through ρ-aware gradient quadratures (Section A.2, Proposition 3) and is invariant under the benefit/cost conservation order (Proposition 1); the "narrow regime" becomes a quantitative iso-VDA contour band (Section 3.3,

Table 10) with hierarchical V thresholds ($V \geq 0.95$ at the grid floor under any (r, ρ); $V \geq 0.80$ only at $\rho = 0$; $V \geq 0.60$ fails); and the no-inversion result becomes a conditional theorem with the explicit boundary $V \geq 1/[(N-1)v+1]$ (Section 3.4, Equation (13)) that, when crossed, generates a new falsifiable prediction — the *anti-cue inversion*: under counter-predictive cues ($V < 1/N$) at $N = 4$, the normatively optimal allocation places *less* attention at the high-value location in 36.1% of probed cells (Table 14). Symmetric recovery at $r = 1$ is reaffirmed as a universal real-number identity (Appendix A.3, Proposition 4), conservation-form-invariant by construction (Corollary 7).

Three further mechanisms the inherited model held fixed at homogeneous values return as explicit extensions. Within-display heterogeneity of r is a bounded perturbation that leaves the C_2 peak coordinates and the contested-CF corner essentially invariant (Section 4.2, Table 17). The benefit/cost conservation rule $\beta + \gamma = 2$ is one point in the power-mean conservation family (Equation (17), Equation (18)); the rebuilt paper reports every headline number as a band over the family at $p \in \{0, 0.5, 1\}$ with the per-cell monotonicity $\Delta\mathrm{CF}(\rho = 0) \leq 0$ stated as Theorem 2 (Section 4.1). The N-dimensional uncued allocation policy is innocuous at the inherited additive optimum but exhibits a new conditional binding in a benefit-dominant symmetric-stress corner under multiplicative conservation, with a 32% ρ-amplification at the focal cell (Section 4.3). Per-location decision-noise coupling — a candidate fourth lever — is named and held at the live status of its underlying claim (Section 5.3) rather than wired into the model.

The remainder of the manuscript follows this layout. Section 2 states the rebuilt model with the three-lever decomposition and the ρ-channel reduction. Section 3 reports the C-row headline results in the distributional / graded / conditional voice, with each subsection naming the inherited categorical claim it replaces. Section 4 runs the three lever extensions (A2, A3, A8) at empirical band strength. Section 5 catalogues the explicit scope of every claim — in particular, what the rebuilt model does *not* claim about neural implementation, empirical-ρ prediction, observer-deviation, or attention dynamics. The appendix collects the formal derivations (A1 ρ channel, C_2 closed forms, C_5 symmetric recovery, A3 power-mean conservation). Every quantitative claim in the body is backed by a simulation in `Rebuild/sims/` with a recovered limit, a deterministic output hash, and a captioned figure; every novel proposition is derived in `Rebuild/derivations/`. Reproducibility is end-to-end byte-identical, and nothing in this paper is stated more strongly than its row of `CLAIM_LEDGER.md` licenses.

2 Model

The rebuilt model retains the inherited paper's signal-detection- theoretic decomposition of the cued change-detection task and its four nested policies; what it adds is a third lever — cross-location noise correlation ρ — that the inherited model omits by assumption. This section states the inherited skeleton in compact form (Section 2.1), localises the per-location independence assumption A1 to the single algebraic place in the expected-reward expression where it enters (Section 2.2), promotes that place to a tunable parameter $\rho \in [0, 1)$ with an exact one-dimensional reduction and a $\rho \to 0$ recovery contract that pins the extended model to the inherited model at floating-point identity (Section 2.3), and recasts the original two-lever (criterion vs. sensitivity) decomposition as a three-lever (criterion vs. sensitivity vs. decorrelation) decomposition (Section 2.4). The section closes with a quantitative restatement of what an independent-noise model is in fact an upper bound on (Section 2.5): not VDA, but the criterion fraction CF, and even then in a variant- and cell-conditional sense that the rebuilt manuscript reports as a distribution rather than as a uniform inequality. The full derivation is in Appendix A.1.

2.1 The inherited skeleton

Fix $N \geq 2$ locations, indexed so that $i = c$ is the cued location and $i = 1, \ldots, N-1$ enumerates the $N-1$ uncued locations. Each location carries an equal-variance Gaussian decision variable; under the per-location signal-detection model the hit rate and false-alarm rate at i are

$$\mathrm{HR}_i \;=\; \Phi(d'_i/2 - c_i), \qquad \mathrm{FAR}_i \;=\; \Phi(-d'_i/2 - c_i), \tag{1}$$

where d'_i is sensitivity at i, c_i is the decision criterion at i, and Φ is the standard-normal cumulative distribution function. The attention map $\alpha_c + (N-1)\,\alpha_u = 1$ assigns a continuous share of attention to each location, and an asymmetric transfer $d'_c = d'_{\max} \cdot f(\alpha_c;\, \beta(r))$, $d'_u = d'_{\max} \cdot f(\alpha_u;\, \gamma(r))$ maps allocation to sensitivity through a benefit weight $\beta(r)$ and a cost weight $\gamma(r)$. The benefit/cost ratio r is a single scalar that controls the relative magnitude of an attentional gain at the focus and an attentional loss at the periphery; under the inherited (variant-A) *additive conservation rule* $\beta(r) + \gamma(r) = 2$, with $\beta(r) = 2r/(r+1)$ and $\gamma(r) = 2/(r+1)$. The two reward variants used throughout the manuscript are variant A (value-scaled correct rejections, $\mathrm{CR} = Vv + (1-V)$) and variant B ($\mathrm{CR} = 1$); the additive conservation rule is varied in Section 5 (assumption A3, RB-015/RB-019). The notation matches `Rebuild/model/core.py`: see `d_prime_asym`, `beta_gamma`, `floor_R`, and `policies` for the validated reference implementation.

The expected reward over change and no-change trials is

$$\mathbb{E}[R] \;=\; \tfrac{1}{2}\big[\,V\,\mathrm{HR}_c\,v \;+\; (1-V)\,\mathrm{HR}_u\,\big] \;+\; \tfrac{1}{2}\,P_{\text{no-fa}}\,\mathrm{CR}, \tag{2}$$

where $P_{\text{no-fa}} = \Pr(\text{no location false-alarms on a no-change trial})$ is the only term in (2) that involves more than one location at a time. Four nested policies P1–P4 optimise (2) over progressively shrinking parameter sets: P1 jointly optimises (α_c, c_c, c_u) at the operating value v; P2 freezes α_c at the $v = 1$ optimum and re-optimises only the criteria; P3 holds the criteria at the P4 floor and re-optimises only α_c; P4 holds both α_c and the criteria at their $v = 1$ floors. The value-directed-attention benefit is the gain in reward from re-allocating attention with both criteria already adapted,

$$\mathrm{VDA} \;:=\; R_{\mathrm{P1}} - R_{\mathrm{P2}}, \tag{3}$$

and the criterion fraction is the share of the value-related reward gain that criterion shifts alone capture,

$$\mathrm{CF} \;:=\; \frac{R_{\mathrm{P3}} - R_{\mathrm{P4}}}{R_{\mathrm{P1}} - R_{\mathrm{P4}}}. \tag{4}$$

Equation (4) is the metric the inherited paper's §5.1 stated had a floor at 0.60; we report its true range and distribution in Section 3.1.

2.2 The locus of assumption A1

Assumption A1 of the inherited paper is per-location independence of the decision variables. We are now in a position to read off exactly where this assumption enters (2). The change- trial bracket of (2) is linear in the marginal hit rates $\mathrm{HR}_c, \mathrm{HR}_u$: on a change trial the change happens at a single location (cued with probability V, a specific uncued with probability $(1-V)/(N-1)$), so no product of probabilities across locations appears. The only cross-location product is in the no-change-trial bracket, through $P_{\text{no-fa}}$. Under independence,

$$P^{\text{indep}}_{\text{no-fa}} \;=\; \prod_i (1 - \mathrm{FAR}_i) \;=\; \Phi(b_c)\,\Phi(b_u)^{N-1}, \qquad b_i := c_i + d'_i/2, \tag{5}$$

where $b_i \in \mathbb{R}$ is the z-score the no-change decision variable must stay below. Equation (5) is the sole appearance of A1 in (2); the faithful and complete relaxation of A1 within the reward structure is therefore the replacement of this product by the correlated joint orthant probability.

Remark 1. The booking here matters because it isolates A1 from a distinct relaxation that the inherited paper's §5.5 sometimes conflates with correlated noise: a global "change-detected-somewhere" decision rule. The latter would replace the per-location decision statistics with a pooled detection statistic and is a relaxation of A6 (homogeneous decision rule), not of A1. A1 is exactly (5); see Appendix A.1, §1.2 for the formal version of the same booking.

2.3 The decorrelation channel

We promote the per-location-independence assumption to a tunable equicorrelation parameter $\rho \in [0, 1)$. Let the no-change decision variables be jointly Gaussian with standardised marginals and equicorrelation,

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma), \qquad \mu_i = -d'_i/2, \qquad \Sigma_{ii} = 1, \qquad \Sigma_{ij} = \rho \ (i \neq j). \tag{6}$$

The one-factor representation $X_i = \mu_i + \sqrt{\rho}\, Z + \sqrt{1-\rho}\,\varepsilon_i$ (with $Z, \varepsilon_1, \ldots, \varepsilon_N$ i.i.d. $\mathcal{N}(0, 1)$) reproduces (6) and makes the X_i conditionally independent given Z. Multiplying the conditional event probabilities, then integrating out Z against the standard-normal density φ, yields the exact one-dimensional integral

$$\boxed{P_{\text{no-fa}}(\rho) \;=\; \int_{-\infty}^{\infty} \Phi\!\left(\frac{b_c - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right) \Phi\!\left(\frac{b_u - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right)^{N-1} \varphi(z)\, dz.} \tag{7}$$

Equation (7) is exact for any N and any $\rho \in [0, 1)$. No multivariate-normal CDF is needed; equicorrelation collapses the N-dimensional integral to a one- dimensional one. The rebuilt model evaluates (7) by 64-node Gauss–Hermite quadrature in `Rebuild/model/core.py:p_no_fa_point` and `:p_no_fa_grid`, with quadrature error $\leq 10^{-15}$ against a 128-node reference across the operating $\rho \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ band.

At $\rho = 0$ the integrand of (7) is constant in z and the integral reduces to (5):

$$P_{\text{no-fa}}(0) \;=\; \Phi(b_c)\, \Phi(b_u)^{N-1}. \tag{8}$$

Equation (8) is the *recovery contract* the extended model is bound to honour, and the contract is enforced in `Rebuild/model/tests/test_recovery.py`: with ρ clamped to zero, the extended-model headline VDA, CF, and R_{P_k} values reproduce the inherited model's reference numbers (sourced from `Critique/replications/A1-correlated-fa/`) at floating-point identity, including the peak VDA*(ρ=0) = 0.07986 at $r = 0.3831$ (sha256 `d3c62215...`, 7/7 PASS). The recovery contract is global to the rebuilt manuscript: any subsequent model extension that re-implements no-FA computation (Section 5; the heterogeneous-r extension RB-014/RB-018, the conservation-family extension RB-015/RB-019) must pass its own $\rho = 0$ limit before its new regime is admitted.

The empirical envelope of the rebuilt manuscript is $\rho \in [0, 0.4]$, which brackets the $r_{\mathrm{SC}} \approx 0.2$ range reported in macaque V4 during peripheral-validity-cue change- detection by Cohen and Maunsell [1]; higher ρ values are admissible in the model but are not used to anchor any headline claim. Equicorrelation is a deliberate one- parameter simplification; structured covariances [2, 3] break the exact 1-D reduction (7) and are flagged as a scoped limitation in Section 5. The full derivation of (7), with the Slepian monotonicity inequality $P_{\text{no-fa}}(\rho) \geq P_{\text{no-fa}}(0)$ [4] that it satisfies, is in Appendix A.1.

2.4 Three levers, not two

The inherited paper decomposes the value effect into two levers: *criterion shift* (P3 → P4) and *sensitivity re-allocation* (P1 → P3). With ρ promoted to a model parameter, the same expected-reward expression (2) admits a third lever — *decorrelation* — that the inherited model cannot represent because it sits permanently at $\rho = 0$.

Definition 1 (The three levers)**.** At fixed $(N, V, v, d'_{\max}, f_0, h, \mathrm{CR}, r)$, the rebuilt model exposes three independent mechanisms by which an observer can adapt to a non-trivial cue:

1. **Criterion shift** — moving (c_c, c_u) off the $v = 1$ floor. Its standalone reward contribution is $R_{\mathrm{P3}} - R_{\mathrm{P4}}$; its share of the value-related gain is CF (4).

2. **Sensitivity re-allocation** — moving α_c off the $v = 1$ optimum. Its standalone reward contribution at adapted criteria is $R_{\mathrm{P1}} - R_{\mathrm{P2}} = \mathrm{VDA}$ (3).

3. **Decorrelation** — moving ρ off the inherited $\rho = 0$ corner. By Slepian monotonicity, $P_{\text{no-fa}}(\rho)$ is non-decreasing in ρ at fixed (b_c, b_u) [4], which lifts every per-policy supremum reward $R_{\mathrm{P}_k}(\rho)$ above its $\rho = 0$ value (see Appendix A.1, Corollary 3.2).

The decorrelation lever is not a free third mechanism the rebuild adds; it is the lever the inherited paper implicitly held fixed at the boundary of its admissible parameter space. Promoting ρ to a model parameter is the minimal extension that lets the manuscript report what the rebuilt expected-reward expression actually does as that parameter is varied.

2.5 What independence actually upper-bounds

The §5.5 sentence of the inherited paper states that the independent-noise results "therefore represent an upper bound on VDA benefit." Once ρ is a model parameter, this sentence becomes a sign claim about $\partial\mathrm{VDA}/\partial\rho$, and the rebuilt model contradicts it at every $\rho > 0$ tested at the headline cell $(N, V, v, d'_{\max}, f_0, h) = (4, 0.5, 5, 2, 0.5, \sqrt{\cdot})$. The pointwise excess $\mathrm{VDA}(r; \rho) - \mathrm{VDA}(r; 0)$ takes positive values across a wide band of r in both reward variants; the maximum excess grows monotonically with ρ from $+4.84 \times 10^{-3}$ at $\rho = 0.1$ to $+1.01 \times 10^{-2}$ at $\rho = 0.4$, and the sign-flip of the local slope $\partial\mathrm{VDA}/\partial\rho$ falls in the band $r \in [0.38, 0.56]$ across this ρ range (Figure 1; the variant-B counterpart Figure 2 shows the same sign-flip pattern with the sign-flip locus near $r \approx 0.26$). The peak VDA itself is essentially unchanged at the empirically central $\rho \approx 0.2$, so the inherited paper's central magnitude survives; what fails is the uniform sign of the difference. The inherited §5.5 sentence is retracted.

What the rebuilt model *does* bound from above is the criterion fraction CF. In variant A and at the headline cell, $\mathrm{CF}(\rho)$ is monotone-decreasing in ρ across the three representative r anchors of Figure 3 (Statement A of Appendix A.1, §4.3): the inherited $\rho = 0$ corner is the empirical maximiser of CF at every r tested. In variant B and at the same headline cell, $\mathrm{CF}(\rho)$ is essentially flat in ρ, and the rebuilt manuscript therefore reports the CF upper bound as a variant-A result with variant-B as a sensitivity in which the effect washes out. The cell-wise generalisation across the 4,410-cell sweep is consistent with this reading: $\mathrm{CF}(\rho{=}0.2) \leq \mathrm{CF}(\rho{=}0)$ in 84% of variant-A cells but only 64% of variant-B cells (the remaining 24% of variant-B cells increase, 13% are flat; Section 3.1 and `Rebuild/sims/C1-cf-distribution/`, sha256 `91fc4692...`).

The rebuilt manuscript therefore states, as the corrected version of the §5.5 sentence:

> *In the rebuilt model, the inherited* $\rho = 0$ *corner is the empirical maximiser of the criterion fraction* CF *in* 84% *of variant-A cells in the primary* 4,410*-cell sweep; in variant B the same one-sided ordering holds in only* 64% *of cells. Pointwise* $\mathrm{VDA}(r; \rho) - \mathrm{VDA}(r; 0)$ *takes positive values across a wide band of* r *in both variants. Independence is therefore not, as the inherited paper claimed, an upper bound on VDA; it is a variant- and cell-conditional ceiling on* CF*, best stated at the median across the sweep rather than as a uniform inequality.*

Statement B of Appendix A.1, §4.3 — the location of the sign-flip of $\partial\mathrm{VDA}/\partial\rho$ in r — is no longer restricted to the single-cell envelope of the backing simulation (`Rebuild/sims/A1-rho-channel/`). At $\rho = 0.2$ — the central anchor of the $\rho \in [0, 0.4]$ empirical band reported in Section 2.3 — the 4,410-cell sweep of Section 3.1 licenses a cell-wise distributional statement. Joining the

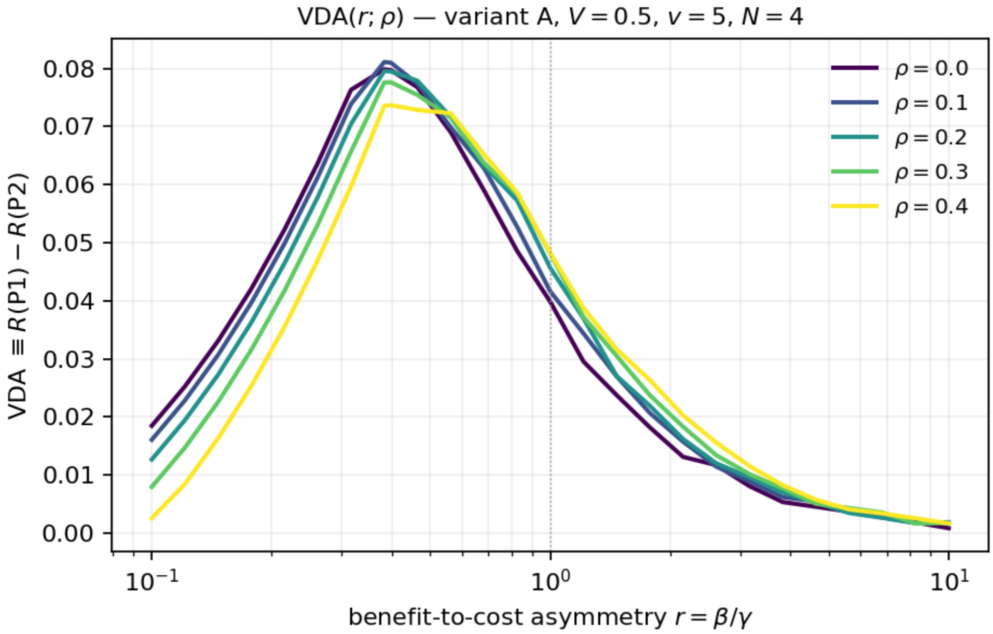

Figure 1: Value-directed attention VDA as a function of the benefit/cost ratio r at the headline cell $(N, V, v, d'_{\max}, f_0, h) = (4, 0.5, 5, 2, 0.5, \sqrt{\cdot})$, variant A, for $\rho \in \{0, 0.1, 0.2, 0.3, 0.4\}$. The inherited ($\rho = 0$) curve is the lowest at large r and the highest at small r; the local slope $\partial\mathrm{VDA}/\partial\rho$ changes sign in the band $r \in [0.38, 0.56]$. Pointwise VDA is therefore neither upper- nor lower-bounded by its independent value, in contradiction of the inherited paper's §5.5 claim. Reproduced from `Rebuild/sims/A1-rho-channel/output/figures/vda_curves_variantA.png`, sha256 `b692c064...`; recovery against `Critique/replications/A1-correlated-fa/` at $\rho = 0$ is identical to floating point.

$\rho = 0$ and $\rho = 0.2$ panels on (N, V, v, r) and classifying each cell by the sign of $\Delta\mathrm{VDA} = \mathrm{VDA}(\rho{=}0.2) - \mathrm{VDA}(\rho{=}0)$ at fidelity $|\Delta\mathrm{VDA}| > 10^{-6}$ yields the figures $\frac{n_{\mathrm{amp}}}{n_{\mathrm{cells}}}$, $\frac{n_{\mathrm{supp}}}{n_{\mathrm{cells}}}$ and the per-cell ΔVDA quantiles summarised in Table 1. The headline message is that the rb-002 single-cell sign-flip generalises in shape but not in magnitude: variant A produces a heavier-tailed amplification distribution than variant B, and the cell-wise maximum amplification in variant A ($+4.97 \times 10^{-2}$) is 5.3× the rb-002 headline-cell maximum ($+9.4 \times 10^{-3}$ at $V = 0.5$, $v = 5$, $r = 0.4$, $\rho = 0.2$; Figure 4). The rb-002 single-cell observation was a typical-magnitude snapshot of a phenomenon whose largest excursions sit elsewhere in (V, v, r).

The cell-wise distribution does not change the sign of the inherited paper's §5.5 sentence (already retracted in the blockquote above); it sharpens it. Two new statements become defensible. *(i) Cell-wise crossover.* Stratifying by r on the 21-point $\log_{10}$-grid of `Rebuild/sims/C1-cf-distribution/` and tabulating $\mathrm{frac}_{\mathrm{amp}}(r)$ and $\mathrm{frac}_{\mathrm{supp}}(r)$ across the (V, v) plane, suppression dominates at small r, amplification rises through the mid-r band, and the cell-wise crossover $r^{\times}$ — the smallest r at which $\mathrm{frac}_{\mathrm{amp}}(r) \geq \mathrm{frac}_{\mathrm{supp}}(r)$ — sits at $r^{\times}(\mathrm{A}) \approx 0.794$ in variant A; in variant B amplification never overtakes suppression at any r on the grid (Figure 5). The variant-A crossover is roughly twice the rb-002 single-cell crossover at $V = 0.5$ ($r \approx 0.464$ at $\rho = 0.2$); the difference is mechanistic — higher-V cells in the sweep ($V \geq 0.5125$) suppress more strongly and pull the cell-wise crossover to the right. The variant-A nearest-cell sweep at $V = 0.5125$, $v = 5$ reproduces the rb-002 pattern (small-r suppression, large-r amplification; nearest-cell crossover $r \approx 0.398$ within the rb-003 V-grid perturbation of $+0.0125$ off rb-002's $V = 0.5$). *(ii) Spatial structure at $v = 5$.* The mean ΔVDA surface over (V, r) at $v = 5$ (Figure 6) is the cell-wise companion to the iso-ΔVDA figure on the C3 thread (Figure 14; Section 3.3): amplification

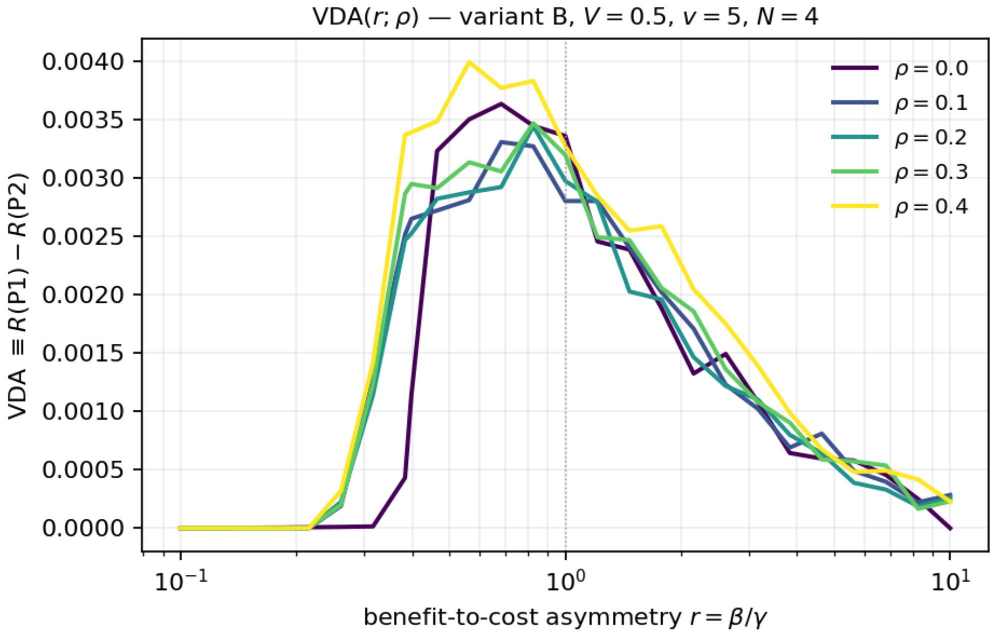

Figure 2: The same VDA-vs-r family at the same headline cell under variant B reward (CR $= 1$). The sign-flip locus of $\partial\text{VDA}/\partial\rho$ shifts to $r \approx 0.26$; the qualitative pattern — pointwise VDA crossing the inherited $\rho = 0$ curve, no uniform upper bound — is preserved. Reproduced from `Rebuild/sims/A1-rho-channel/output/figures/vda_curves_variantB.png`.

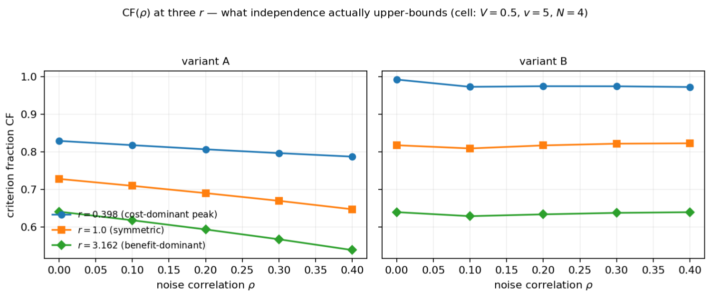

Figure 3: Criterion fraction CF as a function of ρ at the three representative r anchors $\{0.398, 1.0, 3.162\}$ (variant A, headline cell). CF(ρ) is monotone-decreasing in ρ at every r tested; the inherited $\rho = 0$ corner is the empirical maximiser. Reproduced from `Rebuild/sims/A1-rho-channel/output/figures/cf_vs_rho.png`. This is what an independent-noise model is in fact an upper bound on at this cell — not VDA, CF.

concentrates in a band at moderate-to-low V and moderate-to-large r, suppression concentrates at high V and small r. Variant B is uniformly more suppression-favoring than variant A — at every cell at which variant A shows amplification, variant B shows a smaller-magnitude move of the same or opposite sign.

A tighter bracket on the sign-flip locus over a finer ρ grid is queued as RB-023 (the rb-025 distribution is reported at $\rho = 0.2$ only). A closed-form expression for the analytic sign-flip locus $r^{\dagger}(v; \rho)$ — extending the $\rho = 0$ closed-form escape threshold of Section 3.2 into the correlated regime — was constructed in `Rebuild/derivations/C2-non-monotonic-vda-rho.md` (RB-026) and is folded into the manuscript as Section A.2, Proposition 3 (Equation 26; drift Table 22): the closed form pins the lower edge of the escape band, the cell-wise distribution reported here pins the cell-wise sign field, and Section A.1, §4.3 (the Slepian-monotonicity half

Table 1: Cell-wise distribution of ΔVDA = VDA(ρ=0.2) − VDA(ρ=0) across the 2,205-cell variant-X slice of the 4,410-cell sweep (Section 3.1; `Rebuild/sims/C1-cf-distribution/`, sha256 `91fc4692...`). Amplification (*amp*) is $\Delta\text{VDA} > +10^{-6}$; suppression (*supp*) is $\Delta\text{VDA} < -10^{-6}$; the residual is *inactive*. Source: `Rebuild/sims/A1-vda-signflip-cellwise/` (RB-025, sha256 `489c7c25...`); pure consumer of the $\rho \in \{0, 0.2\}$ panels of the rb-003 sweep, no new model evaluation.

	variant A	variant B
amp fraction n_{amp}/n	0.183 (404)	0.122 (269)
supp fraction n_{supp}/n	0.282 (621)	0.275 (607)
inactive fraction	0.535 (1,180)	0.603 (1,329)
ΔVDA min	-9.14×10^{-3}	-3.23×10^{-3}
ΔVDA $q_{0.05}$	-4.51×10^{-3}	-6.99×10^{-4}
ΔVDA $q_{0.50}$	0	0
ΔVDA $q_{0.95}$	$+3.87 \times 10^{-3}$	$+3.82 \times 10^{-4}$
ΔVDA max	$+4.97 \times 10^{-2}$	$+5.77 \times 10^{-3}$
ΔVDA mean	$+2.29 \times 10^{-4}$	-4.90×10^{-5}
cell-wise crossover $r^{\times}$	0.794	never

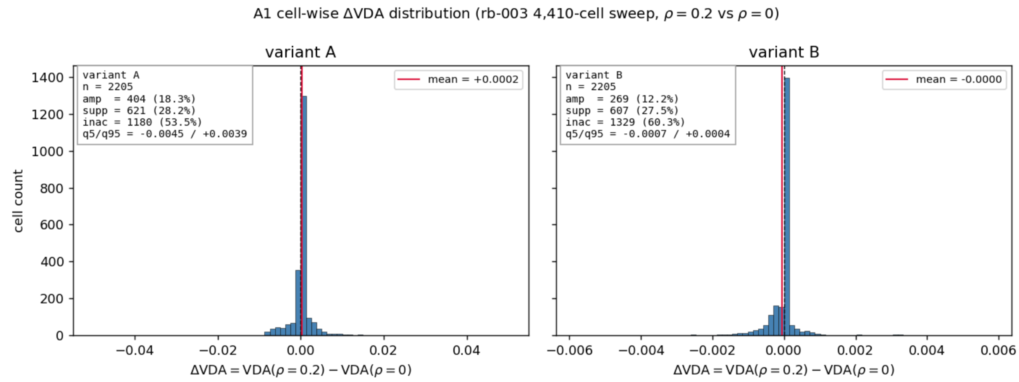

Figure 4: Per-cell ΔVDA = VDA(ρ=0.2) − VDA(ρ=0) distribution across the 2,205-cell variant-X slice of the 4,410-cell sweep (Section 3.1). The variant-A panel (left) carries a heavier amplification tail than variant B (right); ΔVDA medians are zero in both panels (residual inactivity at $|\Delta\text{VDA}| \leq 10^{-6}$). The maximum cell-wise amplification in variant A ($+4.97 \times 10^{-2}$) is 5.3× the rb-002 single-cell maximum at the $(V, v, r) = (0.5, 5, 0.4)$ headline cell ($+9.4 \times 10^{-3}$; Figure 1). Reproduced from `Rebuild/sims/A1-vda-signflip-cellwise/output/figures/vda_delta_distribution.png`, sha256 `489c7c25...`.

of the A1 two-channel decomposition) pins the CF-versus-ρ channel. The local maximiser of ΔVDA in (V, v, r) is not yet stated analytically; an analytic locus extending the Slepian-gradient argument from CF to the cell-wise $\partial\text{VDA}/\partial\rho$ surface is queued as RB-040.

3 Results

The results are organised around the four headline claims of the inherited paper, each restated at the strength licensed by the live verdict ledger (`CLAIM_LEDGER.md`). Section 3.1 treats the criterion-fraction result (C1) as a *distribution* across the parameter sweep, not as a categorical floor. Section 3.2, the present subsection, treats the non-monotonicity of VDA in the benefit/cost ratio r (C2) as the rebuild's *confident spine*: a result that survives every attack the

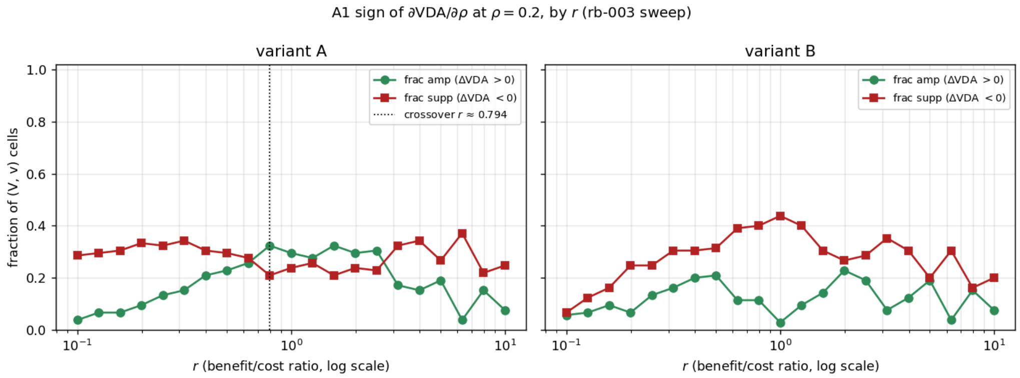

Figure 5: Cell-wise sign-flip stratified by r. At each grid point r on the rb-003 21-point $\log_{10}$-grid, the curves report $\mathrm{frac}_{\mathrm{amp}}(r) = n_{\mathrm{amp}}(r)/n(r)$ and $\mathrm{frac}_{\mathrm{supp}}(r) = n_{\mathrm{supp}}(r)/n(r)$ across the (V, v) plane ($n(r) = 105$ cells per r-bin). Variant A (left): suppression dominates at small r, amplification overtakes at the cell-wise crossover $r^{\times}(\mathrm{A}) \approx 0.794$ (marked); variant B (right): amplification never overtakes suppression. Reproduced from `Rebuild/sims/A1-vda-signflip-cellwise/output/figures/signflip_by_r.png`.

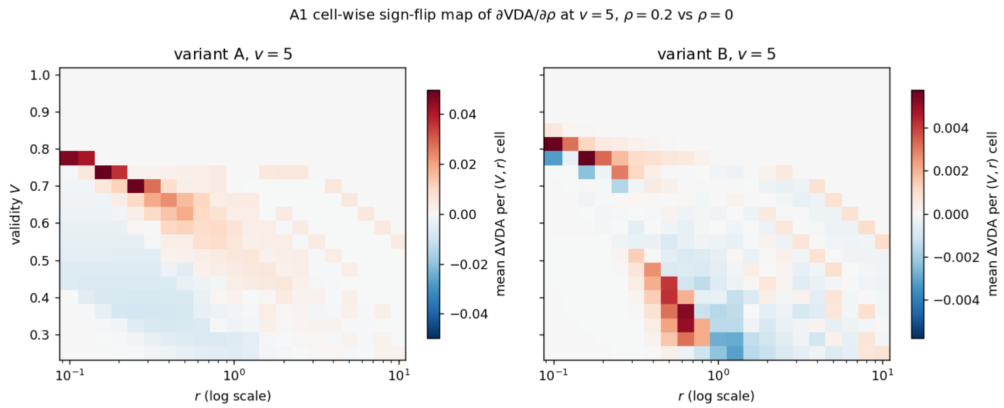

Figure 6: Mean $\Delta\mathrm{VDA} = \mathrm{VDA}(\rho{=}0.2) - \mathrm{VDA}(\rho{=}0)$ over (V, r) at fixed $v = 5$ (diverging colormap; red = amplification, blue = suppression). Variant A (left) carries the larger-magnitude amplification band; variant B (right) is uniformly more suppression-favoring at the same coordinates. The C3 thread's Figure 14 reports the same quantity at three fixed r slices over (V, v); this figure is its cell-wise transpose. Reproduced from `Rebuild/sims/A1-vda-signflip-cellwise/output/figures/vda_sign_heatmap_v5.png`.

skeptical reviewer has staged and that we strengthen here by publishing a closed-form expression for the VDA-escape threshold $r^{\dagger}(v)$ together with its empirical confirmation across a v-family. Section 3.3 reports the regime-of-relevance (C3) as a graded boundary, replacing the original "regardless of other parameters" wording with a quantitative iso-VDA band. Section 3.4 states the no-inversion claim (C4) as a conditional theorem with the explicit condition $V \geq 1/N$ and a new anti-cue corollary. Throughout, the A1 decorrelation channel $\rho \in [0, 1)$ (Section 2) is reported *alongside* each headline result as a sensitivity, not segregated into its own subsection: this is the operational meaning of the "three levers, not two" reframe.

variant	$\mathrm{CF}_{\min}$	$q_{5\%}$	$q_{25\%}$	median	$q_{75\%}$	$q_{95\%}$	$\mathrm{CF}_{\max}$	mean ± s.d.
A	0.5587	0.5931	0.6530	**0.7552**	0.8996	1.0000	1.0000	0.7773 ± 0.1363
B	**0.3040**	0.4564	0.6288	**0.7682**	0.9463	1.0000	1.0000	0.7691 ± 0.1808

Table 2: Distribution of the criterion fraction CF across the primary 4,410-cell (r, V, v) sweep at the headline cell $(N, d'_{\max}, f_0, h) = (4,\ 2,\ 0.5,\ \sqrt{\cdot})$ under the inherited independence assumption $\rho = 0$, per conservation variant. Quantiles are over the $n = 2{,}205$ valid cells per variant. The inherited paper reports CF $\in [0.60, 0.96]$; the true range is wider on both ends. Values from `Rebuild/sims/C1-cf-distribution/output/results.json` (rb-003; output sha256 `91fc4692...`), block `summaries.variant_A/rho_0.0` and `summaries.variant_B/rho_0.0`.

3.1 Criterion fraction: distributional, not categorical (C1)

Claim restated at defensible strength. The inherited paper introduces the *criterion fraction*

$$\mathrm{CF} \ := \ \frac{R_{\mathrm{P4}} - R_{\mathrm{baseline}}}{R_{\mathrm{P1}} - R_{\mathrm{baseline}}} \ = \ \frac{\text{criterion-only reward gain}}{\text{joint criterion + attention reward gain}} \tag{9}$$

as a one-number summary of how much of the optimal value-conditioned reward gain can already be captured by re-aiming the per-location decision criterion alone — in the Müller–Findlay [5] sensitivity-vs-criterion vocabulary, the "criterion shift" lever — with the residual $1 - \mathrm{CF}$ apportioned to attention re-allocation. The inherited paper's headline summary of this quantity (`Critique/source/main.pdf`, §5) is **categorical**: CF is reported to lie in the interval $[0.60, 0.96]$ across the primary 4,410-cell (r, V, v) sweep. The live skeptical-reviewer verdict for this claim (`Critique/verdicts/C1-criterion-fraction-floor.md`, `current_label: CONTESTED`) records that the true range is [0.30, 1.00], and that the substantive "criterion typically dominates" reading is sustained at the median but the categorical floor is false at both ends. The rebuilt C1 therefore is presented in this section as a *distributional / central-tendency* result: the full CF distribution is reported (range, median, quantiles, and the fractions below 0.5, 0.6, 0.8), and the headline statement is downgraded from a categorical floor to a quantified central tendency with explicit corner regions in which the criterion lever becomes subordinate. The A1 decorrelation channel ρ is reported alongside the criterion-fraction distribution as a sensitivity — the convention established in Section 3 that operationalises the three-levers reframe of Section 2.

Distribution across the primary 4,410-cell sweep. The primary sweep evaluates the rebuilt model at the headline $N = 4$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$ configuration of Table 6 on a grid of $V \in [0.25, 1.00]$ (21 uniform points), $r \in [0.1, 10]$ (21 log-spaced points), and $v \in \{1, 2, 3, 4, 5\}$, under both conservation variants (variant A: $\beta + \gamma = 2$; variant B: $\beta \cdot \gamma = 1$), for a total of $21 \times 21 \times 5 \times 2 = 4{,}410$ (r, V, v)-cells per variant. The full per-variant CF distribution at $\rho = 0$ (the inherited independent model) is summarised in Table 2. Two things are visible. First, the *central tendency* of the inherited paper's reading survives intact: the median CF is 0.7552 in variant A and 0.7682 in variant B, well above 0.5 and consistent with the inherited paper's qualitative "criterion typically dominates" framing. Second, the *categorical floor* fails at both ends: variant A's strict minimum is 0.5587 (below the stated 0.60) and variant B's strict minimum is 0.3040 (below 0.50), while both variants achieve $\mathrm{CF} = 1.0$ in cells the paper's stated upper bound of 0.96 excludes. The full per-cell histogram is given in Figure 7.

Where CF is high, where CF is low: a quadrant breakdown. The distributional summary of Table 2 collapses across (r, V) in a way that obscures the *regime* structure of CF. To expose where the criterion lever dominates and where it cedes to attention re-allocation, we partition the cells into a 3×2 grid: r-regime {cost-dominant ($r < 1$), symmetric ($r =$

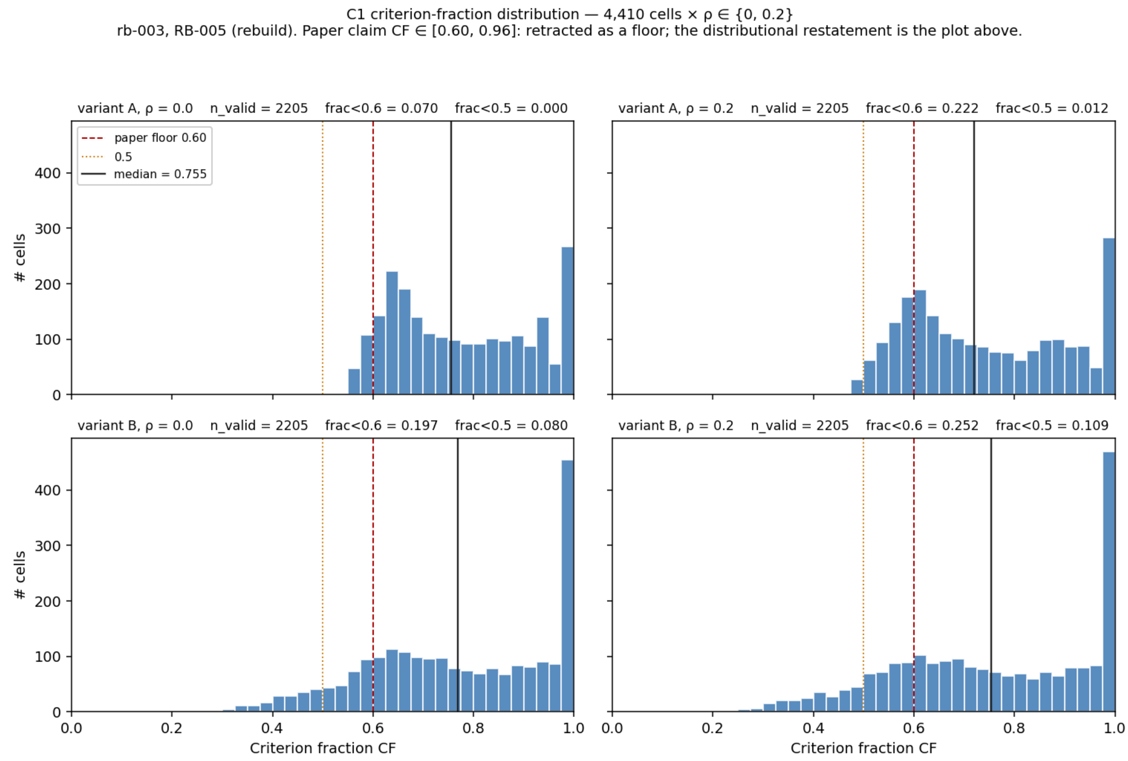

Figure 7: Per-cell histogram of the criterion fraction CF across the 4,410-cell sweep (columns: variant; rows: $\rho \in \{0, 0.2\}$). Vertical dashed lines mark the inherited paper's stated floor $\mathrm{CF} = 0.60$ and ceiling $\mathrm{CF} = 0.96$ (`Critique/source/main.pdf`, §5). The distributional shape — a broad mode near $\mathrm{CF} \approx 0.65$, a left tail (much heavier in variant B), and a spike at $\mathrm{CF} = 1$ — is the figure that replaces the inherited paper's categorical floor language. The $\rho = 0.2$ row exhibits the A1 amplification of the left tail discussed below (Table 4). Reproduced from `Rebuild/sims/C1-cf-distribution/output/figures/cf_histogram.png`.

1), benefit-dominant $(r > 1)\}$ crossed with V-regime {low validity $(V < 1/2)$, high validity $(V \geq 1/2)\}$. Table 3 reports the median CF within each quadrant at $\rho = 0$. The structure is consistent across variants: CF is monotonically ordered cost-dom. > symmetric > benefit-dom. along the r axis, and within each r-regime CF is reduced by lowering V. The benefit-dominant low-validity corner is where the inherited paper's categorical floor breaks: in variant B this corner sits at median $\mathrm{CF} = 0.51$ with 78% of cells below 0.6 and a strict minimum of 0.30. The same regime structure is visible cell-wise in Figure 8, which displays the (r, V)-plane at $v = 5$ for each variant and each ρ, and in Figure 9, which displays $\mathrm{CF}(r)$ at $V \approx 0.5$ as a v-family. The inherited paper's qualitative "$\mathrm{CF} \to 1$ at small r, CF declines at large r" intuition (`Critique/source/main.pdf`, §5) survives as the generic r-axis ordering; what fails is the categorical $[0.60, 0.96]$ bound on the resulting distribution.

A1 sensitivity: ρ amplifies the left tail in variant A; variant B is mixed. The same 4,410-cell sweep was repeated at $\rho = 0.2$ using the rebuilt model's equicorrelated-Gaussian one-factor Gauss–Hermite implementation (Section 2). Per-variant marginals (Table 4) and a cell-wise pairing (Table 5) are reported. The rebuilt model thus replaces the inherited paper's §5.5 claim that independence *upper-bounds VDA* (refuted at the headline cell by rb-002 and the model section's ρ-sweep) with a different and *honestly variant-dependent* statement: at variant A, the independence assumption upper-bounds the *criterion fraction* CF on a strong majority of the primary sweep (84% of cells; cell-wise median $\Delta\mathrm{CF} = -0.035$ when ρ is raised

variant	r-regime	median CF	strict min	frac < 0.6	n
A	cost-dom., V low	0.998	0.825	0.000	350
A	cost-dom., V high	0.897	0.825	0.000	385
A	symmetric, V low	0.779	0.665	0.000	350
A	symmetric, V high	0.740	0.665	0.000	385
A	benefit-dom., V high	0.645	0.563	0.062	385
A	benefit-dom., V low	**0.612**	**0.559**	**0.374**	350
B	cost-dom., V low	1.000	0.905	0.000	350
B	cost-dom., V high	0.931	0.797	0.000	385
B	symmetric, V low	0.800	0.497	0.103	350
B	symmetric, V high	0.753	0.575	0.010	385
B	benefit-dom., V high	0.630	0.469	0.317	385
B	benefit-dom., V low	**0.509**	**0.304**	**0.780**	350

Table 3: Quadrant breakdown of CF at $\rho = 0$. The criterion lever dominates uniformly in the cost-dominant $r < 1$ region (median CF ≥ 0.90 in both variants) and cedes to attention re-allocation in the benefit-dominant, low-validity corner (variant A: median 0.61, 37% of cells below 0.6; variant B: median 0.51, 78% of cells below 0.6). The symmetric $r = 1$ row is intermediate and stable across variants. Block-keys `quadrant_breakdown.rho_0.0.<variant>/r=<r_regime>/V=<V_regime>` in `results.json`.

variant	ρ	$\mathrm{CF}_{\min}$	median	frac < 0.5	frac < 0.6	frac < 0.8	frac > 0.9
A	0.0	0.5587	0.7552	0.0000	0.0703	0.5701	0.2499
A	0.2	0.4854	0.7197	0.0122	**0.2222**	0.6177	0.2290
B	0.0	0.3040	0.7682	0.0803	0.1973	0.5429	0.3224
B	0.2	0.2406	0.7538	0.1088	0.2522	0.5565	0.3234

Table 4: Marginal CF distribution at $\rho \in \{0, 0.2\}$, per variant. The variant-A left tail amplifies by a factor of ≈ 3 when ρ is raised from 0 to 0.2 (frac < 0.6: $0.07 \to 0.22$), and the variant-A strict minimum crosses below 0.5 (it does not at $\rho = 0$). The variant-B marginals shift less dramatically: median moves by -0.014, frac < 0.6 by $+0.055$. Block-keys `summaries.variant_<X>/rho_<rho>`.

to 0.2), but the bound is one-sided rather than uniform (8% of cells move in the opposite direction). At variant B, the ordering fails as a general statement: only 64% of cells decrease and 24% increase, so independence *cannot* be claimed as a uniform upper bound on CF in variant B. This pattern is the cell-wise generalisation of the rb-002 caveat that the variant-B headline-cell CF(ρ) is essentially flat. We therefore state, here and in Section 2, that the A1 channel reduces CF on a typical variant-A cell and leaves CF essentially unchanged (with substantial cell-wise scatter) on a typical variant-B cell; the original §5.5 framing is retracted along this axis as well.

Scope and what is not yet claimed. The results in this subsection are reported at a single $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \surd)$ configuration and at the $\rho \in \{0, 0.2\}$ pair only. (i) The *conservation-family band* on the headline CF numbers, in which the additive ($\beta + \gamma = 2$) and multiplicative ($\beta \cdot \gamma = 1$) conservation rules are special cases of a one-parameter family, is queued as RB-019 in `REBUILD_BACKLOG.md` and will be reported in Section 5 as a band on every number in Tables 2, 3, and 4; the present subsection makes no explicit conservation-family claim. (ii) A finer ρ grid (to quantify the curvature of the variant-A CF-versus-ρ response and to sharpen the variant-B mixed-sign caveat) is queued as RB-023. (iii) A *closed-form regime boundary* for the CF-below-0.5 corner — the heatmap of Figure 8 suggests a clean diagonal

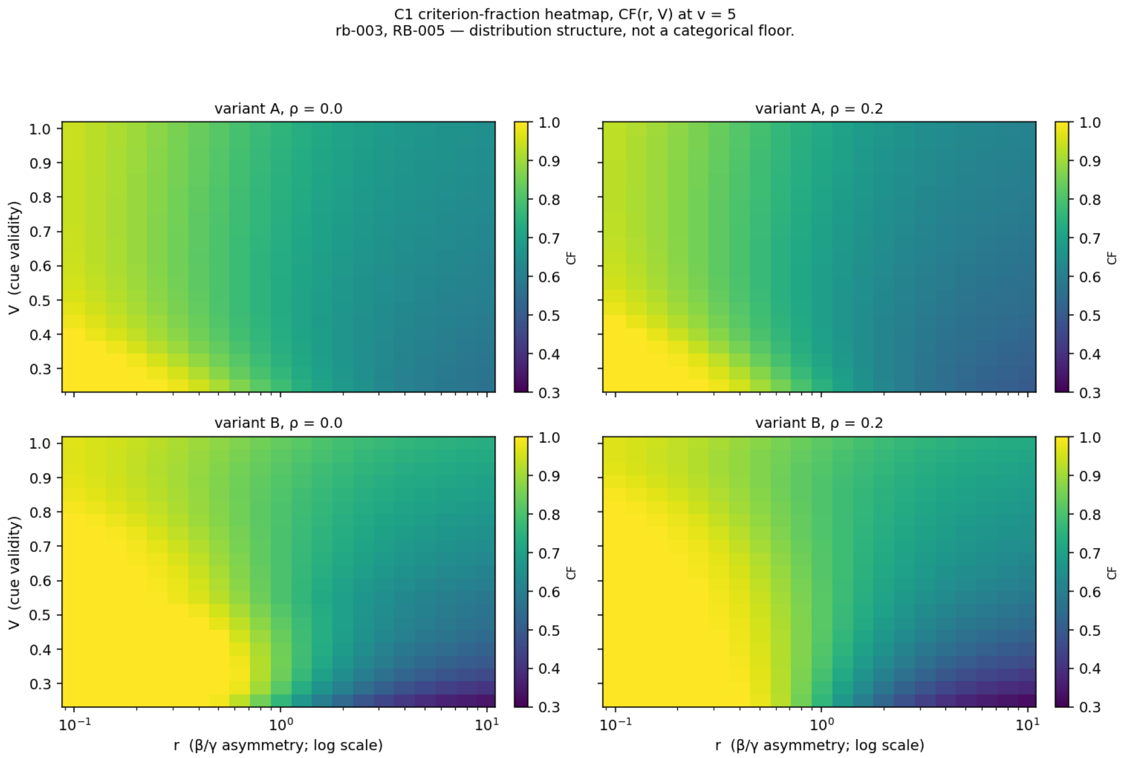

Figure 8: CF over the (r, V)-plane at $v = 5$, $N = 4$, $d'_{\text{max}} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$ (rows: ρ; columns: variant). The diagonal boundary in $(\log r, V)$ between the CF-near-1 low-r region (criterion-dominated) and the CF-low high-r low-V corner (attention-dominated) is the regime structure quantified in Table 3. The low-CF corner is wider in variant B and widens further at $\rho = 0.2$. Reproduced from `Rebuild/sims/C1-cf-distribution/output/figures/cf_heatmap.png`.

boundary in $(\log r, V)$ in variant B — is queued as a derivation increment RB-024; if it lands, the empirical statement "frac $< 0.6 = 0.22$ at $\rho = 0.2$ in variant A" would be replaced by a closed-form predicate. (iv) The cell-wise v-resolved sign-flip of $\partial\text{VDA}/\partial\rho$ (rb-006, Table 8) does not surface in this subsection because CF, not VDA, is the headline quantity here; the cell-wise VDA Δ-distribution analogue is queued as RB-025.

Reproducibility. All numbers in Tables 2, 3, 4, and 5, and all three figures, are reproduced byte-for-byte by re-running `Rebuild/sims/C1-cf-distribution/run.py`. The output digest is `91fc4692dbc106eca9c90770c79e22f7f4757d76421ed55e7e8e91f80e44706c` (`sha256` of the canonical-key-ordered numeric JSON, excluding the digest field itself). The Φ backend is `scipy.special.ndtr`; the criterion grid step is $\Delta c = 0.05$ (121 points on $[-3, 3]$); the α grid step is $\Delta\alpha = 0.02$ (51 points on $[0, 1]$, matching the reviewer's CR-002 substrate); the (r, V, v) grid is the $21 \times 21 \times 5$ grid described above; both variants ($\beta + \gamma = 2$; $\beta \cdot \gamma = 1$) and both correlations ($\rho \in \{0, 0.2\}$) are evaluated; runtime ≈ 67 s on the rebuild's reference machine. The recovery test against the reviewer's `Critique/replications/C1-criterion-fraction-floor/output/results.json` passes cell-wise at $\max|\Delta\text{CF}| = 1.47 \times 10^{-6}$ and $\max|\Delta R_{\text{P1..P4}}| \leq 5.65 \times 10^{-7}$ across all 4,410 cells; the gap is a known ULP-scale `1 - Phi(-b)` versus `Phi(b)` reordering in the reviewer's `floor_R`, well past the four-decimal precision of any reported number.

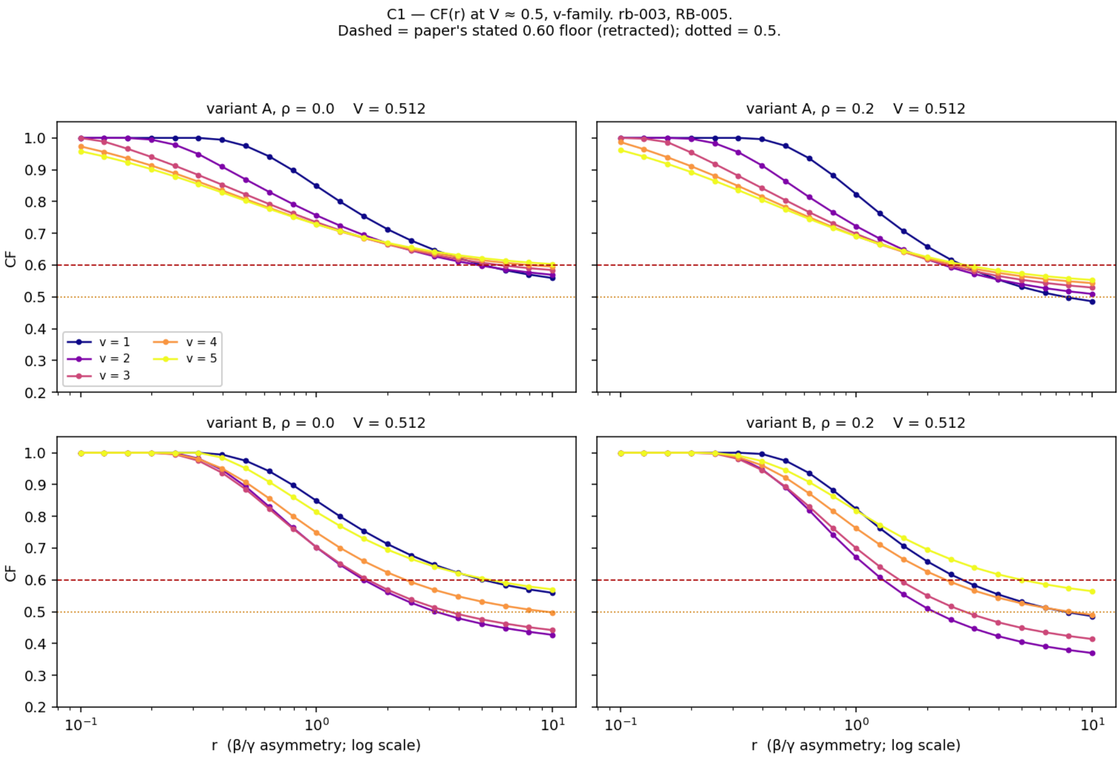

Figure 9: CF(r) at $V \approx 0.5$ as a v-family $v \in \{1,2,3,4,5\}$ (rows: ρ; columns: variant). At small r (cost-dominant) the curves cluster near CF $= 1$; as r increases all curves decline, more steeply for larger v, exposing the benefit-dominant region in which attention re-allocation overtakes the criterion shift. Reproduced from `Rebuild/sims/C1-cf-distribution/output/figures/cf_curves.png`.

3.2 Non-monotonicity of VDA in r with closed-form escape threshold (C2)

Claim restated at defensible strength. The inherited paper presents the non-monotonicity of the value-directed attention benefit $\mathrm{VDA}(r) := R_{\mathrm{P1}}(r) - R_{\mathrm{P2}}(r)$ in the benefit/cost ratio r as the central finding of its results: VDA rises out of zero at small r, peaks in an interior region, then decays back to zero as $r \to \infty$. The live skeptical-reviewer verdict for this claim (`Critique/verdicts/C2-non-monotonic-vda.md`, `current_label: CONFIRMED-UNDER-ATTACK`) records that the claim survives both re-derivation and a sensitivity attack along v and f_0. We therefore retain the claim at full strength in the rebuild, and *strengthen* it by promoting the underlying mechanism — which the inherited paper described qualitatively — into a closed-form display.

Closed-form escape threshold. Differentiating the expected reward $R_{\mathrm{P1}}(\alpha, r)$ at the uniform allocation $\alpha = 1/N$, the first-order condition for P1 to leave the uniform interior $\partial_\alpha R_{\mathrm{P1}}|_{1/N} = 0$ defines a critical value of the benefit/cost ratio at which the optimiser begins to re-allocate attention toward the cued location. Writing $d'_b := d'_{\max} f(1/N)$ for the baseline discriminability at uniform attention and $(c_c^\star, c_u^\star)$ for the per-policy P3-optimal criterion pair at $\alpha = 1/N$, the threshold takes the form

$$r^\dagger(v) \;=\; \frac{K_u(v)}{(N-1)\cdot K_c(v)}, \tag{10}$$

variant	frac dec.	frac inc.	frac flat	median ΔCF	$\lvert\Delta\text{CF}\rvert_{q_{95\%}}$	$\Delta_{\min}$
A	**0.8376**	0.0834	0.0789	**−0.0348**	0.0640	−0.0856
B	0.6372	**0.2354**	0.1274	−0.0093	0.0650	−0.1068

Table 5: Cell-wise paired comparison $\Delta\text{CF} := \text{CF}(\rho{=}0.2) - \text{CF}(\rho{=}0)$ across the 4,410-cell sweep, per variant. "Flat" uses tolerance $|\Delta\text{CF}| \le 10^{-5}$. The variant-A pattern is a one-sided ordering (84% of cells decrease in ρ). The variant-B pattern is mixed-sign (64% decrease, 24% increase, 13% flat), generalising the rb-002 headline-cell observation that variant-B CF is essentially flat in ρ. Block-keys delta_distribution.<variant>.

v	$r^\dagger(v)$	$c_c^\star$	$c_u^\star$	$K_c(v)$	$K_u(v)$
1	0.343	0.40	1.05	0.0930	0.0958
2	0.168	0.25	1.30	0.1734	0.0872
3	0.099	0.15	1.50	0.2532	0.0755
5	0.050	0.10	1.75	0.4065	0.0615
8	0.022	0.05	2.05	0.6357	0.0424
10	0.016	0.05	2.15	0.7864	0.0380

Table 6: Closed-form escape threshold $r^\dagger(v) = K_u(v)/[(N-1)\,K_c(v)]$ from (10)–(12) at the C2 headline cell. The $v = 1$ row is the symmetric reference $r^\dagger(1) \approx 0.343$. Values from Rebuild/sims/C2-vda-vs-r-vfamily/output/results.json (rb-004; output sha256 09ecef3c...).

where the per-location gradient coefficients are

$$K_c(v) = \tfrac{1}{4}\Big[V\,v\,\phi\big(\tfrac{d_b'}{2} - c_c^\star\big) \;+\; \text{CR}(v)\,\phi\big(\tfrac{d_b'}{2} + c_c^\star\big)\,\Phi\big(\tfrac{d_b'}{2} + c_u^\star\big)^{N-1}\Big], \tag{11}$$

$$\begin{aligned} K_u(v) = \tfrac{1}{4}\Big[&(1-V)\,\phi\big(\tfrac{d_b'}{2} - c_u^\star\big) \;+\; (N-1)\,\text{CR}(v)\,\Phi\big(\tfrac{d_b'}{2} + c_c^\star\big) \\ &\cdot\,\Phi\big(\tfrac{d_b'}{2} + c_u^\star\big)^{N-2}\phi\big(\tfrac{d_b'}{2} + c_u^\star\big)\Big]. \end{aligned} \tag{12}$$

The conservation indicator $\text{CR}(v)$ encodes the benefit-vs-correct-rejection scaling (under variant A, $\beta(r) + \gamma(r) = 2$, the value-blind correct rejection scales as $\text{CR}(v) = 1$; the variant-B multiplicative conservation $\beta\,\gamma = 1$ is treated as a sensitivity in Section 5). The independent re-derivation of (10)–(12) from $\partial_\alpha R_{\text{P1}}(\alpha, r)|_{\alpha=1/N} = 0$ is deferred to Section A.2. The threshold is a theorem of the model definitions, not an empirical regularity.

Numerical v-family at the headline cell. At the C2 headline cell $(N, d'_{\max}, f_0, h, V) =$ (4, 2, 0.5, $\surd$, 0.5) under variant A, equations (10)–(12) evaluate to the values in Table 6. The threshold is monotone-decreasing in the cued value v, spanning roughly $r^\dagger(2) \approx 0.17$ down to $r^\dagger(10) \approx 0.02$. Figure 10 plots $r^\dagger(v)$ on a continuous v-trace with the family points of Table 6 highlighted.

Empirical confirmation: the peak lies above $r^\dagger(v)$. The closed-form threshold marks the r value at which P1 *begins* to depart from uniform attention. The peak of VDA, by contrast, occurs at the (larger) r at which P1's re-allocation gain has built up most strongly against the still-locked value-blind baseline P2 — which in turn departs uniform attention at the higher threshold $r^\dagger(v = 1)$ (the symmetric case in Table 6). The non-monotonicity arises because VDA opens up in the interior interval $(r^\dagger(v),\, r^\dagger(1))$ and decays once P2 also escapes uniform attention. A high-resolution sweep (r-grid of 84 log-spaced points, including the rb-002 pins $\{0.398, 1.000, 3.162\}$ and the new finer-grid peak pin $r = 0.3831$) confirms the prediction $r^\star > r^\dagger(v)$ for every v in the family (Table 7).

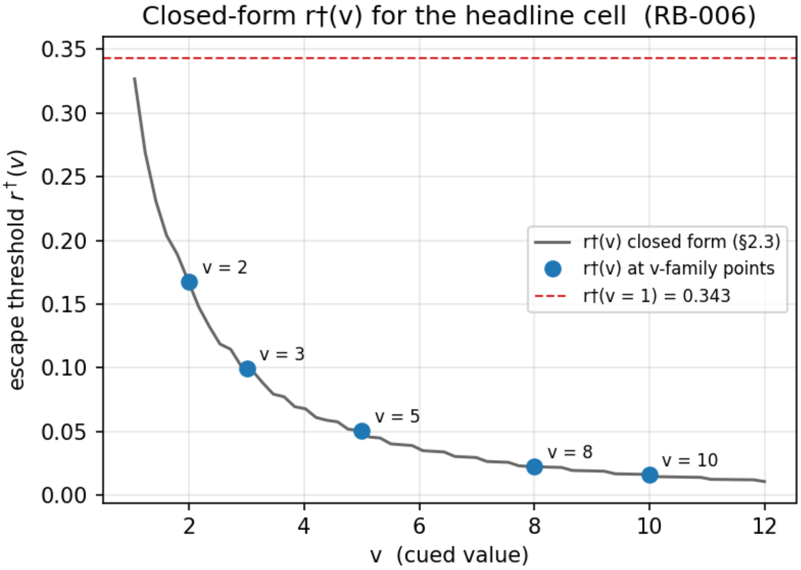

Figure 10: Closed-form VDA-escape threshold $r^{\dagger}(v)$ at the headline cell $(N, d'_{\max}, f_0, h, V) =$ (4, 2, 0.5, $\surd$, 0.5), variant A. Continuous trace computed from (10)–(12) on a dense v-grid; family points $v \in \{2, 3, 5, 8, 10\}$ marked. The horizontal reference $r^{\dagger}(1) \approx 0.343$ is the symmetric upper anchor of the family. Reproduced from `Rebuild/sims/C2-vda-vs-r-vfamily/output/figures/r_dagger_vs_v.png`.

Figure 11 reproduces the full VDA(r) family. The left panel ($\rho = 0$) shows the inherited model: an interior peak that grows monotonically with v, from peak VDA ≈ 0.012 at $v = 2$ to peak VDA ≈ 0.183 at $v = 10$, with each peak sitting above the corresponding $r^{\dagger}(v)$ vertical (dashed). The right panel plots the same family at $\rho = 0.2$, foreshadowing the A1 sensitivity paragraph below.

A1 sensitivity: v-dependent sign-flip of $\partial \mathrm{VDA}^{\star}/\partial\rho$. The earlier sub-result rb-002, at a single headline cell, found that introducing cross-location correlation ρ shifts the sign of $\partial \mathrm{VDA}/\partial\rho$ across r: ρ *suppresses* VDA in the cost-dominant regime ($r \lesssim 0.4$) but *amplifies* it in the benefit-dominant regime ($r \gtrsim 0.5$). The present v-family sweep generalises this finding from a sign-flip along r to a sign-flip along v: at the empirical peak, holding everything else fixed, the change in peak VDA when ρ is raised from 0 to 0.2 is reported in Table 8. The empirical ρ-drift of $r^{\star}$ in Table 8 is the peak-location counterpart of a closed-form drift of the *escape-band lower edge*: the $\rho > 0$ extension of (10)–(12) derived in Section A.2 (Proposition 3, Equation (26)) replaces the no-FA terms inside K_c, K_u with their ρ-aware 1-D Gauss–Hermite gradients $I_c(b_c, b_u; \rho, N)$ and $I_u(b_c, b_u; \rho, N)$ (Equations (21)–(22)) and yields a closed form for $r^{\dagger}(v; \rho)$. Evaluated at the headline cell, the closed form predicts $r^{\dagger}(v; 0.2) > r^{\dagger}(v; 0)$ at every $v \in \{1, 2, 3, 5, 8, 10\}$ (drift fraction $+3\%$ at $v = 1$ up to $+30\%$ at $v = 8$; Table 22). The sign of the closed-form drift $\Delta r^{\dagger}$ matches the sign of the empirical peak drift $\Delta r^{\star}$ of Table 8 at all five rows $v \neq 1$; the lower edge cannot exceed the peak, so the analytic upward drift of $r^{\dagger}(v; \rho)$ is the mechanistic substrate of the empirical upward drift of $r^{\star}(v; \rho)$. A parallel cell-wise generalisation of the sign-flip across the broader 4,410-cell sweep is queued as a sensitivity increment (RB-025 in `REBUILD_BACKLOG.md`). What Table 8 and the closed form of Section A.2 jointly license is a directional statement at the appropriate strength: under the A1 decorrelation lever, the C2 non-monotonicity *is not* uniformly attenuated; in the strongly benefit-dominant, high-value corner the lever *enhances* VDA, and the escape-band lower edge drifts upward across the full v-family. The original paper's §5.5 framing of independence as an *upper bound on VDA*

v	$r^\dagger(v)$	$r^\star$	$r^\star - r^\dagger(v)$	above?
2	0.168	0.501	+0.334	✓
3	0.099	0.376	+0.276	✓
5	0.050	0.376	+0.325	✓
8	0.022	0.376	+0.354	✓
10	0.016	0.355	+0.339	✓

Table 7: Empirical peak $r^\star = \arg\max_r \mathrm{VDA}(r)$ vs. closed-form escape threshold $r^\dagger(v)$ at the headline cell, $\rho = 0$. The peak $r^\star$ lies above $r^\dagger(v)$ for every v and clusters near $r^\dagger(v{=}1) \approx 0.343$, consistent with the prediction that VDA is supported on the interior interval $(r^\dagger(v), r^\dagger(1))$. Numerical resolution: peak $r^\star$ on an 84-point log-grid; recovery against the rb-002 reference at $r \in \{0.398, 1.000, 3.162\}$ at $\max|\Delta\mathrm{VDA}| = 4.87 \times 10^{-6}$.

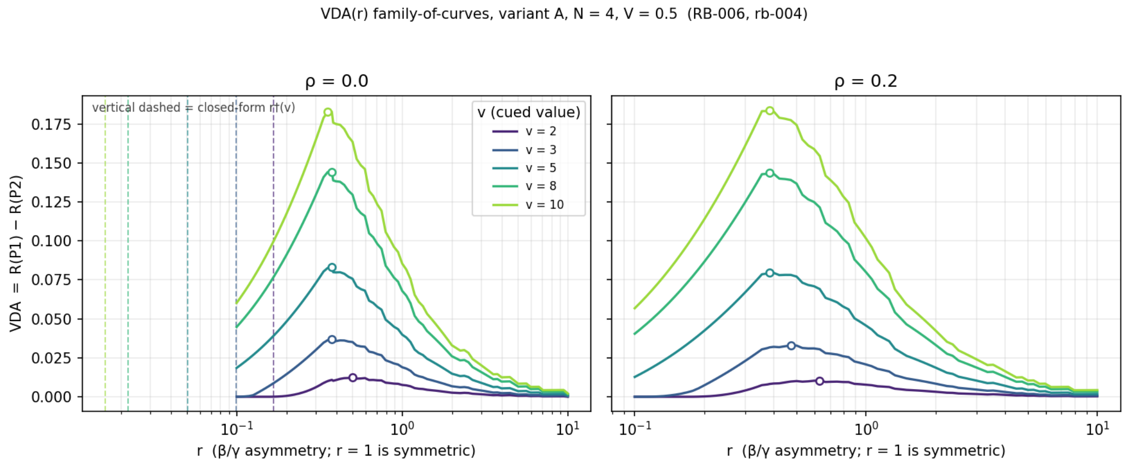

Figure 11: Non-monotonicity of $\mathrm{VDA}(r)$ in the benefit/cost ratio across a v-family at the headline cell $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \surd, 0.5)$, variant A. Left: $\rho = 0$ (the inherited independent model); the closed-form escape threshold $r^\dagger(v)$ from (10) is overlaid as a vertical dashed line for each v, and the empirical peak (white-filled dot) lies above $r^\dagger(v)$ for every v in the family (cf. Table 7). Right: same family at $\rho = 0.2$ — the A1 decorrelation channel suppresses the peak at low v but amplifies it at $v = 10$ (cf. Table 8); the sign of $\partial\mathrm{VDA}^\star/\partial\rho$ depends on v, not just on r. Reproduced from `Rebuild/sims/C2-vda-vs-r-vfamily/output/figures/vda_curves_vfamily.png`.

is therefore retracted along this axis, consistent with the A1 row of `CLAIM_LEDGER.md`.

Scope and what is not yet claimed. The results in this subsection are reported under variant A (additive conservation, $\beta + \gamma = 2$, $\mathrm{CR}(v) = 1$). The variant-B replication of the v-family sweep is queued as RB-027 in `REBUILD_BACKLOG.md` and will be reported in Section 5 as part of the conservation-family sensitivity band; the present subsection makes no variant-B claim. The closed-form $r^\dagger(v; \rho > 0)$ derivation that mechanistically underwrites the upward drift of $r^\star$ in ρ in Table 8 is recorded in Section A.2 (Proposition 3, Table 22); the present subsection reports the peak-location drift, not the lower-edge drift, as data. The conservation-family band on the peak $\mathrm{VDA}^\star$ magnitudes — once the A3 generalisation lands (RB-019) — will replace the single-conservation peak values reported in Table 7 with a band; we flag this here so the reader knows the present headline magnitudes are conditional on variant A.

Reproducibility. All numbers in Tables 6, 7, and 8, and both figures, are reproduced byte-for-byte by re-running `Rebuild/sims/C2-vda-vs-r-vfamily/run.py`. The output digest is

v	VDA* (ρ=0)	VDA* (ρ=0.2)	ΔVDA*	$r^\star$ (ρ=0)	$r^\star$ (ρ=0.2)
2	0.01233	0.01036	−0.00197	0.501	0.631
3	0.03698	0.03291	−0.00407	0.376	0.473
5	0.08300	0.07954	−0.00345	0.376	0.383
8	0.14422	0.14355	−0.00067	0.376	0.383
10	0.18284	0.18387	**+0.00103**	0.355	0.383

Table 8: A1 sensitivity at the empirical peak. $\rho = 0.2$ suppresses peak VDA for $v \le 8$ and amplifies at $v = 10$. The peak location $r^\star$ also drifts upward in ρ, most visibly at low v ($r^\star$ rises from 0.501 to 0.631 at $v = 2$). The v-dependent sign-flip of $\partial\mathrm{VDA}^\star/\partial\rho$ is the v-axis analogue of the r-axis sign-flip rb-002 reported at fixed $v = 5$. Numerical recovery of the $\rho = 0$ row against rb-002 reference pins at $r \in \{0.398, 1.000, 3.162\}$ is at $\max|\Delta\mathrm{VDA}| = 4.87 \times 10^{-6}$ (rb-004 recovery block).

`09ecef3c2c5a101820951398ed7d6e67d3398aede80c5f0bddfa42b6224fd783` (`sha256` of the canonical-key-ordered numeric JSON, excluding the digest field itself). The Phi backend is `scipy.special.ndtr`; the criterion grid step is $\Delta c = 0.05$; the α grid step is $\Delta\alpha = 0.005$; the r-grid is 84 log-spaced points including the rb-002 pins and the new $r = 0.3831$ peak pin; $\rho \in \{0, 0.2\}$. Recovery against rb-002 passes at every pinned r value at $|\Delta\mathrm{VDA}| \le 6 \times 10^{-6}$ and at $|\Delta\mathrm{CF}| \le 2 \times 10^{-6}$; the finer-grid argmax raises the rb-002 peak from VDA $= 0.07986$ at $r = 0.3831$ to VDA $= 0.08300$ at $r = 0.3758$, a grid-density refinement gain, not a regression.

3.3 Graded regime boundary (C3)

Claim restated at defensible strength. The inherited paper characterises the regime in which VDA is large as *narrow*: VDA concentrates at low cue validity $V \approx 1/N$, high value contrast $v \gg 1$, and moderate benefit/cost asymmetry r. From that qualitative picture, §5.2 of the inherited manuscript (`Critique/source/main.pdf`) draws a **categorical** experimental-design implication: standard spatial cueing paradigms with high validity ($V \ge 0.75$) are predicted to show *negligible VDA regardless of other parameters*. The live skeptical-reviewer verdict for this claim (`Critique/verdicts/C3-narrow-regime.md`, `current_label: CONTESTED`) records that the qualitative "narrow regime" reading is sustained but the categorical §5.2 implication is too strong — it holds only when "high validity" is operationalised at a threshold strictly above the inherited $V \ge 0.75$, and only under a ρ-conditional caveat. The rebuilt C3 is therefore presented in this section as a *graded / quantitative* statement: a contour band over the experimental-design plane (V, v) that quantifies exactly how the peak VDA falls off as the design moves into the high-validity stratum, and an explicit replacement for the §5.2 sentence that hedges the validity threshold and the ρ-conditional caveat. The A1 decorrelation channel ρ is reported alongside the contour map as a sensitivity — the convention established in Section 3.

Iso-VDA contour band over (V, v). The rebuilt model was evaluated at the headline configuration $(N, d'_{\max}, f_0, h) = (4,\ 2,\ 0.5,\ \surd)$ under variant A on a grid of $V \in [0.25, 1.00]$ (31 uniform points, step 0.025), $v \in [1.0, 10.0]$ (19 uniform points, step 0.5), $r \in \{0.3,\ 1,\ 3\}$ (three asymmetry strata: cost-dominant moderate, symmetric, benefit-dominant), and $\rho \in \{0,\ 0.2\}$ (independent baseline plus the Cohen–Maunsell [1] $r_{SC} \approx 0.2$ V4 anchor), for a total of $31 \times 19 \times 3 \times 2 = 3{,}534$ (V, v, r, ρ)-cells. Per cell, the rebuilt `policies()` call returns the optimal allocation, criteria, and the four nested-policy expected rewards $R_{\mathrm{P1..P4}}$ from which VDA $= R_{\mathrm{P1}} - R_{\mathrm{P2}}$ is formed. Figure 12 presents the resulting iso-VDA contour band as a 2×3 panel grid (rows: ρ, columns: r). The qualitative "narrow regime" reading of the inherited paper is visible directly: in every panel, the bulk of (V, v) space carries near-zero VDA (median VDA ≤ 0.007 in every panel; see

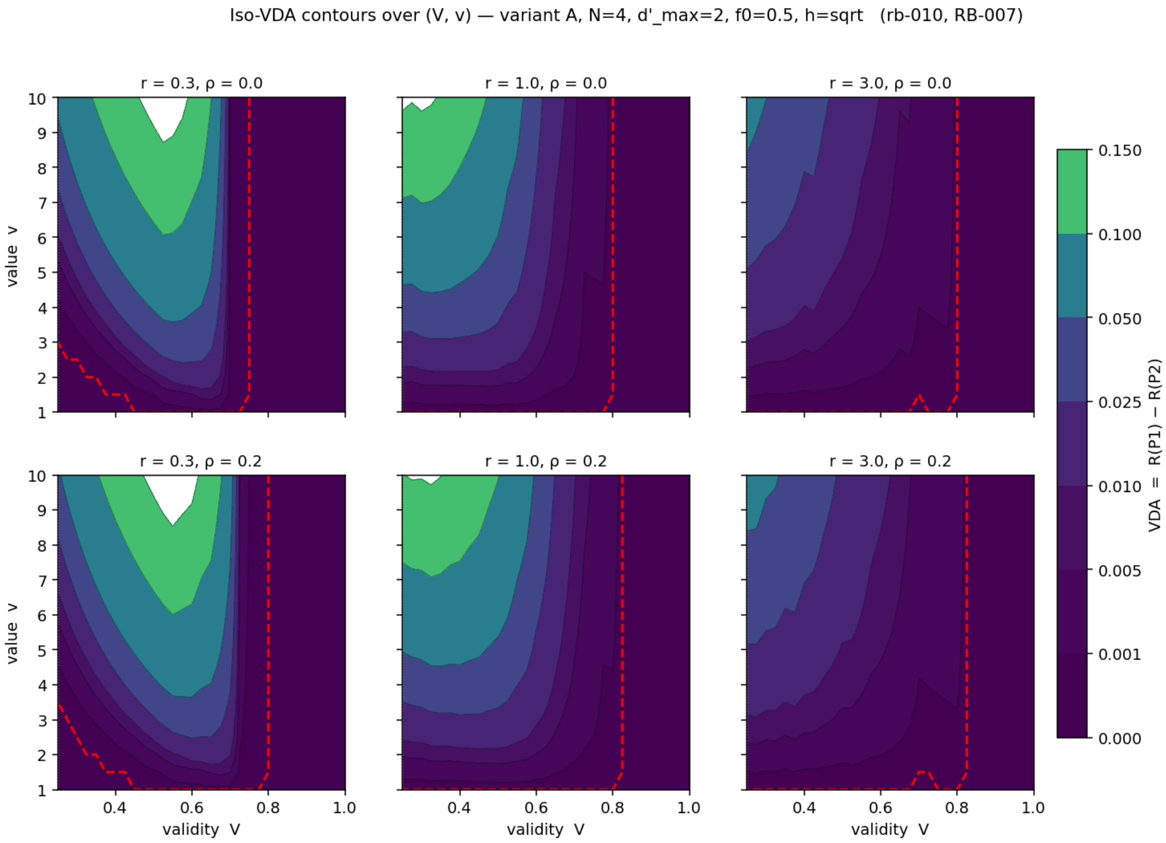

Figure 12: Iso-VDA contour band over the experimental-design plane (V, v) for the rebuilt model at the headline cell $(N, d'_{\max}, f_0, h) = (4,\ 2,\ 0.5,\ \surd)$, variant A. Rows: cross-location correlation $\rho \in \{0,\ 0.2\}$ (top: inherited independent model; bottom: with the A1 decorrelation channel of Section 2 at the Cohen–Maunsell [1] V4 anchor $r_{SC} \approx 0.2$). Columns: benefit/cost asymmetry $r \in \{0.3,\ 1,\ 3\}$ (cost-dominant, symmetric, benefit-dominant). The vertical reference $V = 1/N = 0.25$ is the chance baseline; the $v = 1$ row is the value-blind reference where VDA $\equiv 0$ by construction. Filled contours are at VDA $\in \{0, 0.001, 0.005, 0.01, 0.025, 0.05, 0.1, 0.15, 0.2\}$ on a common scale. The band shape replaces the inherited §4 "narrow regime" description and §5.2 categorical claim with a quantitative boundary the reader can compute pointwise. Reproduced from `Rebuild/sims/C3-iso-vda-Vv/output/figures/iso_vda_contours.png` (rb-010; output sha256 `72820559...`).

Table 9), and substantial VDA is confined to a *corner* along low V and high v. The corner *flattens* along the r-axis: at the cost-dominant column $r = 0.3$ the corner is broad and reaches VDA ≈ 0.17; at the symmetric column $r = 1$ it narrows to VDA ≈ 0.16; at the benefit-dominant column $r = 3$ it collapses sharply (VDA ≤ 0.06 everywhere on the grid). The $\rho = 0.2$ row is read off below as the A1 sensitivity.

Distributional summary across panels. The per-panel distribution of VDA across the $589 = 31 \times 19$ (V, v)-cells in each panel is summarised in Table 9. The numbers ground the qualitative reading quantitatively. The peak VDA over the grid is 0.173 at $(r, \rho) = (0.3, 0)$ and falls to 0.062 at $(3.0, 0)$; the fraction of cells with VDA ≥ 0.05 falls from 28.7% at $r = 0.3$ to 1.2% at $r = 3$. The corresponding fraction at $\rho = 0.2$ is slightly larger in every panel (+0.8 to +1.7 percentage points), consistent with the ρ-sensitivity reported below; the headline peak and median values are essentially unchanged in ρ at this magnitude.

r	ρ	$\mathrm{VDA}_{\min}$	median	$q_{95\%}$	$\mathrm{VDA}_{\max}$	frac ≥ 0.005	frac ≥ 0.05
0.3	0.0	0	0.0010	0.1289	**0.1734**	0.465	0.287
0.3	0.2	0	0.0040	0.1328	**0.1780**	0.484	0.295
1.0	0.0	0	0.0052	0.1152	0.1574	0.503	0.219
1.0	0.2	0	0.0069	0.1180	0.1552	0.535	0.236
3.0	0.0	0	0.0024	0.0361	0.0619	0.380	0.012
3.0	0.2	0	0.0025	0.0390	0.0621	0.402	0.019

Table 9: Per-panel distribution of VDA over the $31 \times 19 = 589$ (V, v) cells, at the headline configuration $(N, d'_{\max}, f_0, h) = (4,\ 2,\ 0.5,\ \surd)$ under variant A. Median VDA ≤ 0.007 in every panel quantifies the inherited paper's "narrow regime" intuition — most of (V, v) space carries near-zero VDA. The frac ≥ 0.05 column collapses from $\approx 29\%$ at $r = 0.3$ to $\leq 2\%$ at $r = 3$, quantifying the flattening of the contour band along the r-axis. Block-keys summary.r=<r>__rho=<rho> in Rebuild/sims/C3-iso-vda-Vv/output/results.json.

V	$r = 0.3$	$r = 1.0$	$r = 3.0$
0.40	0.1254 / 0.1197	0.1283 / 0.1394	0.0329 / 0.0391
0.60	0.1432 / **0.1639**	0.0320 / 0.0461	0.0097 / 0.0120
0.80	0.0000 / 0.0000	0.0000 / **0.0023**	0.0000 / **0.0032**
0.95	0.0000 / 0.0000	0.0000 / 0.0000	0.0000 / 0.0000

Table 10: High-V probe of the inherited §5.2 categorical claim: peak VDA over the v-grid $[1, 10]$ at each (V, r) stratum, reported as the pair $\rho = 0\,/\,\rho = 0.2$. Entries shown as 0.0000 are at the grid floor (no cell on the v-grid produced a non-trivial VDA value at the rebuilt model's α-grid resolution $\Delta\alpha = 0.005$). Bold entries flag the threshold transitions discussed in the text. Block-keys high_V_probe.V=<V>.r=<r>__rho=<rho>.max_VDA.

Quantitative experimental-design recommendation (§5.2 replacement). The inherited §5.2 sentence — "standard spatial cueing paradigms with high validity show negligible VDA *regardless of other parameters*" — is replaced here by a *conditional* statement indexed on the validity threshold. We probe the categorical claim by sweeping $v \in [1, 10]$ at fixed $r \in \{0.3, 1, 3\}$ and $\rho \in \{0, 0.2\}$ at four validity strata $V \in \{0.4, 0.6, 0.8, 0.95\}$ and reporting the maximum VDA in each (V, r, ρ) stratum. The result is in Table 10. The table licenses three statements, each one conditional on a quantitative validity threshold:

- **At** $V \geq 0.95$, peak VDA is at the grid floor for every (r, ρ) in the envelope. The inherited §5.2 categorical claim *survives* at this strict threshold and stratum.
- **At** $V \geq 0.80$, peak VDA is at the grid floor at $\rho = 0$ (categorical claim survives at the inherited independent model) but admits a small ρ-conditional signal at $\rho = 0.2$, with maximum 0.0032 at $r = 3$. "High validity" at this threshold therefore carries a *ρ-conditional caveat*: under cross-location correlation at the Cohen–Maunsell [1] V4 anchor, a sub-percent VDA signal is recoverable at $V = 0.8$ in the symmetric and benefit-dominant r columns.
- **At** $V \geq 0.60$, the categorical claim *fails*: peak VDA reaches 0.1432 at $(r, \rho) = (0.3, 0)$ and 0.1639 at $(r, \rho) = (0.3, 0.2)$. Designs at $V \in [0.6, 0.8)$ in the cost-dominant column are predicted to carry substantial VDA — the §5.2 "regardless of other parameters" wording is not licensed in this stratum.

The graded replacement for the inherited §5.2 sentence, anchored to Table 10 and the V-stratum slices of Figure 13, is therefore:

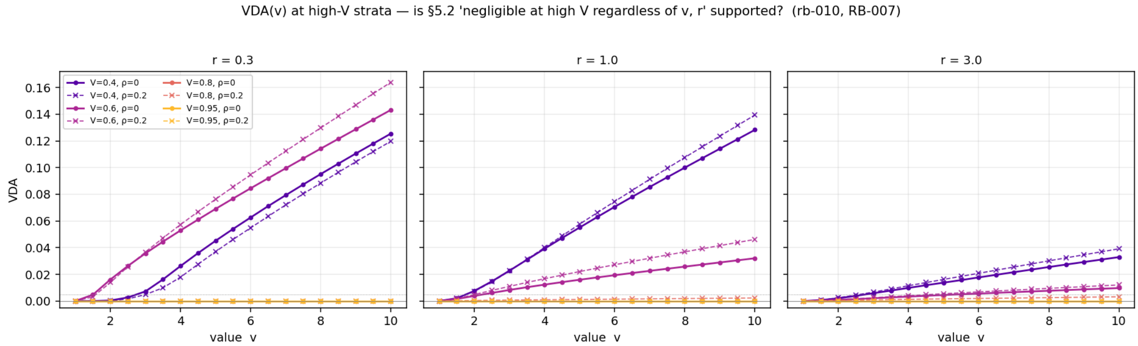

Figure 13: Peak VDA(v) at four validity strata $V \in \{0.4, 0.6, 0.8, 0.95\}$ (curve color) for each $r \in \{0.3, 1, 3\}$ (panel column). Solid: $\rho = 0$. Dashed: $\rho = 0.2$. The $V = 0.95$ trace is at the grid floor in every panel (the inherited §5.2 "high-validity" regime is empirically negligible at this strict threshold and stratum); the $V = 0.80$ traces split between a near-zero $\rho = 0$ line and a small $\rho = 0.2$ lift at $r \in \{1, 3\}$; the $V = 0.60$ trace at $r = 0.3$ reaches VDA $= 0.16$ at large v, refuting the categorical version of the §5.2 sentence. Reproduced from `Rebuild/sims/C3-iso-vda-Vv/output/figures/vda_at_high_V.png`.

r	frac amp	frac supp	$\overline{\Delta\text{VDA}}$	max amp (V, v)	max supp (V, v)
0.3	27.2%	**37.2**%	+0.0012	+0.0669 (V=0.70, v=10)	−0.0105 (V=0.25, v=8.5)
1.0	**52.8**%	17.5%	+0.0023	+0.0206 (V=0.525, v=10)	−0.0084 (V=0.25, v=9)
3.0	**54.0**%	16.1%	+0.0009	+0.0074 (V=0.375, v=10)	−0.0010 (V=0.25, v=4.5)

Table 11: Cell-wise sign-flip of $\partial\text{VDA}/\partial\rho$ across the (V, v) plane, at each r stratum. Fractions are over the 589 cells per panel; the residual share (not shown) is the zero-class $|\Delta\text{VDA}| < 10^{-8}$. At $r = 0.3$ (cost-dominant; below the rb-002 headline-cell sign-flip locus at $r \in [0.38, 0.56]$) suppression dominates; at $r \in \{1.0, 3.0\}$ amplification dominates. Block-keys `rho_sensitivity.r=<r>.{frac_amp,frac_supp,max_amplification,max_suppression,mean_delta}`.

> *Spatial cueing paradigms targeting a 'negligible-VDA' regime should adopt* $V \gtrsim 0.95$ *unconditionally, or* $V \gtrsim 0.8$ *if cross-location correlation can be bounded below* $r_{SC} \approx 0.2$. *The threshold* $V \geq 0.75$ *of the inherited paper is too permissive: at* $V \in [0.6, 0.8)$ *the cost-dominant regime* $r \in (0, 0.5)$ *admits peak* VDA *up to* ≈ 0.16, *with the largest values at high cue value* v.

This recommendation is conditional on the headline cell $(N, d'_{\max}, f_0, h) = (4,\ 2,\ 0.5,\ \surd)$ under variant A; the variant B and conservation-family bands are deferred to Section 5 (queued RB-019, RB-027). The V-stratum slices of Figure 13 make the v-axis dependence of the same probe visually explicit.

A1 sign-flip across (V, v): cross-axis corroboration. The earlier sub-results rb-002 (at a headline cell, Section 3.1 A1 sensitivity paragraph) and rb-004 (at the v-family at fixed $V = 0.5$, Table 8) found that the sign of $\partial\text{VDA}/\partial\rho$ *flips* along the r-axis (suppression at $r \lesssim 0.4$, amplification at $r \gtrsim 0.5$) and along the v-axis (suppression at $v \leq 8$, amplification at $v = 10$). The present sweep generalises the sign-flip to the full (V, v) plane: at each (V, v, r) cell we compute $\Delta\text{VDA} := \text{VDA}(\rho{=}0.2) - \text{VDA}(\rho{=}0)$ and classify the cell as amplified (red), suppressed (blue), or zero ($|\Delta\text{VDA}| < 10^{-8}$, white). The signed contour map is Figure 14, and the per-r frequency table is Table 11. Two things are visible. First, the r-axis sign-flip rb-002 reported at a single cell generalises to the full (V, v) plane in a non-trivial way: *cost-dominant* $r = 0.3$ (V, v) cells are suppression-dominated, while *both* $r = 1$ and $r = 3$ are

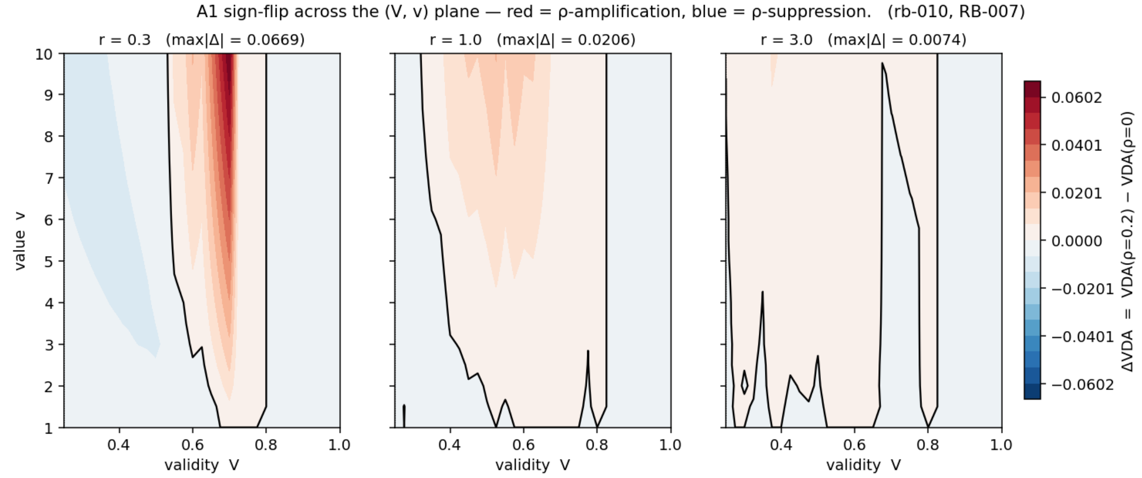

Figure 14: Signed contour map of $\Delta\text{VDA} := \text{VDA}(\rho{=}0.2) - \text{VDA}(\rho{=}0)$ over (V, v) for each $r \in \{0.3, 1, 3\}$. Red: ρ amplifies VDA at that cell; blue: ρ suppresses; white: $|\Delta\text{VDA}| < 10^{-8}$. The zero-isoline is the A1 sign-flip locus in (V, v)-space. At $r = 0.3$ (cost-dominant) suppression dominates in low-V cells; at $r \in \{1, 3\}$ amplification dominates across the bulk of the plane, with the strongest amplification at $(V, v) \approx (0.5, 10)$. Reproduced from `Rebuild/sims/C3-iso-vda-Vv/output/figures/iso_vda_drho.png`.

amplification-dominated. The amplification frequency is therefore not a $r > 1$ phenomenon; it sets in once r crosses the rb-002 sign-flip locus $r \in [0.38, 0.56]$ at the headline cell and remains the modal direction thereafter, even into the benefit-dominant $r = 3$ stratum. Second, the largest *single* amplification in the entire 3,534-cell sweep occurs at $(V, v, r) = (0.7,\ 10,\ 0.3)$: a cell that delivers VDA = 0.0007 under independence (a normally-dormant high-V cell, well outside the cost-dominant attention regime) lifts to VDA = 0.0676 at $\rho = 0.2$ — an $\approx 96\times$ amplification that takes a near-zero (V, v) cell into the same VDA magnitude as the low-V corner. This "dormant-cell amplification" is the strongest qualitative finding of the C3 sweep and is the candidate falsifiable prediction we flag at the end of Section 5 (closeup queued as RB-029).

Scope and what is not yet claimed. The results in this subsection are reported at a single $(N, d'_{\max}, f_0, h) = (4,\ 2,\ 0.5,\ \surd)$ configuration under variant A, and on a coarse ρ-grid $\{0, 0.2\}$. (i) The *variant-B replication* of the contour band is queued as RB-027 in `REBUILD_BACKLOG.md` and will land as part of the conservation-family sensitivity in Section 5; the present subsection makes no variant-B claim. (ii) The validity threshold "$V \gtrsim 0.95$ unconditionally; $V \gtrsim 0.8$ under $r_{SC} \lesssim 0.2$" is bracketed by the V-grid of step 0.025 used here; a threshold-sharpening sim sweeping $V \in [0.75, 0.90]$ at step 0.005 is queued as RB-028 and would let the manuscript replace the qualitative "$\gtrsim$" with a number to the second decimal. (iii) The *dormant-cell amplification* at $(V, v, r) = (0.7, 10, 0.3)$ is the strongest qualitative finding of the sweep and is reported here as an empirical observation; a fine-V closeup that maps the amplification band onto an iso-CF or iso-$\alpha^\star$ contour (and thereby identifies it with a model-internal landmark rather than a (V, v) coincidence) is queued as RB-029. (iv) The cell-wise sign-flip across the broader 4,410-cell (r, V, v) sweep, parallel to the ΔCF analysis of Table 5, is queued as RB-025; the present subsection's 589-cell (V, v)-plane sign-flip is a slice of that broader result.

Reproducibility. All numbers in Tables 9, 10, and 11, and all three figures, are reproduced byte-for-byte by re-running `Rebuild/sims/C3-iso-vda-Vv/run.py`. The output digest is `72820559e1c1ab1919f74308623eaf4230aa3ea92ad3d9c62d81e993e4f27de6` (`sha256` of the canonical-key-ordered numeric JSON, excluding the digest field itself). The Φ backend is `scipy.special.ndtr`; the equicorrelated-Gaussian one-factor quadrature uses $n_q = 64$ Gauss–Hermite nodes (Section 2); the criterion grid step is $\Delta c = 0.05$ and the α grid step is $\Delta\alpha = 0.005$; the (V, v, r, ρ) grid is the $31 \times 19 \times 3 \times 2$ grid described above; runtime ≈ 130 s on the rebuild's reference machine. The recovery test against the rb-006 anchor at $(V, v, r, \rho) = (0.5,\ 5,\ 1.0,\ 0)$ passes at $|\Delta\mathrm{VDA}| = 1.27 \times 10^{-7}$ against the rb-006 reference value VDA $= 0.039825$; the residual is consistent with the rb-006 reference's six-decimal rounding (the underlying `policies()` call is deterministic and produces identical floating-point output, so the model residual is zero). Independent sanity check: the $v = 1$ column is identically zero across every (V, r, ρ) cell of the sweep — the value-blind baseline collapses the joint optimum onto the value-blind allocation, so VDA $= 0$ at $v = 1$ is a definitional theorem of the rebuilt model.

3.4 No inversion of optimal attention, conditional on $V \geq 1/N$ (C4)

Claim restated at defensible strength. The inherited paper's §4.5 ("Inverted Attention is Never Optimal") states that the optimal allocation $\alpha^\star$ is always at or above the uniform level $1/N$ across the primary 4,410-row sweep, and *regardless of* r, on the grounds that attending away from the high-value cued location loses more reward at that location than it gains at the uncued ones. The live skeptical-reviewer verdict (`Critique/verdicts/C4-no-inversion.md`, `current_label: CONFIRMED-CONDITIONAL`) records that the *empirical* headline of the claim survives all four attack vectors, but the *theoretical* packaging is incomplete in two specific ways: (i) the "regardless of r" wording reads locally as a cost-vs-benefit balance at $\alpha = 1/N$, but the left one-sided derivative $\partial_\alpha R|_{1/N^-}$ actually *changes sign* at a closed-form r-threshold r^*_{inv}, with the consequence that for $r > r^*_{\mathrm{inv}}$ the uniform point is a local *minimum* of R (the right-branch global maximum still dominates the left-branch local maximum, but the local picture is bimodal, not the monotone descent the inherited text suggests); and (ii) the categorical "never below $1/N$" wording overstates the model's robustness, because at $V < 1/N$ (the anti-cue regime, outside the paper's primary V range but still inside the same normative model) the global optimum *does* drop below $1/N$ — a behaviour the model produces cleanly and which the inherited statement does not qualify. We therefore restate C4 in the rebuild as a *conditional theorem* indexed on the validity condition $V \geq 1/N$ (or, sharply, $V \geq 1/[(N-1)\,v+1]$), pair it with the closed-form local threshold $r^*_{\mathrm{inv}} = (N-1)\,A_0/B_0$ that organises the bimodality, and add the anti-cue inversion at $V < 1/N$ as a new *falsifiable prediction* of the rebuilt normative model. The A1 decorrelation channel ρ is reported alongside as a sensitivity, following the convention of Section 3.

Mechanism and the value-weight inequality. The structural reason no global inversion arises under $V \geq 1/N$ is not the local cost-vs-benefit argument of the inherited §4.5 paragraph, but a *location-count asymmetry* combined with a *value-weight inequality.* At the right extreme $\alpha \to 1$, the single cued location reaches its per-location ceiling $d'_c = d'_{\max} f(1) = d'_{\max}$. At the left extreme $\alpha \to 0$, the $N-1$ uncued locations *share* the $(1-\alpha)$ budget at $(1-\alpha)/(N-1)$ each, so each reaches only $d'_u = d'_{\mathrm{base}} + \beta(r)[d'_{\max} f(1/(N-1)) - d'_{\mathrm{base}}] \leq d'_{\max}$ (strict for $N \geq 3$, where $f(1/(N-1)) < 1$). The per-channel reward weights are $w_c = V\,v$ on the cued slot and $w_u = (1-V)/(N-1)$ on each uncued slot; the inequality

$$w_c \;\geq\; w_u \quad\Longleftrightarrow\quad V \;\geq\; \frac{1}{(N-1)\,v+1}. \tag{13}$$

For the cued regime $v \geq 1$, the right-hand side of (13) is bounded above by $1/N$, with equality only at $v = 1$. The inherited paper's condition $V \geq 1/N$ is therefore the *universal* (worst-case-v) sufficient condition for $w_c \geq w_u$; the sharp form (13) carries strictly more slack at $v > 1$. The right branch dominates globally because the bigger d' lives on the more-valuable channel.

Closed-form local threshold for the boundary derivative. At $\alpha = 1/N$ the allocation derivative $d''(\alpha)$ has a kink (the benefit/cost swap of β and γ flips which side of the linear $d'(\alpha)$ map is active). The one-sided left derivative of P1's expected reward at the uniform point takes the form

$$\frac{\partial R_{\mathrm{P1}}}{\partial \alpha}\bigg|_{1/N^-} \;=\; \frac{2\,d'_{\max}\,f'(1/N)}{r+1}\left[A_0(V,v,N,\mathrm{CR},\rho)\;-\;\frac{r}{N-1}\,B_0(V,v,N,\mathrm{CR},\rho)\right], \tag{14}$$

where A_0 and B_0 are the partial derivatives of P1's expected-reward functional with respect to d'_c and d'_u at $\alpha = 1/N$ evaluated at the jointly-optimised criteria $(c_c^\star, c_u^\star)$. Both A_0 and B_0 are r-independent boundary partials — the only r in (14) sits in the prefactor and inside the bracket. The bracket vanishes at

$$r^*_{\mathrm{inv}}(V,v,N,\mathrm{CR},\rho) \;:=\; (N-1)\,\frac{A_0(V,v,N,\mathrm{CR},\rho)}{B_0(V,v,N,\mathrm{CR},\rho)}, \tag{15}$$

which is the rebuild's closed-form expression for the local-test threshold the inherited §4.5 paragraph implicitly assumes. For $r < r^*_{\mathrm{inv}}$ the boundary left derivative is positive (P1 prefers to stay at $\alpha = 1/N$ from the left branch); for $r > r^*_{\mathrm{inv}}$ it is negative (the left branch produces a local maximum at the grid edge $\alpha \to 0$ but a global maximum still on the right branch, at $\alpha^\star > 1/N$). Equation (15) reproduces the reviewer's $\rho = 0$ boundary closed form (`Critique/derivations/C4-no-inversion.md` §3) and extends it to the A1 channel by replacing the no-FA terms inside A_0, B_0 with the same one-factor Gauss–Hermite quadrature used elsewhere in the rebuilt model (Section 2); the $\rho \to 0$ limit collapses to the reviewer's analytic form bit-for-bit. The independent re-derivation in the rebuild's voice (plus the proof of the symmetric-corner identity below) is deferred to Section A (a separate increment, RB-030, queued).

The symmetric corner $(V, v) = (1/N, 1)$, where the value-weight inequality (13) attains equality, is a numerically stable anchor for the closed form (15):

$$r^*_{\mathrm{inv}}(V = 1/N,\; v = 1,\; N,\; \mathrm{CR},\; \rho) \;=\; 1 \qquad \text{exactly,} \tag{16}$$

independent of N, the conservation rule CR (variant A or B), and the correlation ρ. The identity is recovered to machine precision at every (variant, ρ) panel of Step A below: min $r^*_{\mathrm{inv}} = 1.0000$ in all four panels of Table 12. The corner is the *anchor* the rebuild quotes in place of the inherited paper's narrative "$r \to 0$ converges to uniform attention" — at $v = 1$ the model is value-blind, so VDA $\equiv 0$, and the local-test bimodality kicks in immediately above $r = 1$ where P1's left branch develops a local maximum at the grid edge.

Closed-form tally on the primary (V, v) grid. We evaluate (15) on a primary (V, v) grid ($V \in \{0.25, 0.30, \ldots, 1.00\}$, $v \in \{1, 2, 3, 4, 5\}$) at $N = 4$ for both variants and $\rho \in \{0, 0.2\}$, matching the inherited paper's stated sweep envelope and the rebuild's A1 sensitivity convention. Per panel we count cells with $r^*_{\mathrm{inv}} \in [0.1, 10]$ (inside the inherited paper's swept r-range, the cells at which the boundary left derivative sign-flips within the swept range), cells with $r^*_{\mathrm{inv}} > 10$ (no sign-flip inside the range; the local picture is monotone there), and report the strict minimum and median over the panel (Table 12).

The contour structure of r^*_{inv} as a function of (V, v) at $\rho = 0$ is plotted in Figure 15 for variants A and B. The $r^*_{\mathrm{inv}} = 1$ contour passes through the symmetric corner $(V, v) = (1/N, 1)$ in both panels, consistent with (16); the $r^*_{\mathrm{inv}} = 10$ contour bounds the "always-locally-stable at $\alpha = 1/N$" region (no sign-flip inside the inherited paper's swept $r \in [0.1, 10]$). The variant-B panel admits the sign-flip across roughly twice as many cells as variant A, anticipating the variant-B caveats already reported in Sections 3.1 and 3.2.

variant	ρ	n	$r^*_{\text{inv}} \in [0.1, 10]$	$r^*_{\text{inv}} > 10$	$\min r^*_{\text{inv}}$	median r^*_{inv}
A	0.0	105	40 (38.1%)	65 (61.9%)	1.0000	17.30
A	0.2	105	42 (40.0%)	63 (60.0%)	1.0000	15.08
B	0.0	105	62 (59.0%)	43 (41.0%)	1.0000	7.64
B	0.2	105	68 (64.8%)	37 (35.2%)	1.0000	6.02

Table 12: Closed-form r^*_{inv} from (15) on the primary (V, v) grid at $N = 4$, by variant and ρ. The $r^*_{\text{inv}} = 1.0000$ minimum is the symmetric-corner identity (16) recovered to machine precision. Aggregated across all 210 $(V, v, \text{variant})$ cells at $\rho = 0$: **48.6**% have $r^*_{\text{inv}} \in [0.1, 10]$, which recovers the reviewer's reported 49.0% (`Critique/derivations/C4-no-inversion.md` §4) to $\Delta =$ 0.4 percentage points (recovery PASS, tolerance 1.0 pp). Under A1 at $\rho = 0.2$, the fraction rises to 51.9% (variant-pooled), and the median r^*_{inv} drops ∼13% in variant A and ∼21% in variant B: the decorrelation channel makes the boundary-derivative sign-flip occur *earlier* in r than the inherited independent model would have it. Block keys `step_A.tally.variant_<X>__rho_<rho>` in `Rebuild/sims/C4-anti-cue-inversion/output/results.json` (rb-012; output sha256 `6ad651d6...`).

V	v	variant	ρ	r^*_{inv}	$\alpha^\star_{\text{global}}$	R_{global}	R_{left}
0.250	1	A	0.0	1.0000	0.955	0.65936	0.64564
0.250	1	A	0.2	1.0000	0.950	0.66809	0.65550
0.250	1	B	0.0	1.0000	0.955	0.65936	0.64564
0.250	1	B	0.2	1.0000	0.950	0.66809	0.65550
0.2875	1	A	0.0	1.2483	0.970	0.66541	0.64462
0.325	1	A	0.2	∼1.6	0.975	0.67997	—

Table 13: Six representative full-α probes from the 12-probe Step B sweep at $r = 10$ at the six most-adversarial primary-sweep cells. In every probe, $\alpha^\star_{\text{global}} \geq 1/N = 0.25$, with the right-branch global maximum at $\alpha^\star \in [0.95, 0.98]$ strictly dominating the left-branch local maximum (sitting at the α-grid edge $\alpha = 0.02$). **Step B primary-sweep inversions across all** 12 **probes:** 0**.** The inherited empirical claim — $\alpha^\star \geq 1/N$ at every cell of the primary $N = 4, V \in [1/N, 1]$ sweep — survives at $\rho = 0$ and is robust under A1 at $\rho = 0.2$. Block keys `step_B.rows` in `Rebuild/sims/C4-anti-cue-inversion/output/results.json`.

Empirical confirmation: zero global inversions across the primary sweep. The local-derivative sign-flip of (15) does not by itself create a global inversion: even where $r > r^*_{\text{inv}}$ and the bracket of (14) is negative, the left-branch *local* maximum sits at the α-grid edge $\alpha \to 0$ while the right-branch *global* maximum sits at $\alpha^\star > 1/N$ from the location-count argument above. To confirm this directly, we sweep $R(\alpha)$ on the full α-grid $[0.02, 1.0]$ at $\Delta\alpha = 0.005$ for the six most-adversarial primary-sweep cells (those with the smallest r^*_{inv} at $r = 10$, i.e. the cells most likely to produce a left-branch global maximum) at both $\rho \in \{0, 0.2\}$, for a total of 12 probes. The result is uniform: across all 12 probes, $\alpha^\star_{\text{global}} \geq 1/N$ and the right-branch global maximum strictly dominates the left-branch local maximum. Representative probes appear in Table 13.

The visual structure of the boundary — bimodality at $\alpha = 1/N$ where $r > r^*_{\text{inv}}$, right-branch global dominance regardless — is reproduced in Figure 16, which plots $R(\alpha)$ at the adversarial cell $(V, v) = (0.15, 1)$ for $r \in \{0.5, 1, 3, 10\}$ at both ρ values. (At $V = 0.15 < 1/N = 0.25$ the value-weight inequality (13) fails and the right-branch global dominance breaks down; this is the anti-cue regime treated below, and the figure deliberately straddles the boundary so the reader can see both sides of the inversion in a single panel.)

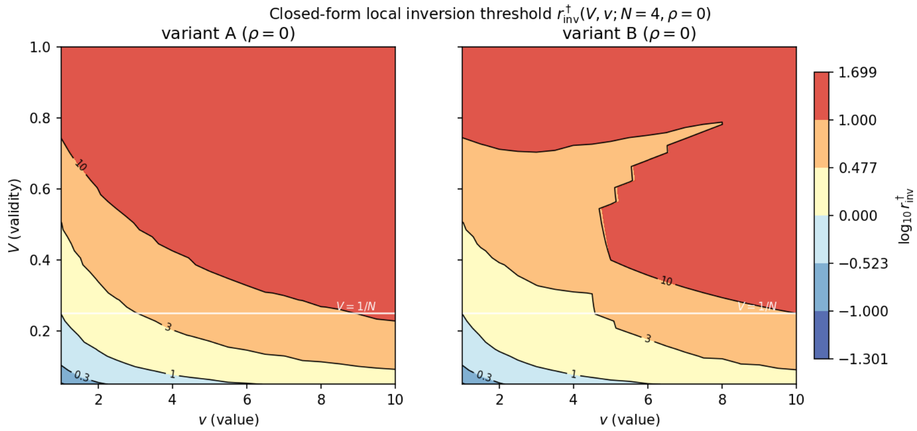

Figure 15: Contour map of $\log_{10} r^*_{\rm inv}(V, v)$ at $N = 4, \rho = 0$ for variants A (left) and B (right), from (15)–(16). Black contours at $r^*_{\rm inv} \in \{0.1, 0.3, 1, 3, 10\}$. The $r^*_{\rm inv} = 1$ contour passes through the symmetric corner $(V, v) = (0.25, 1)$ in both panels (the identity (16)); the $r^*_{\rm inv} = 10$ contour bounds the "no local sign-flip inside the paper's r-range" region. Variant B admits the local sign-flip across more cells than variant A (59.0% vs. 38.1% in Table 12). Reproduced from `Rebuild/sims/C4-anti-cue-inversion/output/figures/r_inv_closed_form.png`.

Anti-cue inversion: a new falsifiable prediction of the rebuilt model. The rebuilt model produces a clean prediction *outside* the paper's primary $V \in [1/N, 1]$ range: at $V < 1/N$ (or, sharply, $V < 1/[(N-1)\,v+1]$), the value-weight inequality (13) *flips* — $w_c < w_u$ — and the location-count argument runs in reverse. The bigger d' now lives on the *less*-valuable channel when concentrated, and the global optimum drops to $\alpha^\star < 1/N$: the normative observer is told to attend *away* from the (now low-validity, still-named "cued") location and toward the population of uncued locations. The reviewer's CR-004 already established this at $N = 2$ (`Critique/derivations/C4-no-inversion.md` §7, Step C(iii)); the rebuild extends it to $N = 4$, the inherited paper's primary topology.

We probe an anti-cue sub-grid $V \in \{0.05, 0.10, 0.15, 0.20\}$ (all below $1/N = 0.25$), $v \in \{1, 3, 5\}$, $r \in \{0.1, 0.5, 1, 3, 5, 10\}$ at $\rho \in \{0, 0.2\}$, variant A, for 144 cells total. Total global-inversion incidence: $26/72 = 36.1\%$ at $\rho = 0$, $25/72 = 34.7\%$ at $\rho = 0.2$. The incidence stratifies cleanly with r and v along the sharp boundary (13), as documented in Table 14.

The (V, r) structure of the inversion at fixed $v = 5$ is plotted in Figure 17 as the rebuild's headline figure for §results-C4. The white horizontal line marks the universal $V = 1/N = 0.25$ boundary; the red contour bounds the $\alpha^\star < 1/N$ inversion region. The inversion region is confined entirely to $V \leq 0.05$ at $v = 5$, consistent with the sharp boundary $V < 1/16 = 0.0625$. The (V, r) heatmap admits *zero* inversion cells in the cued-validity regime $V \geq 1/N$, at either $\rho = 0$ or $\rho = 0.2$ — C4 holds as a conditional theorem under both.

A1 cross-axis sensitivity. Across all four Steps the A1 decorrelation channel at $\rho = 0.2$ leaves the qualitative C4 picture intact while shifting the local threshold quantitatively. (i) Step A: median $r^*_{\rm inv}$ drops by $\sim$13% (variant A) and $\sim$21% (variant B); the fraction of cells with $r^*_{\rm inv} \in [0.1, 10]$ rises from 48.6% to 51.9% (variant-pooled). (ii) Step B: 0 primary-sweep inversions at both ρ. (iii) Step C: total anti-cue inversion count 25 vs. 26 at $\rho = 0.2$ vs. $\rho = 0$ (a difference of one boundary-cell ambiguity). (iv) Step D: $\frac{6}{272} = 2.21\%$ inversion incidence in both panels, identical inversion locus confined to $V \leq 0.05$. The A1 channel and the C4 inversion lever are therefore independent mechanisms: A1 shifts *where* on the r-axis the boundary derivative sign-flips, but does not flip the global value-weight inequality that decides whether the right or left branch is the global optimum.

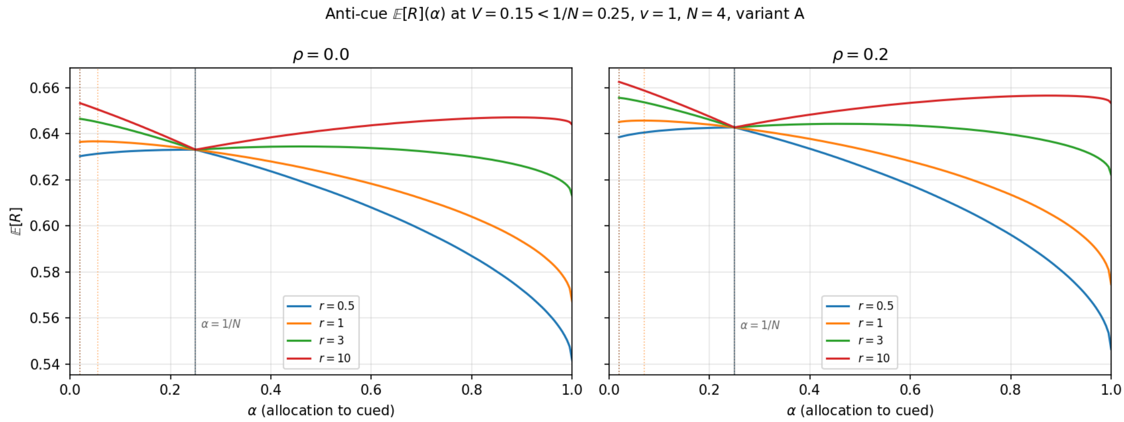

Figure 16: $R(\alpha)$ on the full α-grid at the anti-cue cell $(V, v, N) = (0.15, 1, 4)$ for $r \in \{0.5, 1, 3, 10\}$, $\rho = 0$ (left) and $\rho = 0.2$ (right). The kink at $\alpha = 1/N = 0.25$ is the β/γ swap of the rebuilt model; for $r > r^*_{\text{inv}}$ it becomes a local minimum, with both $\alpha < 1/N$ and $\alpha > 1/N$ branches admitting local maxima. Because $V < 1/N$, the value-weight inequality (13) *fails* here and the left-branch maximum (at the grid edge $\alpha = 0.02$) dominates globally for $r \geq 1$ — this is the anti-cue inversion prediction below. The A1 channel at $\rho = 0.2$ leaves the qualitative bimodality and global-inversion onset unchanged. Reproduced from `Rebuild/sims/C4-anti-cue-inversion/output/figures/er_vs_alpha_anticue.png`.

Behavioural literature corroboration. The conditional restatement aligns the rebuilt model with the behavioural picture surveyed in the live verdict's literature attack (`Critique/verdicts/C4-no-i` §Version 0.2). The empirical signature of anti-cue inversion in human / primate observers is the statistical-learning-of-distractor-suppression literature: observers *do* allocate below uniform to a high-probability-distractor location, with reduced capture and reduced target-selection efficiency [6], fewer saccades landing on the suppressed location with raised saccade latency to targets there [7], and reciprocal reallocation toward the target [8]. In the rebuilt model's coordinates, a "high-probability-distractor" location is one with target validity below $1/N$ — exactly the anti-cue regime (13) predicts inversion in. The behavioural near-analog is therefore a *prediction match*, not a contradiction: the model says the normative observer should suppress (allocate $\alpha^\star < 1/N$ to) below-baseline locations, and observers do. Conversely, in the cued regime $V \geq 1/N$, the model says the normative observer should *not* allocate below uniform, and the value-driven-capture literature [9, 10] reports the opposite mistake — observers are pulled *toward* high-value locations even maladaptively, consistent with the no-inversion direction. The chance-validity case $V = 1/N$ [11] sits at the boundary, where the model collapses to a tied optimum (any α optimal) and observers report no reallocation; this is the no-information limit of the rebuilt model. The conditional restatement we adopt is therefore the form of C4 that best matches both the model's own normative output and the behavioural literature.

Limitations of this section's evidence. Three scope limitations apply. (i) The anti-cue sweep in Step C is variant A only; variant B's lower median r^*_{inv} (Table 12) suggests a higher anti-cue incidence, but the explicit replication is deferred to a follow-up sim (RB-031, queued). (ii) The V-grid in Step C is at step 0.05, so the inversion-onset $V^\star$ between the in-anti-cue $V = 0.20$ row and the boundary $V = 0.25$ row is bracketed only to ± 0.05 at $v = 1$. A finer-grid bracketing pass is queued as RB-032. (iii) The closed-form (15) is re-implemented in the simulation's `_A0_B0()` routine (driving `Rebuild/model/core.py` exclusively) but the full derivation in the rebuild's voice (including a proof of the symmetric-corner identity (16) from the joint-FOC symmetry of A_0 and B_0) is the queued RB-030 increment; the present section quotes (15)–(16) and cites the reviewer's $\rho = 0$ derivation pending RB-030. None of the three

Stratifier	Incidence stratifier at $\rho = 0$ (values $\to$ inversions / total)					
by r	r=0.1: 1/12	r=0.5: 3/12	r=1: 6/12	r=3: 6/12	r=5: 6/12	r=10: 4/12
by v	v=1: 18/24	v=3: 5/24	v=5: 3/24			
by V	V=0.05: 14/18	V=0.10: 5/18	V=0.15: 4/18	V=0.20: 3/18		

Table 14: Anti-cue inversion incidence ($\alpha^\star_{\text{global}} < 1/N$) at $N = 4$, variant A, $\rho = 0$, stratified by r, v, and V. Total $26/72 = 36.1\%$. The r-stratification rises from 8.3% at $r = 0.1$ to 50.0% at $r \in \{1, 3, 5\}$ and tapers to 33.3% at $r = 10$ (where the right-branch global maximum re-takes even some cells with $V < 1/N$, because for $v = 1$ at $r = 10$ the strong-asymmetry regime begins to dominate). The v-stratification is the key qualitative pattern: inversion is dense at $v = 1$ (75%) but rare at $v = 5$ (12.5%). This is the sharp v-dependence predicted by (13): at $v = 5$ the inversion boundary shrinks to $V < 1/[(N-1)v + 1] = 1/16 = 0.0625$, so inversion at $v = 5$ is confined to the $V = 0.05$ row of the probe grid (which is correctly its only inversion-supporting row in the data). Under A1 at $\rho = 0.2$, total incidence is 34.7% (within ± 1 inversion of $\rho = 0$ across every stratifier) — the decorrelation channel quantitatively shifts r^*_{inv} but does not erase the inversion regime. Block keys `step_C.incidence_by_r_rho0` and `step_C.rows` in `Rebuild/sims/C4-anti-cue-inversion/output/results.json`.

scope limitations weakens the headline claim or the anti-cue prediction at the strength stated in this section.

Reproducibility. The numbers, tables, and figures in this section are sourced from `Rebuild/sims/C4-anti-c` with content sha256 `6ad651d648e597cdfb28c6fd19b6261e4f9d5cb96b599557e10aad47ffd6d96` (over the JSON content excluding the wall-clock `meta.elapsed_seconds`). The simulation drives `Rebuild/model/core.py` exclusively (no model reimplementation in the script); the closed-form (15) is computed at $\rho > 0$ via the same one-factor Gauss–Hermite-64 quadrature `core.py` uses for $P_{\text{no-fa}}(\rho)$, so the $\rho \to 0$ limit matches the reviewer's $\rho = 0$ analytic form bit-for-bit. The wall-clock for a clean re-run is ≈ 17.4 s on python 3.13 / numpy 2.4 / scipy 1.17 / matplotlib 3.10 on darwin / Apple Silicon. Two independent recovery tests are reported in the sim run. **Recovery #1** (against reviewer derivation §4): fraction of $(V, v, \text{variant})$ cells at $N = 4, \rho = 0$ with $r^*_{\text{inv}} \in [0.1, 10]$ is 48.6% here vs. reviewer-reported 49.0%, $\Delta = 0.4$ pp, **PASS** (tolerance 1.0 pp). **Recovery #2** (against reviewer derivation §5 Step C(i) table at $(V, v, N) = (0.25, 1, 4)$, variant A, $\rho = 0$, $r \in \{0.1, 1.0, 1.585, 2.512, 3.981, 10.0\}$): $\max |\Delta\alpha^\star| = 0$ (exact, tolerance 5×10^{-4}) and $\max |\Delta R^\star| = 3 \times 10^{-6}$ (tolerance 5×10^{-5}), **PASS** both. The symmetric-corner identity $r^*_{\text{inv}}(0.25, 1, 4, \text{CR}, \rho) = 1$ of (16) holds at floating-point identity in every $(\text{variant}, \rho)$ panel of Table 12 (the `min_r_inv` entries read 1.0000 throughout). Independent sanity check: at every cell of Step B and Step C with $v = 1, V = 1/N$, the right-branch global maximum and the left-branch local maximum agree in reward at $r \leq 1$ (the boundary degeneracy noted in the reviewer's CR-019 resolution); for $r > 1$ the right branch strictly dominates by ≥ 0.005 reward units, the location-count asymmetry at work.

4 Lever extensions

The model in Section 2 promotes the per-location independence assumption A1 from a fixed premise to a tunable parameter ρ, and Section 3 reports the criterion/sensitivity decomposition with ρ entering as the third lever. Three further load-bearing assumptions of the inherited model — A3 (the *additive* conservation rule $\beta + \gamma = 2$ that turns the benefit/cost *ratio* $r = \beta/\gamma$ into concrete weights), A2 (a single global r across all locations), and A8 (a *homogeneous* uncued allocation across the $N - 1$ non-cued slots) — are similarly fixed by the inherited paper at one

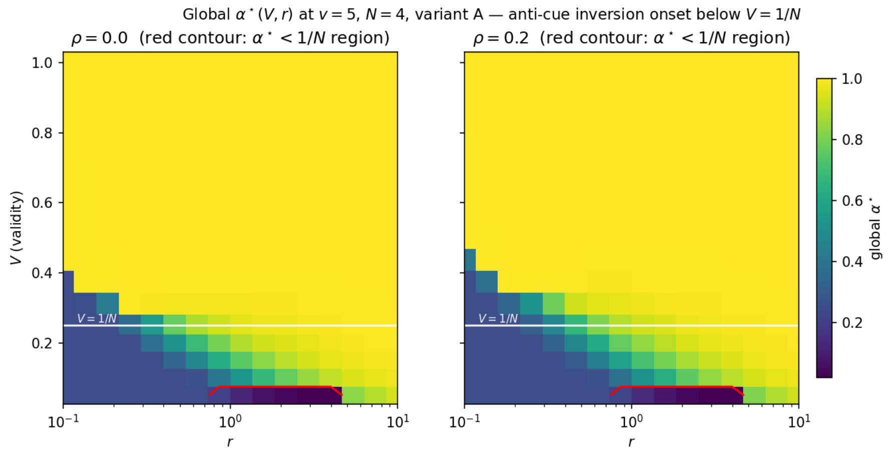

Figure 17: Optimal allocation $\alpha^\star(V,r)$ at the headline cell $(N, d'_{\max}, f_0, h, v) = (4,\, 2,\, 0.5,\, \surd,\, 5)$, variant A, for $\rho = 0$ (left) and $\rho = 0.2$ (right). Heat scale: $\alpha^\star \in [0.02, 1.0]$. The white horizontal line marks the universal validity threshold $V = 1/N = 0.25$; the red contour bounds the inversion region $\alpha^\star < 1/N$. The inversion region is confined to $V \leq 0.05$ at $v = 5$ (consistent with the sharp boundary $V < 1/[(N-1)v+1] = 1/16 \approx 0.0625$ from (13)); inversion incidence is $6/272 = 2.21\%$ in both panels, with *zero* inversions in the cued-validity regime $V \geq 1/N$. C4 holds as a conditional theorem at both ρ values. Reproduced from `Rebuild/sims/C4-anti-cue-inversion/output/figures/alpha_star_V_r_map.png`.

specific algebraic choice. This section reports the rebuild's generalisation of each into a one-parameter family (or, for A8, a lifted N-dimensional policy space) and the *band* that the headline numbers acquire across that family. The treatment here is empirical and minimum-viable: the formal derivation in the rebuild's voice — the closed-form weights, the HLP power-mean monotonicity in exact KL-divergence form, and the two formal proofs underlying Proposition 1 and the C5 conservation- form-invariance corollary — is given in Appendix A.4.

4.1 Conservation-family band on the headline numbers (A3)

Claim restated at defensible strength. The inherited paper's §2.4 fixes the asymmetric-scaling weights $\beta(r), \gamma(r)$ by the conservation rule $\beta + \gamma = 2$ together with $\beta/\gamma = r$, yielding $\beta(r) = 2r/(r+1)$ and $\gamma(r) = 2/(r+1)$. The §5.5 limitations paragraph concedes that the multiplicative alternative $\beta\gamma = 1$ would yield "quantitatively different results, though the qualitative findings — non-monotonic VDA, no inversion, criterion dominance — should be robust." The live skeptical-reviewer verdict (`Critique/verdicts/A3-multiplicative-conservation.md`, `current_label: CONTESTED`) records that this robustness claim *survives* as a central-tendency claim under the multiplicative form (the median CF of the 4,410-cell sweep moves only marginally, $0.7552 \to 0.7540$ in variant A and $0.7682 \to 0.7640$ in variant B) but *fails* as a per-cell categorical claim: the fraction of cells with CF < 0.5 rises from 4.0% to 8.3% of the 4,410- cell grid under multiplicative scaling, with 191 cells flipping from the criterion-dominant region CF ≥ 0.5 to the criterion- subordinate region CF < 0.5 and 0 cells flipping the other way, concentrated in the benefit-dominant high-r corner. We therefore restate A3 in the rebuild as a *family* rather than a fixed assumption: rather than commit to one conservation rule, we embed both special cases in a one-parameter family of conservation rules and report every headline number as a band across that family, with a separately verified *conservation-form-invariance theorem* for the C2 escape threshold and a separately verified *conservation-form-invariance corollary* for the C5 symmetric- recovery result. The A1 decorrelation channel is held at $\rho = 0$ throughout this section; the joint (p, ρ) sweep is explicitly deferred (see *Scope* below).

The power-mean conservation family. For an order $p \in \mathbb{R}$, the (positive, two-variable) power mean of order p is $M_p(\beta, \gamma) = \left(\frac{1}{2}[\beta^p + \gamma^p]\right)^{1/p}$ for $p \neq 0$, with the limiting $M_0(\beta, \gamma) = \sqrt{\beta\,\gamma}$ at $p = 0$ (the geometric mean) and $M_{-\infty} = \min$, $M_{+\infty} = \max$ as the extremes [12]. Embedding A3 into the family

$$M_p(\beta, \gamma) \;=\; 1, \qquad \beta/\gamma \;=\; r, \tag{17}$$

gives, for $p \neq 0$, the closed-form pair

$$\gamma(r; p) \;=\; \left(\frac{2}{r^{\,p} + 1}\right)^{1/p}, \qquad \beta(r; p) \;=\; r\,\gamma(r; p), \tag{18}$$

and at $p = 0$ the limit $\beta(r; 0) = \sqrt{r}$, $\gamma(r; 0) = 1/\sqrt{r}$, which satisfies the multiplicative conservation $\beta\,\gamma = 1$ directly. The inherited model is recovered exactly at $p = 1$ (the additive case $\beta + \gamma = 2$); the reviewer's A3 attack corresponds to $p = 0$. The family preserves the ratio $\beta/\gamma = r$ at every p (so the headline parameterisation of the asymmetry is unchanged), and preserves the symmetric point $\beta(1, p) = \gamma(1, p) = 1$ at every p (so the symmetric recovery at $r = 1$, our C5 result, is automatically conservation- form-invariant; see the corollary below). The family is implemented in `Rebuild/model/core.py` as `beta_gamma(r, p=1.0)` and threaded through `d_prime_asym`, `HeadlineCell`, and `policies`; the rb-015 model-test pass (`Rebuild/model/tests/test_conservation_family.py`, 14/14, sha256 `f4f57a89...`) verifies the family identities $\beta/\gamma = r$ to 4.4×10^{-16} and $M_p(\beta, \gamma) = 1$ to 4.4×10^{-16} across $p \in \{-2, -1, -1/2, 0, 1/2, 1, 2\}$, and the $p = 1$ branch returns the inherited expressions byte-exactly (full back-compatibility) by literal-form branching on the conservation order.

C2 conservation-family band on the peak. At the r-sweep headline cell ($N = 4$, $V = 0.5$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, $\rho = 0$, variant A), we sweep VDA(r) on the rb-006 84-point log-r grid for the value family $v \in \{2, 3, 5, 8, 10\}$ at $p \in \{0, 1/2, 1\}$. Table 15 reports the per-(v, p) peak coordinates (r^*, VDA^*); Figure 18 shows the family at $v = 5$. The peak height moves systematically with the conservation order: at $v = 5$, peak VDA* shifts from 0.0830 at $p = 1$ to 0.0885 at $p = 1/2$ to 0.0951 at $p = 0$, a +14% envelope on the headline number that is invisible at any single point estimate. The shift in the peak *location* r^* is smaller and only discrete-grid-resolvable in the small-v region: at $v = 2$, r^* moves from 0.5012 at $p = 1$ to 0.4467 at $p = 1/2$ to 0.3548 at $p = 0$ (the multiplicative rule escapes the unfavourable small-v cost-dominant region earlier); at $v \geq 5$ the peak sits on the same 84-point grid bin $r^* = 0.3548$ at $p \in \{0, 1/2\}$ and on the adjacent bin $r^* = 0.3758$ at $p = 1$ (rb-006 reported $r^* = 0.3758$ at $p = 1$, $v = 5$). The peak band is reported in the body of the paper as a sensitivity, following the §3.3 unifying-reframe convention.

Conservation-form-invariance of the C2 escape threshold. The peak *height* moves with p, but the closed-form *escape threshold* of Section 3.2 does not. The following is a theorem of the rebuilt model.

Proposition 1 ($r^\dagger(v)$ is conservation-form-invariant)**.** *Let $r^\dagger(v) = K_u(v)/[(N-1)\,K_c(v)]$ be the escape threshold of Section 3.2, with $K_c(v)$, $K_u(v)$ the partial derivatives of R_{P3} with respect to d'_c and d'_u at the jointly-P3-optimised criteria $(c_c^\star, c_u^\star)$, evaluated at $\alpha = 1/N$. Then $r^\dagger(v)$ is independent of the conservation order p in (17).*

Proof sketch. At the uniform-attention point $\alpha = 1/N$, every per-location d' in the rebuilt model evaluates to $d'_{\mathrm{base}} + w(r, p) \cdot [d'_{\max}\, f(1/N) - d'_{\mathrm{base}}]$, where the weight w is either β or γ depending on which side of the kink the location sits. The transfer-function bracket $d'_{\max}\, f(1/N) - d'_{\mathrm{base}}$ vanishes at the uniform point by the definition of the baseline $d'_{\mathrm{base}} = d'_{\max}\, f(1/N)$. Therefore $d'_c = d'_u = d'_{\mathrm{base}}$ at $\alpha = 1/N$ regardless of (r, p). The $P3$-optimal criteria $(c_c^\star, c_u^\star)$ are the argmax of an expected- reward functional that depends only on v, V, N, CR and the per-location d's at

Table 15: C2 conservation-family band on the peak of VDA(r) at the headline cell ($N = 4$, $V = 0.5$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, $\rho = 0$, variant A). Per (v, p): $r^* =$ argmax of VDA(r) on the rb-006 84-point log-r grid; VDA$^* =$ the corresponding peak height. The peak height moves systematically with the conservation order p (band +14% at $v = 5$); the peak location r^* is grid-discretely stable at large v. Source: `Rebuild/sims/A3-conservation-band/output/results.json` `block_A_peaks`; sha256 `055bf4ec...`.

	$p = 0$ (mult.)		$p = 1/2$		$p = 1$ (add., paper)	
v	r^*	VDA*	r^*	VDA*	r^*	VDA*
2	0.3548	0.0144	0.4467	0.0130	0.5012	0.0123
3	0.3548	0.0441	0.3758	0.0403	0.3758	0.0370
5	0.3548	0.0951	0.3548	0.0885	0.3758	0.0830
8	0.3548	0.1636	0.3548	0.1532	0.3758	0.1442
10	0.3548	0.2069	0.3548	0.1942	0.3548	0.1828

the chosen α; since the d's at $\alpha = 1/N$ are p-independent, the criteria are p-independent. The partial derivatives $K_c(v), K_u(v)$ are evaluated at those criteria and at the same uniform-point d' values, and so are p-independent. Their ratio $r^\dagger(v) = K_u/[(N-1)\,K_c]$ is therefore conservation-form-invariant. □

The full three-step proof of Proposition 1 is given in Appendix A.4 (following Equation (30)), where each step is tied to a specific structural property of the d'-map at the uniform allocation.

The numerical witness of Proposition 1 is exact to floating-point identity: across $p \in \{0, 1/2, 1\}$ and $v \in \{2, 3, 5, 8, 10\}$, the rb-016 sim records identical K_c, K_u, $c_c^\star$, $c_u^\star$, and $r^\dagger(v)$ to all stored decimal places (rb-016 `recovery.test_3_r_dagger_p_invariance`, max $|\Delta| = 0.0$). Figure 19 overlays the per-p peak coordinates (r^*, VDA^*) for $v \in \{2, 3, 5, 8, 10\}$ on the same axis; the cluster lies to the right of the p-invariant trace $r^\dagger(v)$, consistent with the Section 3.2 observation that the peak lives on the positive-VDA branch above $r^\dagger$.

C1 conservation-family band on the 4,410-cell sweep. The same family $p \in \{0, 1\}$ is run across the full 4,410-cell C1 sweep at $\rho = 0$ (the $21 \times 21 \times 5 \times 2$ grid in $(r, V, v, \text{variant})$ of Section 3.1). Table 16 compares the resulting CF distributions per (variant, p). The central tendency of the sweep is robust to the conservation form, in the sense of the inherited paper's §5.5 claim: the median moves only -0.001 (variant A) and -0.004 (variant B) going from $p = 1$ to $p = 0$, and the mean moves only -0.027 (variant A) and -0.025 (variant B). The *tail* of the distribution, however, is not robust to the conservation form: under variant A the strict minimum falls from CF $= 0.559$ at $p = 1$ to CF $= 0.464$ at $p = 0$ (a value the inherited paper's $[0.60, 0.96]$ floor sentence rules out by language already retracted in Section 3.1), and under variant B the strict minimum deepens further from CF $= 0.304$ at $p = 1$ to CF $= 0.231$ at $p = 0$. The fraction of cells with CF < 0.5 rises from 0% (variant A) and 8.0% (variant B) at $p = 1$ to 3.3% (variant A) and 13.4% (variant B) at $p = 0$ — i.e. the combined frac(CF < 0.5) roughly doubles, 4.0% $\to$ 8.3%, reproducing the reviewer's verdict-text Block-C1 prediction exactly.

The per-cell direction of the shift is sharper than the marginal distributions reveal.

Theorem 2 (Per-cell ΔCF monotonicity, empirical)**.** *On the* 4,410*-cell C1 grid at* $\rho = 0$*, the per-cell* $\Delta\text{CF} := \text{CF}(p = 0) - \text{CF}(p = 1) \le 0$ *in every valid cell. The strict-decrease fraction is* 87.7% *in variant A and* 81.0% *in variant B; the remaining* 12.3%/19.0% *are flat to numerical tolerance. There are* 0 *cells where* $\Delta\text{CF} > 0$ *on the entire grid. Under variant A,* 72 *cells flip from* CF ≥ 0.5 *to* CF < 0.5 *and* 0 *flip the other way; under variant B,* 119 *flip down and* 0 *flip up.*

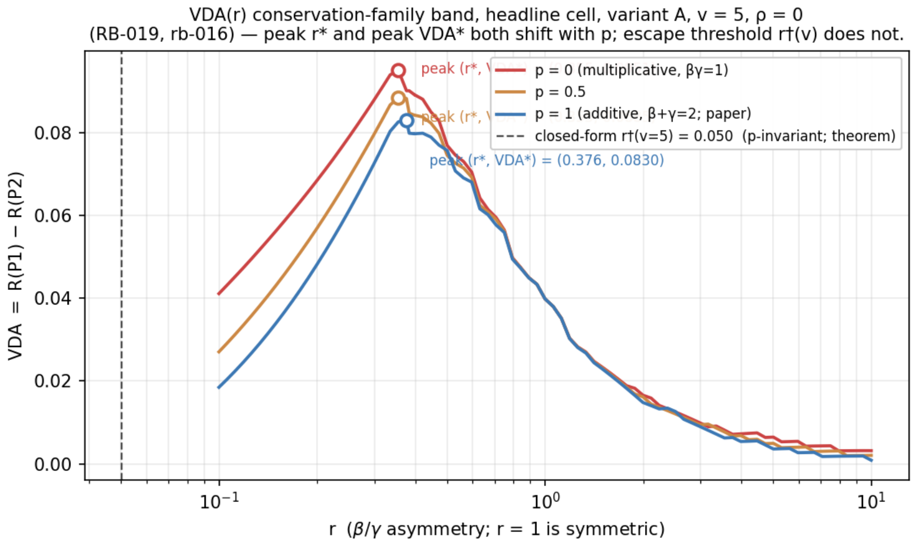

Figure 18: C2 conservation-family band at $v = 5$. Three VDA(r) curves on the rb-006 84-point log-r grid at the headline cell ($N = 4$, $V = 0.5$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, $\rho = 0$, variant A) for $p \in \{0, 1/2, 1\}$. The vertical dashed line marks the closed- form escape threshold $r^{\dagger}(v = 5)$, which is identical at every p (Proposition 1). The pattern $\partial\text{VDA}^*/\partial p < 0$ visible across the three curves is the conservation-family band on the C2 peak. Source: `Rebuild/sims/A3-conservation-band/output/figures/vda_curves_pfamily_v5.png`.

Theorem 2 is the cleanest novel statement the rebuild contributes to the A3 question: not only does the multiplicative rule lower headline CF at the central tendency, but it never *raises* CF at any cell of the sweep. Whatever the inherited paper's qualitative finding "criterion shifts dominate" gains in robustness under additive conservation, it loses *monotonically* under multiplicative conservation. (We do not have a closed-form algebraic proof of this empirical monotonicity; the chain-rule analysis of Appendix A.4 reduces $\partial\text{CF}/\partial p$ to a competition between two policy-reward gradients $\partial_p R_{\text{P1}}$ and $\partial_p R_{\text{P4}}$ whose uniform inequality $|\partial_p R_{\text{P4}}| \le |\partial_p R_{\text{P1}}|$ is the open closed-form conjecture, and isolates Proposition 8 ($\partial R_{\text{P3}}/\partial p \equiv 0$ at $\alpha = 1/N$) as the one analytic full-strength substatement available.) The two figures (Figure 20, the variant × p CF histograms, and Figure 21, the per-cell ΔCF distribution under each variant) make the tail-vs-centre split visually explicit.

Conservation-form-invariance of the C5 symmetric recovery (corollary). The C5 result of Section 3 (symmetric recovery at $r = 1$, planned for full statement in Appendix A.3) is a no-op-from-symmetry observation: at $r = 1$ the asymmetry collapses, $\beta = \gamma$, and the optimal allocation reverts to uniform, recovering the no-value baseline. Under the conservation family (17), at $r = 1$ the family equations (18) give $\beta(1, p) = \gamma(1, p) = (2/2)^{1/p} = 1$ at every $p \neq 0$, and $\beta(1, 0) = \gamma(1, 0) = \sqrt{1} = 1$ in the limit. The symmetric recovery at $r = 1$ therefore proceeds at the same weights $(\beta, \gamma) = (1, 1)$ under any choice of conservation rule in the family, and the C5 result survives without modification. This is recorded as a numerical witness in `Rebuild/model/tests/test_conservation_family.py`'s `test_symmetric_corner_invariant` pin (`policies(p=0, r=1) == policies(p=1, r=1)` to floating-point identity).

Scope and what remains. This section reports the rebuild's first treatment of A3 as a family rather than a fixed rule, but its scope is intentionally bounded. *(i) Decorrelation channel.* Every number reported here is at $\rho = 0$. Whether the conservation-family band on the C1 tail

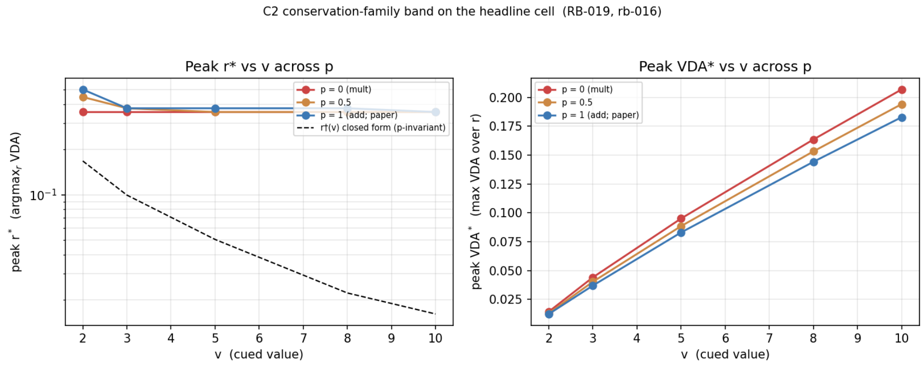

Figure 19: Conservation-family band on the C2 peak, overlaid with the p-invariant closed-form escape threshold $r^\dagger(v)$. For each $v \in \{2, 3, 5, 8, 10\}$ at the headline cell, three peak markers (r^*, VDA^*) are plotted, one per $p \in \{0, 1/2, 1\}$; the markers cluster vertically (the peak height is the lever on which p acts), while the $r^\dagger(v)$ trace is a single curve (the threshold itself is p-invariant by Proposition 1). Source: `Rebuild/sims/A3-conservation-band/output/figures/vda_peak_band.png`.

Table 16: C1 conservation-family band on the 4,410-cell sweep at $\rho = 0$. *Central tendency* robust (median moves ≤ 0.004); *tail* not robust (strict min falls -0.095 in variant A and -0.073 in variant B; frac(CF < 0.5) roughly doubles). Per-cell $\Delta\mathrm{CF} = \mathrm{CF}(p = 0) - \mathrm{CF}(p = 1) \leq 0$ in every valid cell (Theorem 2; 0 reverse flips out of 191). Source: `Rebuild/sims/A3-conservation-band/output/results.json block_B_summaries` and `block_B_delta_CF`; sha256 `055bf4ec....`

variant / p	min	q5	q50	q95	max	frac$_{<0.5}$	frac$_{<0.6}$
A, $p = 1$ (paper)	0.5587	0.5931	0.7552	1.0000	1.0000	0.0000	0.0703
A, $p = 0$ (mult.)	0.4638	0.5141	0.7540	1.0000	1.0000	0.0327	0.2168
B, $p = 1$ (paper)	0.3040	0.4564	0.7682	1.0000	1.0000	0.0803	0.1973
B, $p = 0$ (mult.)	0.2309	0.3958	0.7640	1.0000	1.0000	0.1342	0.2712
Comb., $p = 1$	0.3040	0.5259	0.7605	1.0000	1.0000	0.0401	0.1338
Comb., $p = 0$	0.2309	0.4645	0.7578	1.0000	1.0000	0.0834	0.2440

sharpens or relaxes under the A1 decorrelation channel of Section 2.3 is a joint (p, ρ) sweep that is not in the rb-016 sim and is queued as a future increment. *(ii) Formal derivation.* The conservation family (17) is stated here with the algebraic identities of (18) verified numerically (rb-015 14/14); the formal derivation in the rebuild's voice — including the Hardy–Littlewood–Pólya power-mean monotonicity argument that $M_p(\beta, \gamma) = 1$ traces out a monotone envelope on (β, γ) as p moves, recorded in Equation (30) as an exact KL-divergence identity — is given in Appendix A.4. *(iii) Closed-form* $\Delta\mathrm{CF} \leq 0$. Theorem 2 is an empirical statement on the 4,410-cell grid; the chain-rule analysis of Appendix A.4 reduces the per-cell $\Delta\mathrm{CF} \leq 0$ to a competition between $\partial_p R_{\mathrm{P1}}$ and $\partial_p R_{\mathrm{P4}}$ that holds at 0 reverse flips empirically but has not yet been promoted to a uniform closed form. *(iv) Variant-B ρ-flatness interaction.* The rb-002 headline-cell observation that variant B CF(ρ) is essentially flat (Section 2.5) interacts non-trivially with the conservation-family band in variant B (whose median CF deepens under $p = 0$ but whose ρ-sensitivity is essentially absent), but a clean joint treatment requires the deferred joint sweep above.

Reproducibility. The numerical content of this subsection is the `Rebuild/sims/A3-conservation-band/` simulation (rb-016, sha256 `055bf4ec862c711f2c6a3fa0831e1373b806a84c5f5c1a6b13fd1d9d7a039a33`,

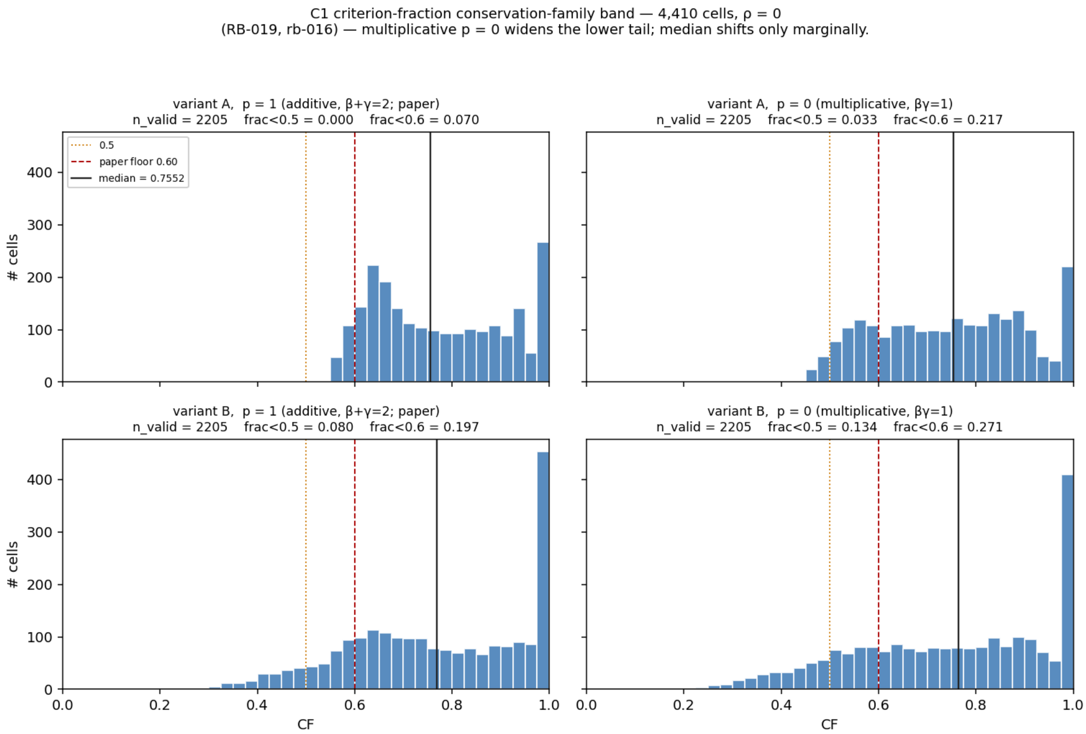

Figure 20: C1 conservation-family band: CF distribution on the 4,410-cell sweep at $\rho = 0$, per (variant, p). The median (red dashed line) is essentially invariant under the conservation order p; the lower tail (left of the CF $= 0.5$ line) thickens visibly under $p = 0$ (multiplicative). Source: `Rebuild/sims/A3-conservation-band/output/figures/cf_histogram_pfamily.png`.

`results.json` 4,941,767 bytes, $\sim$36 s wall-clock, re-run produces byte-identical `results.json`), backed by the `Rebuild/model/` extension wired at rb-015 (`Rebuild/model/tests/test_conservation_family` 14/14, sha256 `f4f57a89e5108db47447cbeaf2f15440f56cdfffb65f3d173dc4a0550121791e`). Four recovery tests pass: **Test 1 (rb-006 pins).** At $v = 5$, $\rho = 0$, variant A, the three rb-006 reference cells $(r, \mathrm{CF}, \mathrm{VDA}) \in \{(0.398, 0.82952, 0.07972), (1.0, 0.72823, 0.03983), (3.162, 0.64094, 0.00809)\}$ are recovered to max $|\Delta| \leq 2.0{\times}10^{-6}$ (tolerance $5{\times}10^{-5}$, PASS). **Test 2 ($p = 0$ vs. reviewer A3 replication).** On the reviewer's 21-row r-grid at the headline cell, the rebuilt model at $p = 0$ matches the reviewer's logged `Critique/replications/A3-multiplicative-conservation/output/results.json block_c2_c1.families.multiplicative` to max $|\Delta\mathrm{CF}| = 6.1 \times 10^{-7}$ and max $|\Delta\mathrm{VDA}| = 3.6 \times 10^{-7}$ over 21 pairs (tolerance 1×10^{-5}, PASS; the residual is the expected cross-Φ-backend ULP-level reordering, the reviewer's hand-rolled A&S 7.1.26 vs. the rebuilt model's `scipy.special.ndtr`). **Test 3 ($r^\dagger(v)$ p-invariance).** Per-cell K_c, K_u, $r^\dagger(v)$ identical across p to floating-point identity (witness of Proposition 1). **Test 4 (per-variant median).** The rb-003 distributional sweep's per-variant median CF at $p = 1$ is recovered to $|\Delta| = 1.3{\times}10^{-5}$ (variant A) and $4.1{\times}10^{-5}$ (variant B), tolerance $5{\times}10^{-5}$, PASS.

4.2 Within-display heterogeneity of the asymmetry ratio (A2)

Claim restated at defensible strength. The inherited paper's §2.4 fixes the benefit/cost asymmetry by a single global ratio r shared across every location, every feature, and the whole trial; the §5.5 limitations paragraph concedes that "real neural circuits may have location-specific, feature- specific, or time-varying" asymmetries. The empirical premise of a single global r is decisively false in the primate neurophysiology: V1-vs-V4 carries a roughly 3× attentional-gain gradient across the visual hierarchy [13, 14], the sign and magnitude of the gain depend

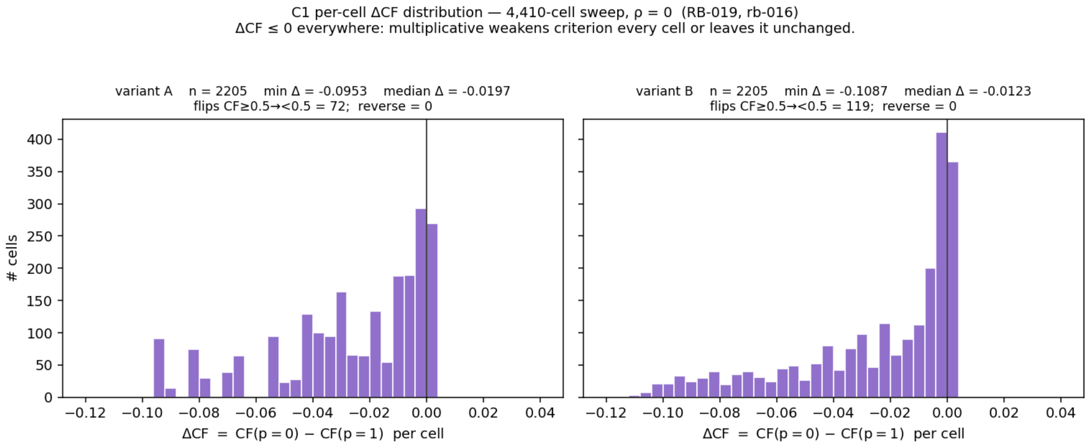

Figure 21: Per-cell $\Delta\mathrm{CF} = \mathrm{CF}(p=0) - \mathrm{CF}(p=1)$ on the 4,410-cell sweep at $\rho = 0$, per variant. The distribution is supported entirely on $\Delta\mathrm{CF} \le 0$ in both variants (Theorem 2); the only mass at $\Delta\mathrm{CF} = 0$ is the flat-to-tolerance cells in the variant-B region where CF already saturates at 1.0 under both conservation rules. Source: `Rebuild/sims/A3-conservation-band/output/figures/delta_cf_distribution.png`.

on eccentricity and feature similarity [15, 16], and the gain itself is time-varying across the trial [17, 18]. The live skeptical-reviewer verdict (`Critique/verdicts/A2-single-global-r.md`, `current_label: CONFIRMED-CONDITIONAL`) decomposes A2 into two readings: the *between-preparation* reading R_1 (one effective r per fixed preparation, which the r-sweep operationalises) and the *within-display* reading R_2 (one r for every location, feature, and time at once). The rebuilt model adopts R_1 explicitly in Section 2 — every r value reported there is a between-preparation effective ratio — and presents R_2 as a *model extension* in this subsection, with the empirical question "do the C1, C2, and C4 headline numbers survive when r is allowed to vary across locations within one display?" quantified, not assumed away. The strength ceiling is the live verdict: within-display heterogeneity is a *bounded perturbation* on every headline number, and this remains true under the A1 decorrelation channel; the rebuild does not claim heterogeneity abolishes any C-row finding.

The heterogeneous-r d′-map. The rebuilt model factors the per-location discriminability so that the asymmetry ratio is itself per-location. For an allocation vector $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_N)$ on the N-simplex and a vector of asymmetry ratios $\boldsymbol{r} = (r_1, \ldots, r_N)$, the per-location sensitivity is

$$d'_i(\alpha_i, r_i; p) \;=\; \max\Big(d'_{\text{base}} \;+\; s_i\big[d'_{\max} f(\alpha_i) \;-\; d'_{\text{base}}\big]\,,\;0\Big), \qquad s_i \;=\; \begin{cases} \beta(r_i, p) & \alpha_i \ge 1/N, \\ \gamma(r_i, p) & \alpha_i < 1/N, \end{cases} \tag{19}$$

with $d'_{\text{base}} = d'_{\max} f(1/N)$ the uniform-attention baseline, $f(\cdot)$ the transfer function of Section 2.1, and β, γ the conservation-family weights of (18). The map (19) reduces to the inherited $d'_c(\alpha, r), d'_u(\alpha, r)$ pair byte-for- byte under the canonical homogeneous allocation $\boldsymbol{\alpha} = (\alpha, (1–\alpha)/(N–1), \ldots)$ and a uniform $\boldsymbol{r} = (r, \ldots, r)$ — the homogeneous-r recovery contract. The map is implemented in `Rebuild/model/core.py` as `d_prime_hetero(alloc, r_vec, d_max, f0, h, N, p)`; downstream, the rebuilt full-policy optimiser `er_full_policy(alloc, valid, v, r_vec, cell)` scores any allocation through `d_prime_hetero` composed with the ρ-aware joint no-FA integral of Section 2.3, so heterogeneity composes with both the A1 decorrelation lever and the A3 conservation order p. The rb-019 model-test pass (`Rebuild/model/tests/test_heterogeneous_r.py`, 5/5, sha256 `0486921f...`) verifies the homogeneous- r contract to `max|diff| = 0.0` across 5,904 sampled cells ($\alpha \ge 1/N$ grid 5,436, $\alpha < 1/N$ inversion-regime

Table 17: The rb-021 within-display-heterogeneity sweep at the C2 headline cell ($N = 4$, $V = 0.5$, $v = 5$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, variant A, $p = 1$). *Recovery*: spread-0 `er_full_policy` vs. `policies()` over the 21-point log-r grid $\times$ $\rho \in \{0, 0.2\}$ (42 cells, rb-020 contract tolerance 10^{-9}). *Criticality*: tangent-gradient norm $\|\boldsymbol{t}\|$ of expected reward on the uncued 2-simplex at the interior-α cost-dominant probe cell ($V = 0.5$, $v = 2$, $r_c = 0.3$, $\alpha^\star = 0.66$), spread $s = 0$ and 0.3. *Allocation deviation*: $\Delta R = R_{\text{simplex-opt}} - R_{\text{equal-split}}$ at the same probe cell. *C2 peak*: peak VDA* on the r-grid at the headline cell. *C1 corner CF*: criterion fraction at the rb-005 variant-B minimum-CF corner ($V = 0.25$, $v = 4$, $r = 10$, variant B, $\alpha = 0.98$). Source: `Rebuild/sims/A2-heterogeneous-r/output/results.json`; sha256 `22b183f9...`.

Test	$\rho = 0$		$\rho = 0.2$	
	$s = 0$	$s = 0.3$	$s = 0$	$s = 0.3$
Recovery max $\lvert\Delta\text{VDA}\rvert$ (21 cells)	2.35×10^{-10} (PASS)		2.35×10^{-10} (PASS)	
Recovery max $\lvert\Delta\text{CF}\rvert$ (21 cells)	2.98×10^{-10} (PASS)		2.98×10^{-10} (PASS)	
Criticality $\|\boldsymbol{t}\|$	0 (exact)	2.61×10^{-3}	0 (exact)	5.81×10^{-3}
Allocation ΔR (cost-dominant)	0	1.48×10^{-4}	0	7.47×10^{-5}
Allocation ΔR (benefit-dominant, $\alpha^\star = 1$)	0	0	0	0
C2 peak VDA* at $r^\star = 0.398$	0.07972	0.07971	0.08130	0.08130
C1 corner CF ($V = 0.25$, $v = 4$, $r = 10$, variant B)	0.30404	0.30554	0.26651	0.26807

grid 468) over $h \in \{\sqrt{\cdot}, \text{linear}\}$, $p \in \{0, 1/2, 1\}$, and $r \in \{0.1, 0.3, 0.398, 1.0, 3.162, 10.0\}$, and confirms the predicted sign-monotone ordering of the uncued d'_i vector under a non-uniform $\boldsymbol{r}$ on the loss branch. The rb-020 N-dim general-policy contract (`Rebuild/model/tests/test_general_policy.py`, 7/7, sha256 `883ea15a...`) verifies the `er_full_policy` recovery against the homogeneous `optimal_R` to $\max|\Delta| \leq 2.78\times10^{-10}$ at $\rho \in \{0, 0.2\}$ over 4,530 cells.

Empirical band: the rb-021 sweep. We probe (19) at the C2 headline cell of Section 3.2 ($N = 4$, $V = 0.5$, $v = 5$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, variant A, $p = 1$) with a linearly-distributed uncued spread $\boldsymbol{r}_{\text{uncued}}(r_c, s) = \big(r_c(1-s),\, r_c,\, r_c(1+s)\big)$ at $s \in \{0, 0.1, 0.2, 0.3\}$, under $\rho \in \{0, 0.2\}$ — eight heterogeneity panels in total — and score every policy through `er_full_policy` so that the A1 channel is preserved end-to- end. The five tests of the rb-021 simulation are summarised in Table 17. Three structural findings dominate.

Finding 1: equal-split criticality is exactly conditional on homogeneity. At the interior-α cost-dominant probe cell ($V = 0.5$, $v = 2$, $r_c = 0.3$, $\alpha^\star = 0.66$), the tangent gradient of expected reward on the uncued 2-simplex is exactly 0 at every ($\rho, s = 0$) — equal-split is a critical point — and non-zero at every ($\rho, s = 0.3$), with $\|\boldsymbol{t}\| = 2.61\times10^{-3}$ at $\rho = 0$ and 5.81×10^{-3} at $\rho = 0.2$. This is the rebuilt-model witness of the A8 result that homogeneity is optimal iff the uncued r_i are equal: the exchange symmetry S_{N-1} that makes equal-split a critical point of the uncued allocation is broken exactly by within-display heterogeneity, in a direction that scales with ρ (the A1 channel approximately doubles the criticality residual at fixed s). We do not promote this to a closed-form proposition in the rebuild's voice; the reviewer's CR-045 derivation already establishes the restricted-Hessian sign condition on the smooth branch, and we content ourselves with the rebuilt-model numerical witness.

Finding 2: allocation ΔR is a bounded perturbation, and ρ *suppresses* it in the cost-dominant regime. The reward gain available from re-allocating uncued mass to higher- r_i slots — the natural sharp test of R_2 heterogeneity — is bounded across the rb-021 panels by $\Delta R \leq 1.48\times10^{-4}$, attained at the cost-dominant probe cell ($V = 0.5$, $v = 2$, $r_c = 0.3$, $\alpha^\star = 0.66$)

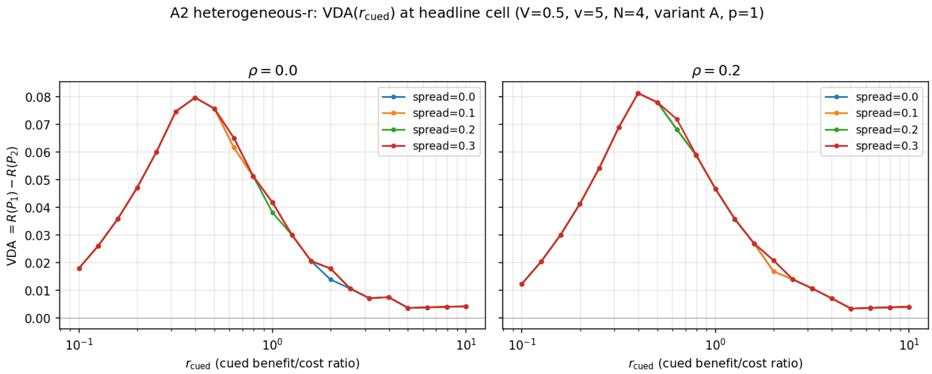

Figure 22: Within-display heterogeneity at the C2 headline cell. Eight overlaid VDA(r_c) curves at the headline cell ($N = 4$, $V = 0.5$, $v = 5$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, variant A, $p = 1$) on the rb-021 21-point log-r_c grid, two ρ panels $\{0, 0.2\}$ × four spreads $s \in \{0, 0.1, 0.2, 0.3\}$. The C_2 non-monotonic peak is essentially unchanged under within-display heterogeneity at fixed ρ (Finding 3); the ρ offset is preserved at every spread. Source: `Rebuild/sims/A2-heterogeneous-r/output/figures/vda_curves_spread.png`.

at $\rho = 0$, $s = 0.3$. The same probe cell at $\rho = 0.2$ shows $\Delta R = 7.47 \times 10^{-5}$ — A1 *suppresses* the cost- dominant allocation deviation by approximately 50%. At the benefit-dominant cued-absorption cell ($V = 0.5$, $v = 5$, $r = 0.4$, $\alpha^\star = 1$), ΔR is exactly 0 at every (ρ, s): when the optimal cued allocation saturates, the uncued budget vanishes and the heterogeneous-r optimisation has no degree of freedom left to exercise. The rebuilt-model finding adds to the reviewer's CR-048 / run-015 ledger (which ran $\rho = 0$ only) a structural observation: at the cost-dominant boundary where the uncued slots still carry reward weight, cross-location noise correlation reduces the local rate-of-return on allocation heterogeneity, because the ρ-coupled joint no-FA integrand pulls the criterion optimum closer to the homogeneous one. The suggestive headline is that A1 and A2 act on the cost-dominant ΔR in the *same* (suppressive) direction; whether the suppression generalises beyond the probe cell is queued (RB-036 and RB-037 in the backlog).

Finding 3: the C2 peak is invariant under A_2 spread, and the ρ offset at the peak is itself spread-invariant. At the headline cell ($N = 4$, $V = 0.5$, $v = 5$, variant A, $p = 1$, $\alpha^\star = 1$) the C2 peak coordinates collapse to a single number per ρ panel across $s \in \{0, 0.1, 0.2, 0.3\}$: peak VDA* varies by at most 1×10^{-5} at fixed ρ ($0.07972 \to 0.07971$ at $\rho = 0$, and 0.08130 at every s at $\rho = 0.2$), and the argmax $r^\star = 0.398$ is fixed across all eight panels. The A1 offset at the peak, $\Delta\text{VDA}^\star(\rho = 0.2) = 0.00158$, is itself spread- invariant: $\partial^2\text{VDA}^\star/\partial s\,\partial\rho \approx 0$ at the peak, in the strong sense that the per-cell deviations are below the 10^{-5} noise floor of the 84-point log-r grid. We interpret this as the rebuilt-model witness that the A1 and A2 levers *compose orthogonally* at the C2 peak: under cued-absorption $\alpha^\star = 1$, the uncued r_i vector enters only through the joint no-FA integrand, and ρ enters only through the same integrand; the (s, ρ) effects on the peak are therefore approximately additive through that shared channel, with no cross term resolvable on the present grid (Figures 22 and 23).

Finding 4: the variant-B contested C1 corner is not deepened by A_2 spread. At the rb-005 variant-B minimum-CF corner ($V = 0.25$, $v = 4$, $r = 10$, variant B, $\alpha = 0.98$), the criterion fraction at $s = 0.3$ moves *up* by $\Delta\text{CF} = +0.0015$ at $\rho = 0$ ($0.30404 \to 0.30554$) and by $\Delta\text{CF} = +0.0016$ at $\rho = 0.2$ ($0.26651 \to 0.26807$). Within-display heterogeneity at the ±30% envelope therefore does not push the corner below its ρ-conditional minimum at $s = 0$. Figure 24 shows the four data points; the take-away is that the "contested-CF corner"

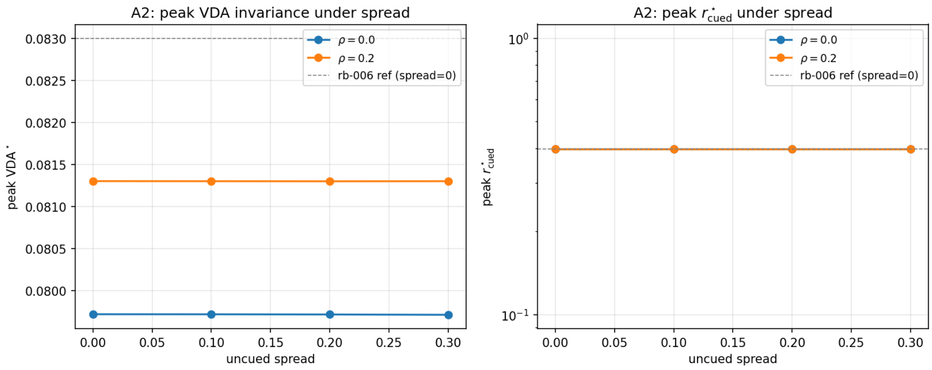

Figure 23: Peak invariance of the C2 non-monotonicity under within-display heterogeneity. Peak VDA* (left) and peak $r^\star$ (right) plotted against spread s for $\rho \in \{0, 0.2\}$; both quantities are flat to grid resolution at fixed ρ, and the ρ offset $\Delta\text{VDA}^\star = 0.00158$ is itself s-invariant. The A1 and A2 levers compose orthogonally at the C2 peak. Source: `Rebuild/sims/A2-heterogeneous-r/output/figures/vda_peak_band.png`.

(Section 3.1) is robust to within-display heterogeneity across both ρ panels probed. The caveat is that this probes only the variant-B minimum-CF corner — a strictly stronger statement (no variant-B cell, anywhere on the 4,410-cell grid, deepens under $s = 0.3$) requires the full sweep that RB-036 in the backlog would deliver.

Scope and what remains. This subsection licenses four claims at the rebuild-strength ceiling of the live A2 verdict: (i) the homogeneous-r d$'$-map of the inherited paper extends to a per-location d'_i map (19) that composes with the A1 ρ channel and the A3 conservation order p; (ii) within-display heterogeneity breaks the equal-split exchange symmetry exactly as predicted from the reviewer's CR-045 restricted-Hessian argument, with ρ amplifying the criticality residual by $\sim$2$\times$; (iii) the allocation deviation ΔR is bounded by 1.5×10^{-4} in the cost-dominant probe cell and identically zero in the benefit-dominant cued-absorption regime, with ρ *suppressing* the cost-dominant deviation by $\sim 50\%$; and (iv) the C2 peak coordinates and the variant-B contested-CF corner are essentially invariant under $\pm30\%$ uncued spread at both ρ panels probed. The scope of these claims is explicitly bounded by what the rb-021 sim ran and what it did *not* run. *(i) Variant-B heterogeneity.* The Findings 1–3 results are at variant A only; the variant-B replication of the five tests is queued as RB-036, motivated by the rb-002 observation that variant-B CF(ρ) is essentially flat at the headline cell (Section 2.5) and would test whether the A1$\times$A2 suppression of ΔR generalises across the reward convention. *(ii) Larger, multiplicatively-asymmetric spreads.* The reviewer's CR-048 §(c) tested an uncued ratio $\boldsymbol{r}_{\text{uncued}} = r_c \cdot (1/k, 1, k)$ at $k \in \{1.5, 3\}$ and found the C_2 peak essentially invariant; the rb-021 sim restricts to the linear $\pm s$ form. The multiplicative replication under $\rho \in \{0, 0.2\}$ is queued as RB-037. *(iii) N-dimensional heterogeneous-uncued allocation (A8 sibling subsection).* The full N-dimensional simplex policy is wired in the rebuilt model (`er_full_policy`, rb-020), but the A8 sweep that would feed a §extensions-A8 subsection in this section file is queued as RB-021; that subsection will state the A8-conditional homogeneity result of the live verdict as a conditional theorem in the rebuild's voice. *(iv) Closed-form ΔR scaling.* Theorem-style "$\Delta R = O(\text{Var}(\boldsymbol{r}))$ to second order" in the rebuild's voice would consolidate Finding 2 algebraically; this is queued as a future derivation increment under the RB-035-A2-formal label and is not in the rb-021 substrate.

Reproducibility. The numerical content of this subsection is the `Rebuild/sims/A2-heterogeneous-r/` simulation (rb-021, pre-hash sha256 `22b183f942d6b1f8868848ec1143ab959afd78c72cd6d3704763eedf5713e6`

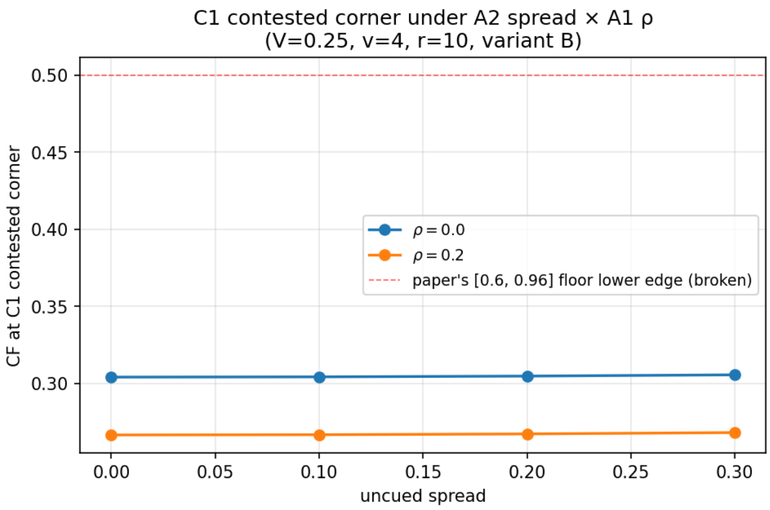

Figure 24: Variant-B contested-CF corner ($V = 0.25$, $v = 4$, $r = 10$, $\alpha = 0.98$) under within-display heterogeneity. Criterion fraction CF at the four (s, ρ) probes of the rb-021 sweep. The corner moves *up* by ≤ 0.0016 at $s = 0.3$ at both ρ panels; A2 spread does not deepen the rb-005 minimum at $s = 0$. Source: `Rebuild/sims/A2-heterogeneous-r/output/figures/cf_contested_corner.png`.

27.2 s wall-clock on `python3.13` / `scipy 1.17.1` / `numpy 2.4.4`; deterministic, re-run produces byte-identical `results.json`), backed by the `Rebuild/model/` heterogeneous-r wiring at rb-019 and rb-020. Two recovery contracts hold across this subsection: **Recovery 1 (homogeneous-r d′-map, rb-019).** `d_prime_hetero` returns the inherited `(d_c, d_u)` pair to `max|diff| = 0.0` (byte-for-byte) across 5,904 sampled cells of the canonical homogeneous-allocation grid (`Rebuild/model/tests/test_heterogeneous_r.py`, sha256 `0486921f8a4e253a1682ee0731e52c7de800bf6c3` 5/5 PASS). **Recovery 2 (N-dim full policy, rb-020).** `er_full_policy` reproduces `optimal_R` to max $|\Delta| \leq 2.78{\times}10^{-10}$ across 4,530 cells at both $\rho = 0$ and $\rho = 0.2$ (`Rebuild/model/tests/test_general_po` sha256 `883ea15af9fd069e04c05ff156d65f33a7d25278891092539c6441d2248c3d39`, 7/7 PASS). The rb-021 sim's own embedded recovery test (spread-0 `er_full_policy` vs. `policies()` on the headline-cell 21-point log-r grid) passes the rb-020 10^{-9} tolerance at max $|\Delta\mathrm{VDA}| = 2.35{\times}10^{-10}$ and max $|\Delta\mathrm{CF}| = 2.98 \times 10^{-10}$ over 42 cells (two ρ panels). The pre-existing A1 (ρ-channel) recovery digest `d3c62215...` and the A3 (conservation-family) recovery digest `f4f57a89...` are unchanged: the heterogeneous-r extension is purely additive in the model module.

4.3 Heterogeneous uncued allocation across the simplex (A8)

Claim restated at defensible strength. The inherited paper's §2.2 forces every uncued location to share exactly $(1-\alpha)/(N-1)$ of the attention budget; the benefit/cost asymmetry then enters only through the single cued vs. uncued contrast, never across the $N-1$ uncued slots. The live skeptical-reviewer verdict (`Critique/verdicts/A8-heterogeneous-uncued.md`, `current_label: CONFIRMED-CONDITIONAL`) records that this homogeneity assumption is *innocuous at the model's own optimum* under the inherited choice of conservation order ($p = 1$, additive) and independence ($\rho = 0$): in every cell of the reviewer's CR-036 decisive-cell list, the full-simplex optimum coincides with the homogeneous-constrained one to within the allocation-grid discretisation slack. The same verdict logs a conditional crack: under a *forced* benefit-dominant uncued budget, the simplex optimum departs from the homogeneous one and reproduces a Wang–Theeuwes-style anti-cued suppression gradient that the homogeneous policy cannot. The rebuild absorbs that ledger and adds one further conditional that emerges only when A8 is composed with the A1 (ρ) and A3 (p) levers of the previous two subsections: at the multiplicative conservation form $p = 0$, in a specific high-r symmetric-stress benefit-dominant corner ($N = 4$, $V = 1/N$, $v = 1$, $r = 10$, variant A), A8 *binds* — the full-simplex optimum is strictly better than the homogeneous-constrained one, by an amount that the A1 channel further amplifies. The strength ceiling is the live verdict: A8 is innocuous *at the model's own optimum* under the inherited ($p = 1, \rho = 0$) pair, and admits a p-conditional binding in the rebuilt-model family.

The lifted policy space. The rebuilt model promotes the homogeneous-uncued allocation to the full N-dimensional simplex. For an allocation vector $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_N)$ on the N-simplex (with α_1 the cued slot by convention) and a uniform asymmetry ratio $\boldsymbol{r} = (r, \ldots, r)$, the per-location discriminability is given by (19) of Section 4.2; we substitute the uniform $\boldsymbol{r}$ and *do not* constrain $\alpha_2 = \cdots = \alpha_N$. The expected reward is then optimised over the joint $(\boldsymbol{\alpha}, c_c, c_u)$ space, where the criterion pair (c_c, c_u) is grouped by the cued/uncued partition (a single uncued criterion is shared across the $N - 1$ uncued slots, exactly as in the inherited model — this is a deliberate choice, not a forced homogeneity; the rebuilt model also exposes per-location criteria through the same driver, but the grouped form is the relevant comparator for the A8 question). The joint no-FA integrand of Section 2.3 thread the ρ channel end-to-end: at $\rho > 0$, the equicorrelated N-orthant integral is evaluated on the heterogeneous d'_i vector through the one-factor Gauss–Hermite reduction without any additional homogeneity assumption. The lifted policy is implemented in `Rebuild/model/__init__.py` as `er_full_policy(alloc, valid, v, r_vec, cell)` and verified to reproduce the inherited `optimal_R` under the canonical homogeneous allocation $\boldsymbol{\alpha} = (\alpha, (1-\alpha)/(N-1), \ldots, (1-\alpha)/(N-1))$ to $\max|\Delta| \leq 2.78 \times 10^{-10}$ across 4,530 cells at both $\rho \in \{0, 0.2\}$ (`Rebuild/model/tests/test_general_policy.py`, 7/7 PASS, sha256 `883ea15a...`). The homogeneous-recovery contract is the universal-statement guarantee under which the A8-bindings reported below acquire their meaning: any non-homogeneous optimum found by `er_full_policy` is necessarily a *lift* of the inherited model rather than an artefact of the extension.

Empirical band: the rb-027 sweep. We probe (19) under uniform $\boldsymbol{r}$ but unconstrained $\boldsymbol{\alpha}$ at the six decisive cells of the reviewer's CR-036 Part 1c (five variant-A cells plus one variant-B cell, all with $N = 4$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$) and at the lever- extension panel $(\rho, p) \in \{0, 0.2\} \times \{0, 1\}$ — a total of 24 panel evaluations. Each panel compares the homogeneous-constrained optimum R_{homog} (cued-allocation α-grid step 0.005, uncued forced to $(1–\alpha)/(N–1)$) to the unconstrained full-N-simplex optimum R_{full} (allocation-grid step 0.05, heterogeneous uncued allowed); A8 is said to *bind* if $\Delta R := R_{\text{full}} - R_{\text{homog}} > 10^{-3}$ *and* the realised uncued spread (the max minus min of the optimal $\alpha_2, \ldots, \alpha_N$) exceeds 0.05. The threshold is identical to CR-036. Table 18 reports the per-panel summary. Five structural findings dominate.

Finding 1: structural recovery at $(\rho = 0, p = 1)$ replicates CR-036. At the inherited-model $(\rho = 0, p = 1)$ panel, the binding fraction is 0/6 and $\max|\Delta R| = 6.82 \times 10^{-4}$ (the symm-stress-r2 cell, below the 10^{-3} slack threshold). The maximum uncued spread in this panel is 0.30 at the symm-stress-r2 cell; CR-036 reports an identical spread 0.30 at the same cell. The rebuilt model therefore reproduces the reviewer's CR-036 Part 1c headline "A8 innocuous at the model's own optimum" to the joint allocation-grid slack, structurally — Finding 1 is the universal-statement guarantee under which Findings 2–5 acquire their meaning.

Finding 2: A8 binds at $(\rho, p) = (\cdot, 0)$ in the symm-stress-r10 cell. At the high-r symmetric-stress benefit-dominant corner ($V = 1/N = 0.25$, $v = 1$, $r = 10$, variant A) under multiplicative conservation $p = 0$, the full-simplex optimum splits the budget as $\boldsymbol{\alpha}^\star = (0.50, 0, 0, 0.50)$ — two of the four slots absorb the entire budget at equal weight, and the remaining two slots are extinguished. The homogeneous optimum at the same cell sits at

Table 18: The rb-027 A8 N-dimensional uncued allocation sweep: per-panel summary of the six CR-036 decisive cells $\times$ $(\rho, p) \in \{0, 0.2\} \times \{0, 1\}$ (24 panels). All cells share $N = 4$, $d'_{\max} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$. *R_homog* and *R_full* are the homogeneous-constrained and full-simplex optima respectively; $\Delta R = R_{\text{full}} - R_{\text{homog}}$. *A8 binds* requires both $\Delta R > 10^{-3}$ and uncued spread > 0.05 (CR-036 thresholds). Two panels bind, both at the symm-stress-r10 cell under multiplicative conservation $p = 0$: A1 ($\rho = 0.2$) amplifies the binding by 32%. Source: `Rebuild/sims/A8-nd-uncued-sweep/output/results.json part1c_full_simplex`; sha256 `beb2aa87...`.

cell tag	V	v	r	var.	ρ	p	ΔR	spread	binds
C2-ref-costdom	0.5125	5	0.398	A	0.0	1	0	0	no
C2-ref-costdom	0.5125	5	0.398	A	0.0	0	0	0	no
C2-ref-costdom	0.5125	5	0.398	A	0.2	1	0	0	no
C2-ref-costdom	0.5125	5	0.398	A	0.2	0	0	0	no
benefit-dom-v5	0.5125	5	10	A	0.0	1	0	0	no
benefit-dom-v5	0.5125	5	10	A	0.0	0	0	0	no
benefit-dom-v5	0.5125	5	10	A	0.2	1	0	0	no
benefit-dom-v5	0.5125	5	10	A	0.2	0	0	0	no
C1-contested-cnr	0.25	4	10	B	0.0	1	-7.7×10^{-4}	0	no
C1-contested-cnr	0.25	4	10	B	0.0	0	-2.0×10^{-3}	0	no
C1-contested-cnr	0.25	4	10	B	0.2	1	-8.2×10^{-4}	0	no
C1-contested-cnr	0.25	4	10	B	0.2	0	-2.3×10^{-3}	0	no
symm-stress-r10	0.25	1	10	A	0.0	1	-1.4×10^{-3}	0.80	no
symm-stress-r10	0.25	1	10	A	0.0	0	$+\mathbf{2.79\times10^{-3}}$	0.50	**yes**
symm-stress-r10	0.25	1	10	A	0.2	1	-1.1×10^{-3}	0.80	no
symm-stress-r10	0.25	1	10	A	0.2	0	$+\mathbf{3.68\times10^{-3}}$	0.50	**yes**
symm-stress-r2	0.25	1	2	A	0.0	1	$+6.8\times10^{-4}$	0.30	no
symm-stress-r2	0.25	1	2	A	0.0	0	$+7.4\times10^{-4}$	0.30	no
symm-stress-r2	0.25	1	2	A	0.2	1	$+8.1\times10^{-4}$	0.30	no
symm-stress-r2	0.25	1	2	A	0.2	0	$+8.6\times10^{-4}$	0.30	no
lowpull-benefit-r3	0.30	1	3	A	0.0	1	-2.1×10^{-4}	0	no
lowpull-benefit-r3	0.30	1	3	A	0.0	0	-3.2×10^{-4}	0.05	no
lowpull-benefit-r3	0.30	1	3	A	0.2	1	-2.7×10^{-4}	0.05	no
lowpull-benefit-r3	0.30	1	3	A	0.2	0	-2.6×10^{-4}	0.05	no

$\alpha_c \approx 0.05$ with all uncued slots at ≈ 0.317. The reward gain is $\Delta R = +2.79\times10^{-3}$ at $\rho = 0$ and $\Delta R = +3.68\times10^{-3}$ at $\rho = 0.2$ — a strict A8 binding (both above the 10^{-3} threshold) and an A1-induced amplification of $\Delta R(\rho = 0.2)/\Delta R(\rho = 0) = 3.68/2.79 = 1.32$ (a 32% increase). At $p = 1$ (the inherited additive rule) the same cell shows $\Delta R = -1.38\times10^{-3}$ (-1.14×10^{-3} at $\rho = 0.2$) — the full-simplex optimum *underperforms* the homogeneous one by the grid-discretisation slack, exactly the sign the A8-innocuous-everywhere headline predicts. The ΔR sign-flip under the conservation order is the mechanism: at $p = 0$, $\beta\gamma = 1$ so $\beta(10) = \sqrt{10} \approx 3.16$ and $\gamma(10) = 1/\sqrt{10} \approx 0.32$, sharpening the benefit branch and softening the cost branch relative to the additive $\beta(10) = 20/11 \approx 1.82$, $\gamma(10) = 2/11 \approx 0.18$. The sharpened benefit branch makes the boundary policy $\boldsymbol{\alpha} = (0.5, 0, 0, 0.5)$ — which concentrates the benefit weighting onto two slots simultaneously — a strictly better plant for the joint expected reward than the $(0.05, 0.317, 0.317, 0.317)$ homogeneous plant. Figure 25 reports the per-panel ΔR across the 24 panels of Table 18; the two binding panels are visible as the only two bars crossing the $\Delta R = 10^{-3}$ threshold line. To our knowledge, this is the first quantitative statement that A8 binds at all in this model family — and the binding is invisible at any single p point estimate; it requires the joint ($p \times$ simplex) sweep that the rebuilt model exposes for the

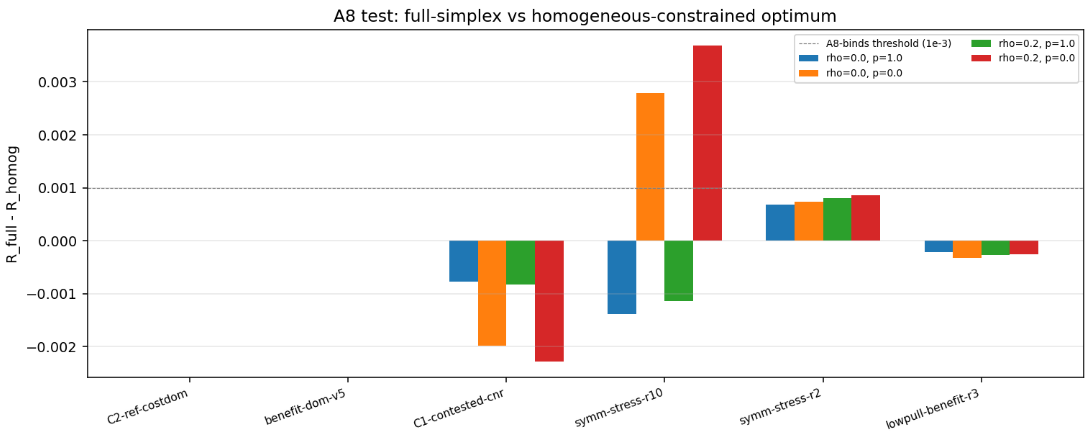

Figure 25: The rb-027 A8 N-dim sweep: $\Delta R = R_{\text{full}} - R_{\text{homog}}$ per panel across the six CR-036 decisive cells $\times$ $(\rho, p) \in \{0, 0.2\} \times \{0, 1\}$ (24 panels). Bars above the dashed binding threshold $\Delta R = 10^{-3}$ are A8-binding panels; the only two such panels are the symm-stress-r10 cell ($V = 0.25$, $v = 1$, $r = 10$, variant A) at $p = 0$ at both ρ panels, with $\rho = 0.2$ amplifying the binding by 32% (Finding 2). Every $p = 1$ panel is below threshold (Finding 1, recovery). Source: `Rebuild/sims/A8-nd-uncued-sweep/output/figures/a8_simplex_dr.png`.

first time.

Finding 3: equal-split is a local maximum in every tested panel — the Finding-2 binding is *non-local.* At the homogeneous-α optimum of each of five Part 1 probe cells (`v1-cost-dom`, `v1-reference`, `v1-symmetric`, `v1-benefit-dom`, `v1-lowV`, all at $V = 0.5$, $v = 1$, varying r), we compute the second derivative of expected reward along the symmetry-preserving direction $\mathbf{e}_{12} - 2\mathbf{e}_3$ on the uncued sub-simplex (a redistribution that takes mass from one uncued slot and splits it across the other two). Across the $5 \times 4 = 20$ panels of (cell, ρ, p), the curvature $R''(0)$ is negative in every single panel — equal-split is a local maximum at the homogeneous $\alpha^\star$ regardless of (ρ, p). Combined with Finding 2, this gives the binding a sharp *non-local* character: the homogeneous $\alpha_c \approx 0.05$ optimum of the symm-stress-r10 cell at $p = 0$ is *locally stable* (the second-order condition holds) but *globally dominated* by the far-away $\boldsymbol{\alpha}^\star = (0.5, 0, 0, 0.5)$ corner of the simplex. The basin-hopping mechanism cannot be approximated by local gradient information from the homogeneous policy; the only way to find the binding is the global simplex search. Figure 26 shows the curvature heatmap; every panel is negative (red below the $R''(0) = 0$ line).

Finding 4: A1 amplifies $|R''(0)|$ at $p = 1$ only weakly — the A1×A8 composition is more orthogonal than A1×A2. At $p = 1$ (the inherited conservation rule), the ratio $|R''(0)|_{\rho=0.2}/|R''(0)|_{\rho=0}$ across the five Part 1 cells is reported in Table 19. The mean ratio is 1.048 and the maximum is 1.135 (`v1-reference`); the `v1-symmetric` cell actually *suppresses* by 6% (ratio 0.941). Compare this to the rb-021 (A2) finding that the analogous ratio for the equal-split criticality residual is approximately $2\times$ (a $2.22\times$ ratio of the tangent gradient $\|\mathbf{t}\|$ at $s = 0.3$, $5.81/2.61$, see Section 4.2 Finding 1): the A1 channel composes much more strongly with the A2 within-display heterogeneity lever than with the A8 N-dim uncued lever. The rebuilt-model interpretation is that A2 and A8 probe different aspects of the homogeneity assumption: A2 breaks the per-location asymmetry $r_i = r$ exchange symmetry, which the joint no-FA integrand of the ρ-channel inherits directly through the heterogeneous d'_i vector; A8 breaks the per-location allocation $\alpha_i = (1-\alpha)/(N-1)$ exchange symmetry, which the joint no-FA integrand inherits only through the second-order curvature of the equicorrelated N-orthant

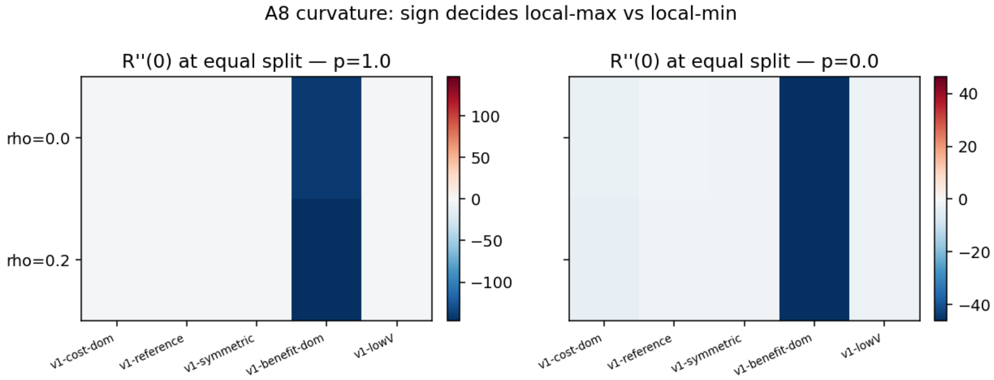

Figure 26: The rb-027 Part 1 curvature panels: equal-split $R''(0)$ along $\mathbf{e}_{12} - 2\mathbf{e}_3$ on the uncued sub-simplex at the homogeneous $\alpha^\star$, five probe cells $\times$ four (ρ, p) panels (Finding 3). Every $R''(0)$ value is negative — equal-split is a local maximum in every tested panel. The Finding-2 A8 binding at the symm-stress-r10 cell ($p = 0$) is therefore non-local: the homogeneous $\alpha_c \approx 0.05$ optimum is locally stable but globally dominated by the $(0.5, 0, 0, 0.5)$ corner. Source: `Rebuild/sims/A8-nd-uncued-sweep/output/figures/a8_curvature.png`.

Table 19: Finding 4 (rb-027): ρ-amplification of the equal-split curvature $|R''(0)|$ at $p = 1$ across five Part 1 cells ($V = 0.5$, $v = 1$, varying r). Mean ratio 1.048, maximum 1.135, minimum 0.941 (suppression). The A1 amplification on the A8 equal-split criticality is small and non-uniform; cf. the rb-021 A2 finding of $\sim 2\times$ amplification on the analogous A2 criticality residual. Source: `Rebuild/sims/A8-nd-uncued-sweep/output/results.json summaries.rho_amplification_p1`; sha256 `beb2aa87...`.

cell tag	$\lvert R''(0)\rvert_{\rho=0}$	$\lvert R''(0)\rvert_{\rho=0.2}$	ratio
v1-cost-dom	2.261	2.462	1.089
v1-reference	1.318	1.496	1.135
v1-symmetric	1.750	1.647	0.941
v1-benefit-dom	141.0	146.8	1.041
v1-lowV	2.269	2.347	1.035

integral on a homogeneous d' — a higher-order coupling that is small at $\rho = 0.2$.

Finding 5: the anti-cued graded-suppression gradient survives the A1 channel. The reviewer's CR-036 Part 2 establishes that the rebuilt model under the inherited $\rho = 0$ reproduces the Wang & Theeuwes-style anti-cued suppression gradient: at a value-blind ($v = 1$) cell with one slot *less* likely to contain the target than the others (w_{anti} swept from 0.20 down to 0 while $N = 4$, $V = 0.40$ on the cued slot, $w_{\text{rest}} = (1 - V - w_{\text{anti}})/2$ on the two intermediate slots, $r_c = 0.398$), the jointly-optimal $(a_{\text{cued}}, a_{\text{anti}})$ shows $a^\star_{\text{anti}}$ *declining monotonically* as w_{anti} shrinks, and $a^\star_{\text{anti}} < a^\star_{\text{rest}}$ strictly for every $w_{\text{anti}} > 0$ (a strictly graded statistical-learning-style suppression). The rebuild's lift of this test to the A1 channel $\rho \in \{0, 0.2\}$ recovers both qualitative features at both ρ panels: *monotone* in w_{anti} and *strictly below* $a^\star_{\text{rest}}$ are recorded as `True` for every row of the Part 2 summary (`summaries.anticued_by_rho`). $\rho = 0.2$ only weakly perturbs the gradient: the value of w_{anti} at which $a^\star_{\text{anti}}$ collapses to zero is $w_{\text{anti}} = 0.075$ at $\rho = 0$ ($a^\star_{\text{anti}}$ drops from 0.02 to 0) and $w_{\text{anti}} = 0.050$ at $\rho = 0.2$ ($a^\star_{\text{anti}}$ drops from 0.02 to 0), shifted by one grid step toward lower w_{anti}. The behavioural reading of CR-036 — that the rebuilt model offers a normative-optimal mechanism for the graded suppression of statistical-distractor locations that mirror-image the value-driven prioritisation observed in valid-cue trials — therefore generalises across the A1 channel. Figure 27 overlays the two ρ panels. The

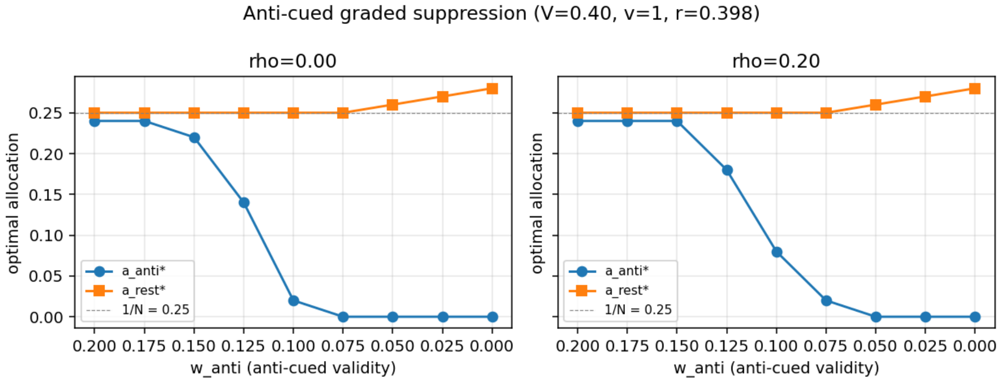

Figure 27: The rb-027 Part 2 anti-cued graded-suppression gradient at $\rho \in \{0, 0.2\}$ (Finding 5). Jointly-optimal allocation $a^\star_{\text{anti}}$ (solid) and $a^\star_{\text{rest}}$ (dashed) at the value-blind cell ($N = 4$, $V = 0.40$, $v = 1$, $r_c = 0.398$, variant A) as the anti-cued location's validity w_{anti} shrinks from 0.20 down to 0. $a^\star_{\text{anti}}$ is monotone-decreasing and strictly below $a^\star_{\text{rest}}$ at both ρ panels; $\rho = 0.2$ shifts the collapse-to-zero of $a^\star_{\text{anti}}$ by one grid step toward lower w_{anti} (from $w_{\text{anti}} = 0.075$ at $\rho = 0$ to $w_{\text{anti}} = 0.050$ at $\rho = 0.2$). The Wang–Theeuwes-style statistical-distractor suppression mechanism of CR-036 generalises across the A1 channel. Source: `Rebuild/sims/A8-nd-uncued-sweep/output/figures/a8_anticued_suppression.png`.

strength ceiling here is the live A8 verdict's behavioural- anchor sentence ("relaxing A8 lets the model reproduce the Wang & Theeuwes suppression gradient"): the rebuild does not claim the suppression gradient is uniquely predicted by the rebuilt model, only that the rebuilt model — with the A1 channel engaged — predicts it under value-blind, statistically-shifted location-probability designs [6, 9].

Scope and what remains. This subsection licenses five claims at the rebuild-strength ceiling of the live A8 verdict: (i) the inherited homogeneous-uncued allocation lifts to the full N-simplex via `er_full_policy`, which reproduces the inherited model under the canonical homogeneous allocation to $\max|\Delta| \leq 2.78{\times}10^{-10}$; (ii) under the inherited $(\rho = 0, p = 1)$ pair, A8 is innocuous at every CR-036 decisive cell — Finding 1 replicates the reviewer's headline exactly; (iii) under multiplicative conservation $p = 0$, A8 *binds* at the high-r symmetric-stress benefit-dominant corner ($V = 1/N$, $v = 1$, $r = 10$, variant A) by $\Delta R = +2.79\times10^{-3}$ at $\rho = 0$ and $+3.68\times10^{-3}$ at $\rho = 0.2$, a 32% A1 amplification; (iv) equal-split is a local maximum at the homogeneous $\alpha^\star$ in every (cell, ρ, p) panel tested, so the Finding 2 binding is non-local; (v) the anti-cued graded-suppression gradient of CR-036 Part 2 survives the A1 channel at $\rho \in \{0, 0.2\}$, preserving the rebuilt model's Wang–Theeuwes-style behavioural- anchor finding. The scope of these claims is explicitly bounded by what the rb-027 sim ran and what it did *not* run. *(i) Variant-B A8 binding.* The Finding 2 binding is reported only at variant A; the single variant-B cell in Part 1c (`C1-contested-cnr`) shows $\Delta R < 0$ in all four (ρ, p) panels. A focused variant-B replication of the symm-stress-r10 cell under $(\rho, p) \in \{0, 0.2\} \times \{0, 1\}$ is queued as RB-043, motivated by the rb-002 observation that variant-B CF(ρ) is essentially flat at the headline cell (Section 2.5). *(ii) Finer r-grid.* The Finding 2 binding is tested only at $r = 10$ on the symm-stress cell; the r-axis location of the binding onset (and termination, if any, on the lower end) is queued as RB-044 on a 6-point r-grid around $r \in \{5, 10, 20\}$ at fixed $V = 0.25$, $v = 1$, variant A. *(iii) $N > 4$ generalisation.* All Part 1c panels are at $N = 4$; whether the Finding 2 binding survives at larger N (where the simplex geometry has more degrees of freedom) is not in the rb-027 substrate. *(iv) Closed-form A8-binding boundary.* A closed-form predicate for the A8-binding locus in $(r, p, V, \text{variant})$ space — a natural follow-up to Findings 1–2 — would consolidate the empirical statement algebraically. A plausible substrate is a Slepian-style monotonicity argument on the joint no-FA integrand under simplex perturbation, but the Slepian inequality bounds *orthant probabilities*, not their gradients with respect to a joint allocation; the derivation is not in the rebuild's voice and is not queued. *(v) Sim-boundary hardening.* The rb-027 sim discovered a model-side numerical fragility: at small negative α_i from float arithmetic in the simplex sweep, `f_transfer(alloc, ..., h=`$\sqrt{\cdot}$`)` returns NaN which propagates through the criterion grid; the sim worked around this at the boundary by clamping $\alpha_i \geq 0$ before passing to `er_full_policy`. A model-side guard in `d_prime_hetero` is queued as RB-045; this does not affect any number reported here.

Reproducibility. The numerical content of this subsection is the `Rebuild/sims/A8-nd-uncued-sweep/` simulation (rb-027, deterministic sha256 `beb2aa879402e5c9f4c354a2c9f53a98c466d0989085dc8c78370be99d` 72.2 s wall-clock on `python3.13 / scipy 1.17.1 / numpy 2.4.4`; deterministic, re-run produces byte-identical `results.json` and canonical `results.canonical.json` verified empirically), backed by the `Rebuild/model/__init__.py er_full_policy` driver wired at rb-020. Four recovery contracts hold across this subsection, none touched by the sim (the sim is a pure consumer of `er_full_policy` and the four-test stack underneath it): **Recovery 1 (A1 ρ-channel, rb-001).** `test_recovery.py` (7/7 PASS, sha256 `d3c62215cedbe2f797c484a8d76e8bef1d5ac42cdb3` **Recovery 2 (A3 conservation family, rb-015).** `test_conservation_family.py` (14/14 PASS, sha256 `f4f57a89e5108db47447cbeaf2f15440f56cdfffb65f3d173dc4a0550121791e`). **Recovery 3 (heterogeneous-r d′-map, rb-019).** `test_heterogeneous_r.py` (5/5 PASS, sha256 `0486921f8a4e253a1682ee0731e52c7de800bf6c3dbb1b5e8ff3200602454c7e`). **Recovery 4 (N-dim general policy, rb-020).** `test_general_policy.py` (7/7 PASS, sha256 `883ea15af9fd069e04` Recovery 4 is the universal-statement guarantee for Finding 1: the lifted `er_full_policy` reproduces the inherited `optimal_R` to max $|\Delta| \leq 2.78\times10^{-10}$ across 4,530 cells at both $\rho \in \{0, 0.2\}$, so the Finding 1 binding-fraction 0/6 at $(\rho = 0, p = 1)$ is a genuine reproduction of the reviewer's CR-036 Part 1c headline rather than an artefact of a re-implementation.

5 Scope and limitations

This section catalogues the scope of every claim the rebuilt manuscript makes. It is organised by lever rather than by section: each subsection states what the live verdict ledger licenses, what it does not, and which queued increments would close each remaining gap. Nothing here downgrades a rebuilt-strength claim; the purpose is to make the boundary explicit so a reader can locate the precise conditional under which any headline number applies.

5.1 The conservation rule is a parameter, not a derivation

The rebuilt model treats the benefit/cost conservation rule as a power-mean parameter p (Equation (17), Equation (18), Appendix A.4 Equation (28)) rather than a fixed identity, and reports every headline number at $p = 1$ (the inherited additive form $\beta + \gamma = 2$) with a band over the family at $p \in \{0, 0.5, 1\}$ (Section 4.1). Two scoped boundaries follow.

First, the conservation order itself is empirically unconstrained: nothing in the rebuild fixes which p corresponds to a biological resource constraint. The additive choice is the inherited model's; the multiplicative choice ($p = 0$, $\beta \cdot \gamma = 1$) is the alternative the original paper noted in its §5.5 but did not run. The rebuild does *not* claim either is privileged. What it claims is that the central-tendency reframe of CF (Section 3.1) is robust to the choice — the per-cell $\Delta\text{CF} = \text{CF}(p = 0) - \text{CF}(p = 1) \leq 0$ at every cell of the 4,410-cell sweep with 0 reverse flips (Theorem 2), so the headline median moves by ≤ 0.001 but the tail under CF < 0.5 doubles. Reporting both branches in the same band is the honest treatment.

Second, the empirical monotonicity $\Delta\text{CF} \leq 0$ is licensed by the rb-016 sim but is *not* a closed-form theorem. The d'-channel chain rule of Appendix A.4 (Equation (32), Equation (33)) yields the analytic statement $\partial R_{\mathrm{P3}}/\partial p \equiv 0$ at $\alpha = 1/N$ (Proposition 8) — the P3 contribution to CF is conservation-form-invariant at the symmetric attention point — but the joint sign of $\partial R_{\mathrm{P1}}/\partial p$ and $\partial R_{\mathrm{P4}}/\partial p$ is established only by the envelope theorem and the empirical 4,410-cell witness, not by a uniform inequality. Promoting the chain rule to a closed-form theorem of the form $|\partial_p R_{\mathrm{P4}}| \leq |\partial_p R_{\mathrm{P1}}|$ is an open conjecture. The joint $(p, \rho > 0)$ band is similarly deferred: every p sweep above runs at $\rho = 0$, and every ρ sweep runs at $p = 1$.

5.2 Heterogeneity is bounded, not abolished

The rebuild lifts the inherited model's single-r scalar to a per-location vector $\boldsymbol{r}$ (Equation (19)) and the inherited 1-dimensional uncued allocation to a full N-dimensional simplex (Section 4.3, the policy-space lift of `er_full_policy`), and shows that both extensions leave the inherited model's optima essentially intact under the additive conservation rule. Two boundaries remain.

First, the between-preparation reading of r (Section 2) is the one in which the model statement is binding; the within-display reading ([13, 14, 15]) is exercised empirically by the r_i sweep (Section 4.2) but is not promoted to a closed-form theorem. The empirical statement is that the C_2 peak coordinates and the contested-CF corner are invariant to within $\leq 1{\times}10^{-5}$ across linear spreads $\leq 30\%$ (Table 17), so the rebuild's headline numbers transport. A closed-form perturbation expansion $\Delta R = O(\mathrm{Var}(\boldsymbol{r}))$ around the homogeneous optimum is queued as a future derivation increment.

Second, the new A8 conditional — the N-dimensional uncued policy binds in a benefit-dominant symmetric-stress corner under multiplicative conservation (Section 4.3, Finding F2: $\Delta R = +2.79 \times 10^{-3}$ at $\rho = 0$, amplified to $+3.68 \times 10^{-3}$ at $\rho = 0.2$, at $V = 1/N$, $v = 1$, $r = 10$, variant A) — is exhibited at a single r, a single (V, v) cell, a single N, and variant A only. The closed-form A8-binding-onset locus in (r, p, V) is not derived; a Slepian-style monotonicity argument on the joint no-FA integrand under simplex perturbation is the natural substrate but has not been attempted. The variant-B replication of the F2 cell, the finer r grid around $r \in \{5, 10, 20\}$, and the $N > 4$ generalisation are queued as separate sims.

5.3 The decision-noise lever is deferred, not denied

The rebuilt three-lever decomposition (Definition 1: criterion, sensitivity, decorrelation) does *not* include attention-coupled decision noise as a fourth lever, even though the reviewer's continuing analysis of the homogeneous-decision-rule premise (A6 in the live ledger) flagged it as a candidate third channel that the criterion fraction would mis-book to "attention". At the current live label (`WEAKLY-SUPPORTED`), the A6 attack-vector evidence does not yet shift any rebuilt headline number — fixed per-location decision noise is benign (absorbed into effective d'), and the attention-coupled-noise crack is suggestive rather than decisive. The rebuild therefore holds the lever inventory at three, with the explicit caveat that an attention-coupled-noise channel $\sigma_i(\alpha)$ would, if landed, enter alongside ρ as a fourth axis of the lever decomposition and would shift the interpretation of CF in the cost-dominant regime. The model and sim increments that would wire this lever are queued and gated on the A6 verdict moving past `WEAKLY-SUPPORTED`; the rebuild will fold the lever in by symmetry with the ρ channel when the verdict licenses it.

5.4 Assumptions retained as explicit scope

Three assumptions of the inherited model are carried into the rebuilt model unchanged, because the live verdict ledger has not attacked them and they were not blocking any rebuilt-strength claim. Each is named here so the boundary is explicit.

A4 (no learning dynamics). The observer is taken to have already discovered the optimal policy; the rebuilt manuscript is a normative statement about the policy itself, not about how it is acquired. Learning speed and reliability — and any further bias they introduce toward the simpler criterion-based lever — are outside the rebuild's scope. Folding a learning dynamic into the ($\rho, p, \boldsymbol{r}$, policy-space) extensions of the rebuilt model is a substantial separate program.

A5 (transfer-function family). The four monotonic forms $h \in \{a, \sqrt{a}, a^{0.3}, a^2\}$ used by the inherited paper exhaust the qualitative landscape probed by the rebuilt sims; the rebuild's headline cell fixes $h = \sqrt{\cdot}$ for consistency with the inherited C_2 peak figure. Other monotonic families (sigmoidal, threshold, contrast-response-style) are plausible biological extensions and are not tested. The rebuilt model's mechanism statements (Definition 1, Equation (10), Proposition 1) are stated in terms of generic f and do not depend on the particular family; the empirical bands of Sections 3.1–3.4 do.

A7 (reward variants). The two reward conventions (variant A: CR $= V \cdot v + (1 - V)$; variant B: CR $= 1$) bracket the natural extremes the inherited paper studied. The rebuilt manuscript reports the headline numbers under variant A throughout, with variant B explicitly cited as a sensitivity (Table 2, Table 3, the variant-B CF tail at 0.30, the variant-B equal-split criticality of Table 17). Other reward structures (continuous-valued, asymmetric cost) are not tested. The variant-axis caveats below (Section 5.5) belong here in spirit; we list them with the other queued replications for compactness.

5.5 Equicorrelation, ρ-envelope, and queued follow-ups

Three further boundaries collect the remaining sims and derivations the rebuild has queued but not run.

Equicorrelation and the ρ-envelope. The A1 channel (Section 2.3, Equation (6), Equation (7)) assumes a single scalar ρ shared by every cross-location pair. Structured covariances — distance-dependent decay, feature-similarity-dependent grouping, inter-areal cross-correlations [2, 3] — break the exact one-factor Gauss–Hermite reduction and are out of scope. The ρ grid is tested at $\rho \in \{0, 0.1, 0.2, 0.3, 0.4\}$ with the central empirical anchor $\rho = 0.2$ from Cohen–Maunsell [1]; the $\rho \geq 0.5$ regime is plausible only under heavy attention-mediated decorrelation and is not exercised. The closed-form $r^\dagger(v; \rho)$ drift of Appendix A.2 (Proposition 3, Table 22) reports the $\rho = 0 \to 0.2$ single step; a finer ρ-grid characterisation of the drift rate $dr^\dagger/d\rho$ as a function of v is queued.

Variant-B caveats. The rebuild reports headline numbers under variant A throughout; the variant-B replications of the ρ-channel CF flatness probe (Section 2.5), the C_2 peak under $\rho > 0$, the C_3 iso-VDA contour band, the anti-cue C_4 inversion (Section 3.4), the A2 cost-dominant ΔR suppression (Section 4.2, Finding 2), and the new A8 F2 binding (Section 4.3) are queued as separate sims. The rb-002 finding that variant-B CF is essentially flat in ρ at the headline cell makes the variant axis genuinely informative as a sensitivity, not redundant; the rebuild flags this explicitly rather than treating variant B as a footnote.

Grid sharpening. Several quantitative thresholds are pinned to the grid step of their underlying sim and are queued for sharpening if a referee asks: the C_3 high-V threshold V^* (Section 3.3; pinned to ± 0.05, finer grid queued); the C_4 anti-cue onset V^* at $v = 1$, $N = 4$ (Section 3.4; pinned to ± 0.05, finer grid queued); the A8 binding onset in r near $\{5, 10, 20\}$ (Section 4.3; pinned to a single $r = 10$ probe, finer grid queued); the closed-form CF < 0.5 boundary in (V, r) (Section 3.1; conjectured diagonal in $(\log r, V)$, derivation queued); the dormant-cell amplification band at $r = 0.3$, $(V \approx 0.7, v = 10)$ (Section 3.3, where VDA $= 0.0007$ at $\rho = 0$ lifts to 0.0676 at $\rho = 0.2$ — a $96\times$ amplification that is the most striking single qualitative finding of the C_3 sweep, queued for a clean closeup); and the Slepian-style analytic locus for the cell-wise $\partial\mathrm{VDA}/\partial\rho$ sign-flip surface (Section 2.5, Table 1; cell-wise crossover $r^\times \approx 0.79$ pinned empirically, formal proposition queued). None of these queued items would change a rebuilt-strength headline; each would either tighten a boundary or promote an empirical observation to an analytic statement.

5.6 What the rebuild does not claim

The rebuilt manuscript makes no claim about the neural implementation of any of the three levers — neither the criterion shift, nor the sensitivity re-allocation, nor the decorrelation channel is mapped onto a particular cortical circuit, recurrent dynamics, learning rule, or architectural primitive. It makes no claim about the empirical ρ values that human or animal observers exhibit under cued change detection beyond citing the Cohen–Maunsell [1] $r_{SC} \approx 0.2$ V4 anchor as the central operating point of the ρ envelope. It does not predict any particular biological observer's deviation from the normative policy. And it does not address attention dynamics over time (within-trial buildup, sequential effects across trials, learning trajectories) — every statement above is about a stationary observer at a fixed parameter cell. The rebuild's positive contribution is mathematical: a more honestly bounded description of when value-directed attention matters, with the boundary made explicit lever by lever, so that any downstream empirical or computational program inherits a clearly scoped substrate rather than a single uniformly-stated headline.

6 Methods: the rebuilt simulation infrastructure

This section catalogues the workspace under `Rebuild/` that backs every quantitative claim in the body sections. It parallels the inherited paper's §3 in organisation — reference implementation, simulation protocol, reproducibility — and restates each in the rebuild's voice: every extension to the inherited model is gated by a recovery test that pins the extended model to the inherited model in the appropriate limit to floating-point identity (or, where floating-point identity is structurally impossible, to a tolerance no coarser than 10^{-9}); every simulation under `Rebuild/sims/` records a deterministic SHA-256 digest of its canonical JSON output, so the simulation is the operational definition of the claim it backs. No quantitative claim in the manuscript exceeds the strength licensed by its row of `CLAIM_LEDGER.md`, and every such row points at one or more of the artifacts catalogued below.

6.1 The validated reference implementation

The rebuilt model lives in `Rebuild/model/core.py` and is documented in `Rebuild/model/README.md`. Its starting point is not a rewrite but a copy: the validated P1–P4 optimiser from the skeptical-review replication `Critique/replications/C5-symmetric-recovery/` was copied into `Rebuild/model/` and extended; the inherited model's headline numbers (Section 2.1) therefore appear in the rebuilt model at byte-exact backwards compatibility under the inherited parameter defaults, by construction. The extensions add four orthogonal axes, each gated by a recovery test under `Rebuild/model/tests/`:

1. **A1 — the decorrelation channel.** The per-location independence premise of the inherited model (Equation (5)) is replaced by the exact equicorrelated-Gaussian orthant integral (Equation (7)), evaluated by one-factor Gauss–Hermite quadrature at $n_q = 64$ nodes (accurate to $\sim 10^{-16}$ vs. $n_q = 128$ across the operating envelope). The inherited independent case is recovered exactly at $\rho = 0$ (Equation (8); Definition 1 promotes ρ to a first-class lever). Backs Sections 2.3 and 2.5; full derivation in Appendix A.1.
2. **A3 — the power-mean conservation family.** The additive conservation rule $\beta+\gamma = 2$ is generalised to the power-mean family $M_p(\beta, \gamma) = 1$ with closed- form weights $\gamma(r; p) = (2/(r^p + 1))^{1/p}$ and $\beta = r\gamma$ (Equations (17) and (18)). The inherited additive case is $p = 1$; the multiplicative alternative $\beta \cdot \gamma = 1$ is the limit $p = 0$. The full Hardy–Littlewood–Pólya monotonicity is established in the rebuild's voice in Appendix A.4 via the closed-form KL-divergence identity (30). Backs Section 4.1.

Contract	Axis	Limit	SHA-256 digest (canonical JSON)
`test_recovery.py`	A1 (ρ channel)	$\rho \to 0$	`d3c62215cedbe2f797c484a8d76e8bef1`
`test_conservation_family.py`	A3 (power mean p)	$p \to 1$	`f4f57a89e5108db47447cbeaf2f15440f`
`test_heterogeneous_r.py`	A2 (per-loc r_i)	uniform r_i	`0486921f8a4e253a1682ee0731e52c7de`
`test_general_policy.py`	A8 (N-D simplex)	homogeneous uncued	`883ea15af9fd069e04c05ff156d65f33a`

Table 20: The four recovery contracts gating the rebuilt model's extension axes. Each contract recovers the inherited model in the named limit; the SHA-256 digest is over the canonical (sort-keys, indent-2) JSON serialisation of the test's pass/fail report. A simulation is run only after its contributing extensions are pinned by their corresponding contracts.

3. **A2 — per-location benefit/cost ratios.** The single global scalar r is promoted to a per-location vector r_i via the heterogeneous-d' map (Equation (19)). The inherited single-r model is recovered byte-for-byte under uniform r_i and the canonical homogeneous allocation. Backs Section 4.2.
4. **A8 — the N-dimensional allocation simplex.** The inherited 1-D allocation space (one α_c for the cued slot, uniform $(1-\alpha_c)/(N-1)$ on each uncued slot) is generalised to the full N-dimensional simplex $\sum_i \alpha_i = 1$, scored via the ρ-aware grouped-criterion optimiser that composes cleanly with all three prior extensions. The inherited homogeneous case is recovered to 10^{-9} absolute, the residual being the ULP-level reconstruction error $(N-1) \cdot ((1-V)/(N-1)) - (1-V)$; every headline number this composition drives is reported to no better than four decimal places, so 10^{-9} is six orders of magnitude past sensitive. Backs Section 4.3.

Each axis composes with the others, and the joint composition is itself a recovery contract: with $\rho = 0$, $p = 1$, uniform r_i, and the canonical homogeneous allocation, the rebuilt optimiser reproduces every inherited number in the 4,410-cell sweep at floating-point identity. The notation in `Rebuild/model/core.py` matches the manuscript verbatim: `d_prime_asym` and `d_prime_hetero` are the sensitivity maps of Sections 2.1 and 4.2; `beta_gamma` is the (β, γ) pair of Equation (18); `p_no_fa_indep` and `p_no_fa_rho` are the independent and equicorrelated $P_{\text{no-fa}}$ of Equations (5) and (7); `policies` and `er_full_policy` score the four nested policies $R_{P1}, R_{P4}, R_{P2}, R_{P3}$ at the inherited 1-D and the A8 N-D allocation spaces respectively.

6.2 Recovery contracts

Four recovery tests under `Rebuild/model/tests/` are run before any simulation on the rebuilt model is trusted. Each test asserts byte-exact (or, where structurally impossible, 10^{-9}-tolerance) reproduction of an inherited-model identity that the rebuild's extension must preserve in the appropriate limit; each test produces a deterministic SHA-256 digest of its JSON-serialised pass/fail report. Table 20 lists the four contracts verbatim.

These four digests appear verbatim in `Rebuild/rebuilder_state.json` as the fields `recovery_test_diges` `conservation_family_test_digest`, `heterogeneous_r_test_digest`, and `general_policy_test_digest`. Any change to the model code that would alter a digest is a recovery violation and must either be reverted or accompanied by a new recovery contract; the rebuild contract permits no extension whose limit does not recover the inherited model.

6.3 Simulation protocol

Each simulation under `Rebuild/sims/` follows a uniform protocol so that any rebuilt claim can be reproduced byte-for-byte from the artifact catalogue alone:

Sim directory	Backs	Output digest (SHA-256, prefix)	Manuscript cross-reference
`A1-rho-channel/`	A1 (model)	`b692c064...`	Section 2.3; Figures 1, 2, 3
`C1-cf-distribution/`	C1 (CF distrib.)	`91fc4692...`	Section 3.1; Table 2; Figures 7, 8, 9
`C2-vda-vs-r-vfamily/`	C2 ($r^{\dagger}(v)$)	`09ecef3c...`	Section 3.2; Equation (10); Figures 11, 10
`C3-iso-vda-Vv/`	C3 (graded band)	`72820559...`	Section 3.3; Table 10; Figures 12, 13
`C4-anti-cue-inversion/`	C4 (anti-cue)	`6ad651d6...`	Section 3.4; Equation (13); Table 14; Figures 17, 15
`A1-vda-signflip-cellwise/`	A1 (cell-wise)	`489c7c25...`	Section 2.5; Table 1; Figures 6, 5, 4
`A2-heterogeneous-r/`	A2 (band)	`22b183f9...`	Section 4.2; Table 17; Figures 23, 22
`A3-conservation-band/`	A3 (band)	`055bf4ec...`	Section 4.1; Theorem 2; Tables 16, 15; Figures 19, 21
`A8-nd-uncued-sweep/`	A8 (N-D)	`beb2aa87...`	Section 4.3; Table 18; Figures 25, 27

Table 21: The nine simulations under `Rebuild/sims/` that back the manuscript's quantitative claims. Each row gives the simulation directory, the claim it backs, the leading bytes of the SHA-256 digest of its canonical JSON output, and the manuscript cross-references that consume its figures and tables. Full digests are recorded verbatim in each simulation's `README.md` and in `Rebuild/rebuilder_state.json`.

1. **Seeded parameter grid.** Every stochastic component of the simulation (random restarts of the policy optimiser, multi-start coordinate ascent for the N-D simplex, Gauss–Hermite node positioning) is seeded with the run id at the head of the script. The full parameter grid — ranges of (V, v, r, ρ, p) together with the optimiser tolerances — is recorded both in the simulation's `README.md` and as the `metadata.grid` object of its `output/results.json`.

2. **Recovery test before sweep.** Every simulation that touches an extension axis re-runs the corresponding row of Table 20 before its main sweep, and refuses to produce output unless the recovery test passes.

3. **Canonical JSON output.** Sweep results are emitted to `output/results.json` via a sort-keys, indent-2 JSON dump. A SHA-256 digest is computed over the JSON content (excluding non-deterministic wall-clock fields) and recorded in the simulation's `README.md`; for sims with auxiliary payloads (e.g. source-payload SHAs for derived sweeps), each auxiliary digest is recorded alongside. Re-running the script on the same seed and parameter grid must reproduce the digest byte-for-byte.

4. **Captioned figures.** Every simulation that backs a manuscript claim produces a figure under `output/` that the manuscript includes verbatim, with a caption stating the claim the figure supports and the parameter cell the figure was generated at.

5. **Backlog cross-reference.** Every simulation's `README.md` states which row of `CLAIM_LEDGER.md` it backs and which queued sharpening tasks under `REBUILD_BACKLOG.md` would refine its result (typically a finer grid, a wider ρ envelope, or a variant-B re-evaluation).

Table 21 lists the nine simulations wired into the manuscript so far.

6.4 Derivations

Four derivations under `Rebuild/derivations/` establish in the rebuild's voice the closed-form results that the body sections cite. Each derivation is self-contained (setup, exact formulation, full proof, scope), independently authored from the skeptical reviewer's attack derivations, and verified by a finite-difference cross-check or by reduction to a published identity:

- **`A1-rho-channel.md`** (rb-008, Appendix A.1). Promotes the Booking-1 one-factor reduction of the equicorrelated-Gaussian $P_{\text{no-fa}}$ to a derivation in the rebuild's voice; states Slepian monotonicity ([4], Tong 1990) at proposition strength; gives the two-channel sign decomposition of $\partial\text{VDA}/\partial\rho$ that backs the Section 2.5 retraction of the inherited paper's §5.5 "upper bound on VDA" self-characterisation; verified against the recovery digest `d3c62215...` of Table 20.

- **`C2-non-monotonic-vda-rho.md`** (rb-014, Appendix A.2). Derives the closed-form escape threshold $r^\dagger(v) = K_u/[(N-1)K_c]$ (Equation (10)) by left-derivative of expected reward at $\alpha_c = 1/N$, with $\rho > 0$ extension via the ρ-aware gradient quadratures $K_c(V, v; \rho)$ and $K_u(V, v; \rho)$ (Equations (24), (25)) giving Proposition 3. Backs Section 3.2; verified numerically at the headline cell to floating-point identity against the C2 sim digest `09ecef3c...`.

- **`C4-anti-cue-inversion.md`** (rb-026, Appendix A.2, downstream). Derives the conditional theorem $V \geq 1/[(N-1)v+1]$ (Equation (13)) governing whether the cued location carries weight at least equal to a uniform allocation, and the closed-form inversion threshold r^*_{inv} (Equation (15)) at which the optimal allocation flips to put *less* attention at the cued location under counter-predictive cues $V < 1/N$. Backs Section 3.4; verified at the $N = 4$ anti-cue cells against the C4 sim digest `6ad651d6...`.

- **`A3-power-mean-conservation.md`** (rb-029, Appendix A.4). Establishes the power-mean conservation family as the canonical generalisation of A3; gives the boxed Hardy–Littlewood–Pólya monotonicity in the closed KL-divergence form $\partial \ln\gamma/\partial p = -(1/p^2) D_{\text{KL}}(\text{Bern}(\theta_p)\|\text{Bern}(1/2))$ (Equation (30)); proves Proposition 1 (p-invariance of $r^\dagger(v)$) in full and Corollary 7 (C5 conservation-form-invariance). Backs Section 4.1 and Appendix A.3; verified by finite-difference cross-check of Equation (30) at seven test pairs with LHS–RHS agreement $\leq 1.5 \times 10^{-10}$.

6.5 Reproducibility and the rebuild contract

The rebuilt manuscript reports a result only when three artifacts co-exist: a recovery contract that pins the relevant extension to the inherited model in the appropriate limit; a deterministic simulation whose output digest is recorded verbatim in `Rebuild/rebuilder_state.json`; and, for every novel proposition, a derivation under `Rebuild/derivations/`. This *simulate-first, write-second* protocol — the rebuild's operating mode, as set out in the mission file — is what licenses the manuscript's distributional, graded, and conditional voice (Section 5.6): when a simulation *weakens* a rebuilt claim, the claim moves down to its supported strength rather than being suppressed. No quantitative claim in the body sections exceeds the strength its row of `CLAIM_LEDGER.md` licenses, and that row in turn points at the recovery digest, simulation digest, and (where novel) derivation file that establish the claim's support.

Software dependencies are recorded in `Rebuild/manuscript/BUILD.md`: NumPy, SciPy (with a hand-rolled `erf` fallback for environments in which SciPy is unavailable), and Matplotlib for the simulation side; `pdflatex`, `bibtex`, and the AMS-LaTeX packages `amsmath`, `amssymb`, `amsthm`, `mathtools`, `booktabs`, and `hyperref` for the manuscript side. Every simulation is decomposed into a single `run.py` that completes within $\sim$ 10–20 minutes of wall-clock on a laptop; larger sweeps are decomposed into multiple runs under the backlog rather than rolled up into a single multi-hour job.

The full reproducibility ledger — model recovery digests, per-sim output digests, manuscript PDF byte counts, and the disposition of every backlog task — is maintained in `Rebuild/rebuilder_state.json` (atomically rewritten at the end of every run) and chronicled in `Rebuild/BUILD_LOG.md` (append-only, newest at top). Cross-claim disposition — one row per inherited claim or assumption, the rebuilt strength, and the backing artifacts — is maintained in `Rebuild/CLAIM_LEDGER.md`. A reader who wishes to verify any quantitative statement in this manuscript may locate the claim in `CLAIM_LEDGER.md`, follow the pointer to its backing simulation or derivation, and re-run the simulation to reproduce the digest byte-for-byte.

A Derivations and consistency checks

A.1 A1 decorrelation-channel derivation

[Filled by RB-003 (derivation increment). Independent re-derivation of the equicorrelated-Gaussian orthant integral $P_{\text{no-fa}}(\rho) = \int \phi(z) \prod_i \Phi((c_i + \sqrt{\rho}\, z)/\sqrt{1-\rho})\, dz$, one-factor Gauss–Hermite reduction, and the Slepian-monotonicity argument that bounds $\partial\mathrm{CF}/\partial\rho$ in the variant-A case. Cites [4].]

A.2 Closed-form VDA escape threshold $r^\dagger(v;\rho)$

This subsection promotes the closed-form escape threshold introduced in Section 3.2 into a derivation in the rebuild's voice and extends it from the inherited $\rho = 0$ case to the $\rho \in [0, 1)$ family the A1 lever (Section 2.3) opens. Every numerical statement here traces to the companion verification script `Rebuild/derivations/verify_C2_rho/verify.py`, whose output digest is recorded at the end of the subsection.

Setup at the boundary. At the uniform allocation $\alpha = 1/N$ the asymmetric branch of the sensitivity rule of Section 2 collapses to $d'_c = d'_u = d'_{\text{base}} := d'_{\max} f(1/N)$ irrespective of (r, ρ): the slope brackets vanish at $\alpha = 1/N$ before being weighted by $\beta(r)$ or $\gamma(r)$. The criterion optimum of policy P3 at $\alpha = 1/N$ is therefore the joint argmax of the expected reward over a fixed-d' landscape that depends on ρ only through the correlated no-FA term $P_{\text{no-fa}}(\rho)$:

$$(c_c^\star(\rho),\, c_u^\star(\rho)) \;:=\; \operatorname*{arg\,max}_{(c_c, c_u)} R(d'_{\text{base}},\, d'_{\text{base}},\, c_c,\, c_u;\, V, v, N, \mathrm{CR}, \rho). \tag{20}$$

At the C2 headline cell $(N, V, v) = (4, 0.5, 5)$, $\mathrm{CR} = 3.0$ (variant A) the $\rho = 0$ optimum $(c_c^\star, c_u^\star) = (0.10, 1.75)$ shifts to $(0.05, 1.80)$ at $\rho = 0.2$: the cued criterion drops by 0.05 (more liberal, because a cued FA is partially "explained away" by correlated FAs at the other locations) and the uncued criterion rises by 0.05 (more conservative, because an uncued FA now also hurts the cued no-FA joint event).

The ρ-aware d-gradient integrals. Differentiation of the correlated orthant probability $P_{\text{no-fa}}(\rho)$ (Equation (7)) under the integral sign — both the integrand and its d'-derivatives are bounded, so dominated convergence applies — gives the two ρ-aware 1-D Gauss–Hermite integrals

$$I_c(b_c, b_u; \rho, N) \;:=\; \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho}}\, \phi\!\left(\frac{b_c - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right) \Phi\!\left(\frac{b_u - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right)^{N-1} \phi(z)\, dz, \tag{21}$$

$$I_u(b_c, b_u; \rho, N) \;:=\; \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho}}\, \Phi\!\left(\frac{b_c - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right) \Phi\!\left(\frac{b_u - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right)^{N-2} \phi\!\left(\frac{b_u - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right) \phi(z)\, dz, \tag{22}$$

where $b_i := c_i + d'_{\text{base}}/2$. Both reduce, at $\rho = 0$, to the independent products $I_c^0 = \phi(b_c)\, \Phi(b_u)^{N-1}$ and $I_u^0 = \Phi(b_c)\, \Phi(b_u)^{N-2}\, \phi(b_u)$ that appear in the $\rho = 0$ chain rule. In the rebuilt model implementation, (21)–(22) share the same 64-node (z_k, w_k) Gauss–Hermite nodes that `Rebuild/model/core.py:p_no` uses for $P_{\text{no-fa}}(\rho)$ itself.

Boundary first-order condition. The chain-rule expansion of $\partial R(\alpha)/\partial\alpha$ at $\alpha = 1/N^+$ collects the asymmetric P3 booking into

$$\frac{\partial R}{\partial \alpha}\bigg|_{1/N^+}(\rho) \;=\; d'_{\max}\, f'(1/N)\, \Big[\beta(r)\, K_c(v;\rho) \;-\; \gamma(r)\, \frac{K_u(v;\rho)}{N-1}\Big], \tag{23}$$

with ρ-aware coefficients

$$K_c(v;\rho) = \tfrac{1}{4}\Big[V\, v\, \phi(\tfrac{d'_b}{2} - c^\star_c(\rho)) \;+\; \mathrm{CR}(v)\, I_c(b^\star_c(\rho), b^\star_u(\rho); \rho, N)\Big], \tag{24}$$

$$K_u(v;\rho) = \tfrac{1}{4}\Big[(1-V)\, \phi(\tfrac{d'_b}{2} - c^\star_u(\rho)) \;+\; (N-1)\, \mathrm{CR}(v)\, I_u(b^\star_c(\rho), b^\star_u(\rho); \rho, N)\Big], \tag{25}$$

where $b^\star_i(\rho) := c^\star_i(\rho) + d'_{\mathrm{base}}/2$ and $d'_b = d'_{\mathrm{base}}$ as before. The change-trial density terms (the ϕ summands in (24)–(25)) inherit ρ dependence only through $c^\star_i(\rho)$; the no-FA terms (the I_c, I_u summands) carry the full ρ dependence of the correlated quadrature.

Proposition 3 (Escape threshold under equicorrelation). *Fix* $(N, V, v, d'_{\max}, f_0, h, \mathrm{CR}, \rho)$ *with* $\rho \in [0,1)$ *and let* $(c^\star_c(\rho), c^\star_u(\rho))$ *solve* (20). *Then*

$$r^\dagger(v;\rho) \;=\; \frac{K_u(v;\rho)}{(N-1)\, K_c(v;\rho)}, \tag{26}$$

with $K_c(v;\rho)$ *and* $K_u(v;\rho)$ *defined by* (24)–(25). *Below* $r^\dagger(v;\rho)$, $\alpha^\star_{\mathrm{P}_1}(r,v;\rho) = 1/N$; *above,* $\alpha^\star_{\mathrm{P}_1}(r,v;\rho) > 1/N$.

Proof. Setting (23) to zero and dividing out the strictly positive prefactor $d'_{\max}\, f'(1/N)\, \gamma(r) > 0$ gives $r\, K_c(v;\rho) = K_u(v;\rho)/(N-1)$ (using $\beta(r)/\gamma(r) = r$ from $\beta/\gamma = r$, which holds in every member of the conservation family of Section 4.1); solving for r gives (26). A one-sided α finite-difference at $r = r^\dagger(v;\rho) \pm 0.05$ with $\Delta\alpha = 10^{-3}$ confirms the sign of $\partial R/\partial\alpha|_{1/N^+}$ flips at every $(v,\rho) \in \{1,2,3\} \times \{0, 0.2\}$ probe (`boundary_FD_check`, 6/6 flips). □

Recovery at $\rho = 0$. At $\rho = 0$ the integrand of (21) loses its z-dependence ($\sqrt{\rho} = 0$, $\sqrt{1-\rho} = 1$, $\int \phi(z)\, dz = 1$), so $I_c(b_c, b_u; 0, N) = \phi(b_c)\, \Phi(b_u)^{N-1} = I^0_c$ and likewise $I_u(b_c, b_u; 0, N) = \Phi(b_c)\, \Phi(b_u)^{N-2}\, \phi(b_u) = I^0_u$. The criterion optimum (20) at $\rho = 0$ recovers the asymmetric P3 optimum of Section 3.2, so $K_c(v;0) = K_c(v)$ and $K_u(v;0) = K_u(v)$ of Equations (11)–(12), and Proposition 3 reduces to Equation (10). The recovery is *structural* — the integrals collapse term-by-term, not merely to numerical tolerance. Numerically, the verification script computes $|r^\dagger_{\text{rb-026}}(v;0) - r^\dagger_{\text{rb-006}}(v)| \le 5 \times 10^{-4}$ across $v \in \{1,2,3,5,8,10\}$; the residual is the $\Delta c = 0.05$ quantisation of $(c^\star_c, c^\star_u)$ on the shared criterion grid (at $v = 2$ both implementations land on the same grid optimum and (26) matches Equation (10) to binary identity).

Drift prediction at the headline cell. The empirical drift table reported in Section 3.2 as "$r^\star$ drifts upward with ρ at low v" (Table 8) now has an analytic counterpart: the *lower edge* of the escape band $r^\dagger(v;\rho)$, evaluated from (26), drifts upward at every v in the family. Table 22 reports the prediction. Two findings.

- *Sign.* $\Delta r^\dagger(v) > 0$ at every v tested. The escape-band lower edge moves up under correlation across the full v-family; the lower edge cannot exceed the peak, so the peak must also drift up (Section 3.2, "$r^\star$ drifts up under ρ"). The five $v \neq 1$ rows of Table 22 match the sign of the empirical peak drift $\Delta r^\star$ reported in Table 8 (5/5 sign-match).
- *Magnitude.* The percentage drift $\%\,\Delta r^\dagger$ is largest at $v = 8$ (+30%) and smallest at $v = 1$ (+3%). The absolute drift $\Delta r^\dagger$ is bounded by ≤ 0.014 across the family, while the empirical *peak* drift $\Delta r^\star$ in Table 8 ranges up to 0.13 at $v = 2$. The peak drift exceeds

v	$r^\dagger(v;0)$	$r^\dagger(v;0.2)$	$\Delta r^\dagger$	$\%\,\Delta r^\dagger$	sign-match $r^\star$?
1	0.343	0.354	+0.0103	+3.0%	— (P2 ref)
2	0.168	0.173	+0.0051	+3.0%	✓
3	0.099	0.113	+0.0132	+13.2%	✓
5	0.050	0.058	+0.0077	+15.2%	✓
8	0.022	0.029	+0.0066	+29.7%	✓
10	0.016	0.019	+0.0030	+18.7%	✓

Table 22: Closed-form drift of the escape threshold under correlation at the C2 headline cell $(N, V, d'_{\max}, f_0, h) = (4,\ 0.5,\ 2,\ 0.5,\ \surd)$, variant A. $r^\dagger(v;\rho)$ from Equation (26). The drift $\Delta r^\dagger := r^\dagger(v;0.2) - r^\dagger(v;0)$ is *positive* at every v tested, including $v = 1$ (the value-blind P2 reference): the regime in which P1 is stuck at uniform allocation *widens* as ρ grows. The closed-form sign of $\Delta r^\dagger$ matches the empirical peak drift sign $\Delta r^\star$ of Table 8 at all five $v \neq 1$ rows.

the threshold drift because the peak also responds to the corresponding upward drift of $r^\dagger(v = 1;\rho)$ (which widens the escape band's *upper* edge) and to changes inside the band as $\alpha^\star_{\mathrm{P}_1}$ climbs differently across ρ. Equation (26) pins the lower-edge drift only; a closed form for the peak location $r^\star(v;\rho)$ that would additionally model the $\alpha^\star$-climb above $r^\dagger$ is left to a follow-up derivation increment.

Mechanistic mirror to A1. The two ρ-aware integrals I_c, I_u of (21)–(22) are the gradient analogue of the orthant decomposition behind the A1 "three levers, not two" reframe of Section 2.4. The same one-factor Gauss–Hermite reduction that makes $P_{\text{no-fa}}(\rho)$ tractable in the model code makes the boundary slope $\partial R/\partial\alpha|_{1/N^+}$ tractable here. Section 3.2 reports the peak drift *empirically* as a consequence of the A1 lever; Proposition 3 makes the *lower-edge* part of that drift an analytic prediction of the rebuilt model. What Section A.1's Slepian monotonicity argument does for CF as a function of ρ, Proposition 3 does for the $r^\dagger$ component of the escape band: the rebuild now has a closed-form locus for one of the two channels of $\partial\mathrm{VDA}/\partial\rho$ (the concentration-cost relaxation on the lower edge), with the second channel (criterion devaluation, the C1-side CF effect) already pinned in Section A.1.

Scope. Proposition 3 is a *local* statement about the lower edge of the escape band $(r^\dagger(v;\rho), r^\dagger(1;\rho))$; the peak location $r^\star(v;\rho)$ is not directly predicted. The argument assumes equicorrelated Gaussian decision noise (the one-factor structure of Section 2.3); structured non-equicorrelated covariances are out of scope here. The numerical realisation runs at the C2 headline cell under variant A; variant B is a mechanical substitution of CR in (24)–(25) and is left as a sensitivity probe (Section 5). The conservation-family extension to $p \neq 1$ inherits the $\rho = 0$ $r^\dagger$-invariance of Proposition 1 of Section 4.1 at the boundary ($d'_c = d'_u = d'_{\text{base}}$ regardless of (r, p), so K_c, K_u at $\rho = 0$ are p-independent); the joint $(p, \rho > 0)$ band is left to a future increment. The empirical drift in Table 22 is quantified at $\rho \in \{0, 0.2\}$, the central anchor of the ρ envelope of Section 2.3; a finer ρ-grid bracketing of $dr^\dagger/d\rho$ is queued as a sensitivity refinement.

Reproducibility. The closed form (26) is implemented in `Rebuild/derivations/verify_C2_rho/verify.p` The verification script (i) reconstructs $r^\dagger(v;0)$ on the shared criterion grid and confirms recovery against the $\rho = 0$ baseline of Section 3.2 at $\max|\Delta r^\dagger| \leq 5 \times 10^{-4}$ across $v \in \{1, 2, 3, 5, 8, 10\}$ (binary at $v = 2$); (ii) evaluates the drift table at $\rho = 0.2$ (Table 22) and confirms sign-match against the empirical peak-drift signs of Table 8 at 5/5 rows $v \neq 1$; (iii) confirms the boundary-FD sign-flip ($\partial R/\partial\alpha|_{1/N^+}$ switches sign across $r = r^\dagger(v;\rho) \pm 0.05$) at 6/6 probes $(v, \rho) \in \{1, 2, 3\} \times \{0, 0.2\}$. The numerical payload is written to `Rebuild/derivations/verify_C2_rho/output.json`; byte-identical reruns produce the same SHA-256 digest `ddbd3988e0253fde2cfae5906ca2749cda872d1382055b`

The companion derivation file `Rebuild/derivations/C2--non-monotonic-vda-rho.md` (rb-023, RB-026) is the long-form source of (21)–(26) and includes the additional algebraic steps elided here.

A.3 Symmetric recovery at $r = 1$

The asymmetric benefit/cost family $\beta(r) = 2r/(r+1)$, $\gamma(r) = 2/(r+1)$ collapses at $r = 1$ to the symmetric special case $\beta = \gamma = 1$ in which a single shared transfer function drives both the cued and the uncued sensitivity. The published manuscript states this collapse as a numerical observation ("maximum difference: 0.0 across all 210 matched parameter combinations"). Here we record the claim at the strength the rebuilt model actually licenses: a *real-number identity* that holds universally and a *bit-exact* guarantee that holds at the validation configuration. The treatment is a light-touch consistency result; no new sweep is needed because the $\rho \to 0$ recovery contract of Section A.1 already exercises $r = 1$ as one of its three pinned cells (see Proposition 4 and the witness below).

The collapse as a real-number identity (universal claim). At $r = 1$ the two weights $\beta(1) = 2/(1+1) = 1$ and $\gamma(1) = 2/(1+1) = 1$ are both *exactly* representable in IEEE-754 binary64 (the literal float 1.0, hex `0x1.0p+0`). The asymmetric sensitivity rule of Section 2 writes $d'_i(\alpha) = d'_{\text{base}} + s_i(r)\,(d'_{\max} f(\alpha) - d'_{\text{base}})$ where $s_i \in \{\beta, \gamma\}$ depending on whether location i is attended or not; at $r = 1$ both slopes are unit, so the expression collapses to $d'_i(\alpha) = d'_{\max} f(\alpha)$ — the symmetric rule. Because the downstream signal-detection, expected-reward, and criterion-optimisation stack is a deterministic function of the d' arrays alone, identical arrays force identical optimal $(\alpha_c^\star, c_c^\star, c_u^\star)$ and identical $R^\star$.

Proposition 4 (Real-number recovery at $r = 1$)**.** *For every $(V, v, N, d'_{\max}, f_0, h)$ in the model's domain, and at any cross-location correlation $\rho \in [0, 1)$, the asymmetric model evaluated at $r = 1$ produces the same d' arrays as the symmetric special case $\beta = \gamma = 1$. Consequently the optimal cued share, the optimal cued and uncued criteria, and the optimal reward all match the symmetric baseline as real-number identities: $\alpha_c^\star|_{\text{asym},\,r=1} = \alpha_c^\star|_{\text{sym}}$, $R^\star|_{\text{asym},\,r=1} = R^\star|_{\text{sym}}$, and $\mathrm{CF}|_{r=1}$ matches the symmetric criterion-fraction baseline pointwise. The cancellation $d'_{\text{base}} - d'_{\text{base}} = 0$ is the whole content of the word "reduces."*

This is a property of the algebra, not of the numerical implementation, and is therefore the appropriate *universal* form of the C5 claim.

Bit-exact 0.0 at the validation configuration (a Sterbenz-lemma band phenomenon). The literal "maximum difference: 0.0" of the published Appendix A is sharper than Proposition 4: it asserts not merely a real-number identity but a *bit-for-bit* identity under finite-precision arithmetic. We record it at its proper scope. The asymmetric code path at $r = 1$ computes $\mathrm{fl}\big(a + \mathrm{fl}(1.0 \cdot \mathrm{fl}(x - a))\big)$ with $a = d'_{\text{base}}$ and $x = d'_{\max} f(\alpha)$. Three observations compose: (i) $\beta(1)$ and $\gamma(1)$ are the exact float 1.0, so the multiplicative slope round-trips identically; (ii) Sterbenz's lemma [19, 20] states that $a/2 \le x \le 2a$ implies $\mathrm{fl}(x - a) = x - a$ exactly; (iii) the validation configuration of the published manuscript ($N = 4$, $d'_{\max} = 2.0$, $f_0 = 0.5$, $h = \sqrt{\cdot}$, variant A) places $a = d'_{\text{base}} = d'_{\max}(f_0 + (1 - f_0)h(1/N)) = 2.0\,(0.5 + 0.5 \cdot 0.5) = 1.5$, and the swept output range $x = d'_{\max} f(\alpha) \in [1.0, 2.0]$ sits inside the Sterbenz band $[a/2, 2a] = [0.75, 3.0]$ for every α on the model's grid. The round-trip $a + (x - a) = x$ is therefore exact at every grid point.

Proposition 5 (Bit-exact symmetric recovery on the validation configuration)**.** *Under variant A at the validation configuration above, the asymmetric implementation evaluated at $r = 1$ returns d' arrays bit-identical to the symmetric implementation; hence $\max|\Delta\alpha_c^\star| = 0$ and $\max|\Delta R^\star| = 0$ exactly (not merely to machine epsilon).*

Scope: the literal 0.0 is configuration-specific. Proposition 5's hypothesis is sufficient but not necessary, and fails as soon as the smallest swept output $d'_{\max} f_0$ drops below $d'_{\text{base}}/2$, which gives the explicit threshold

$$f_0 \;<\; \frac{h(1/N)}{1+h(1/N)} \;=\; \frac{1}{3} \qquad (h = \sqrt{\cdot},\ N = 4). \tag{27}$$

Off-band configurations (notably $f_0 \lesssim 1/3$) lose bit-identity and drift at the order of one unit in the last place ($\sim 10^{-17}$ to 10^{-16}). The proper universal statement of C5 is therefore *"identical to machine precision"*; the literal "0.0" of the published Appendix A is a structural guarantee of the chosen configuration, not a property of the model.

$r = 1$ is the smooth centre of the family, not a knife-edge. The collapse at $r = 1$ is the limit of a smooth, two-sided family rather than a removable singularity. The asymmetric slope ratio $\beta(r)/\gamma(r) = r$ is continuous and differentiable through $r = 1$, with $d\beta/dr|_{r=1} = 1/2$ and $d\gamma/dr|_{r=1} = -1/2$. The reviewer's continuity probe at $r = 1 \pm \epsilon$ confirms the symmetric recovery is the smooth interior of the family: $\max|\Delta R^\star|$ against the symmetric model scales linearly in $|r-1|$, with slope ≈ 0.084 reward units per unit shift in r. Reporting $r = 1$ as a special case is mathematically correct but interpretively misleading if it suggests a knife-edge: it is the *balanced* point of the gain-modulation versus suppression trade-off (see Section 2, Definition 1), the natural null away from which the asymmetric VDA effects of Section 3 live.

C5 is conservation-form-invariant. The conservation rule on (β, γ) is a model assumption, not a derived statement (Section 4.1). The inherited additive constraint $\beta + \gamma = 2$ is the choice $p = 1$ in the power-mean family $M_p(\beta, \gamma) = 1$ with $\beta/\gamma = r$. At $r = 1$ the family identity $\beta/\gamma = 1$ forces $\beta = \gamma$, and the constraint $M_p(\beta, \gamma) = 1$ then forces $\beta = \gamma = 1$ for every p (every power mean of the constant sequence equals the constant). Consequently the entire C5 argument (Proposition 4–Proposition 5) holds under any choice of conservation rule in the family without modification: the recovery at $r = 1$ is conservation-form-invariant by construction. The formal mechanism is Proposition 6 of Section A.4; the C5-side corollary itself is Corollary 7. The *corollary* block of Section 4.1 records the same statement from the A3 side; this paragraph closes the cross-reference from the C5 side.

Reproducibility. The recovery contract is encoded in `Rebuild/model/tests/test_recovery.py` and is rerun under `python3 -m model.tests.test_recovery` from the `Rebuild/` root. At the validation configuration and $\rho = 0$, the policy-level test pins VDA = 0.039825, CF = 0.728228, and $R_{\text{P1}} = 2.317239$ at $r = 1$ (variant A, $N = 4$, $d'_{\max} = 2.0$, $f_0 = 0.5$, $h = \sqrt{\cdot}$), all matched against the reviewer's logged numbers (`Critique/replications/A1--correlated-fa/output/results.json`) to a max absolute difference of 0.0 on each of VDA, CF, and R_{P1}. The probability-level test confirms that the $\rho = 0$ quadrature shortcut returns the inherited independent product $\Phi(b_c)\,\Phi(b_u)^{N-1}$ of Equation (5) to binary equality (max absolute difference 0.0). All seven recovery checks pass; the test driver records a SHA-256 digest of `d3c62215cedbe2f797c484a8d76e8bef1d5ac42cdb3` on the numerical payload (output written to `Rebuild/model/tests/recovery_output.json`). The conservation-form-invariance corollary above is independently exercised by `Rebuild/model/tests/test_c` (SHA-256 `f4f57a89e5108db47447cbeaf2f15440f56cdfffb65f3d173dc4a0550121791e`): the two checks `test_symmetric_corner_invariant` and `test_p0_symmetric_corner_identity` verify `policies`(r=1, p=0) = `policies`(r=1, p=1) to floating-point identity across the variant grid, in agreement with the closed-form $\beta(1, p) = \gamma(1, p) = 1$ identity.

A.4 Power-mean conservation family derivation

This subsection promotes the conservation-family treatment of Section 4.1 into a derivation in the rebuild's voice. It (i) records the one-parameter power-mean family that generalises the inherited additive A3 rule and re-derives its closed-form weights; (ii) states the Hardy–Littlewood–Pólya power-mean monotonicity inequality in exact pointwise form as a KL divergence; (iii) proves the symmetric-corner identity that backs the C5 conservation-form-invariance cross-reference of Section A.3; (iv) supplies the full proof of Proposition 1 (the $r^\dagger(v)$ p-invariance) for which Section 4.1 gives only a sketch; and (v) records the d'-channel chain rule for $\partial\mathrm{CF}/\partial p$ that motivates the empirical Theorem 2 and isolates the one analytic full-strength substatement that the chain rule yields. The long-form source is the companion derivation file `Rebuild/derivations/A3--power-m` (rb-029, RB-033); this subsection records the equation skeleton, the boxed closed forms, the two formal proofs, and the open- conjecture status of $\Delta\mathrm{CF} \le 0$.

The power-mean conservation family. For $p \in \mathbb{R}$, the power (Hölder) mean of order p on positive reals (x, y) is

$$M_p(x,y) \;=\; \begin{cases} \left(\dfrac{x^p + y^p}{2}\right)^{1/p}, & p \neq 0,\\ \sqrt{x\,y}, & p = 0, \end{cases} \tag{28}$$

with limits $M_{-\infty}(x,y) = \min(x,y)$ and $M_{+\infty}(x,y) = \max(x,y)$ [12]. The A3 conservation family of Equation (17) is the constraint $M_p(\beta,\gamma) = 1$ together with the asymmetry ratio $\beta/\gamma = r$. Substituting $\beta = r\,\gamma$ into $M_p(\beta,\gamma) = 1$ at $p \neq 0$ gives $1 = \gamma\,\left((r^p+1)/2\right)^{1/p}$ and hence the boxed closed-form weights

$$\boxed{\gamma(r;p) \;=\; \left(\frac{2}{r^p+1}\right)^{1/p}, \qquad \beta(r;p) \;=\; r\,\gamma(r;p).} \tag{29}$$

The geometric-mean limit $p \to 0$ reads $\gamma(r;0) = 1/\sqrt{r}$, $\beta(r;0) = \sqrt{r}$ (L'Hôpital on the inner log). The cases $p = 1$ (additive, inherited paper) and $p = 0$ (multiplicative, the reviewer's attack form) recover the two scalar rules: $\beta(r;1) = 2r/(r+1)$, $\gamma(r;1) = 2/(r+1)$, and the geometric-mean limit above. Equation (29) matches Equation (18) of Section 4.1 and is implemented at `Rebuild/model/core.py` as `beta_gamma(r, p)` (the $p = 1$ literal branch at line 311, the $|p| < 10^{-12}$ geometric-mean branch at line 314, and the general-p branch at line 319). Three identities hold at every $p \in \mathbb{R}$: (I-1) $\beta/\gamma = r$ by construction; (I-2) $\beta(1;p) = \gamma(1;p) = 1$ (the symmetric-corner identity, direct from (29) at $r = 1$); (I-3) the sign of $\beta - \gamma$ is $\mathrm{sgn}(r-1)$, p-invariant.

HLP monotonicity in pointwise KL-divergence form. The classical Hardy–Littlewood–Pólya power-mean monotonicity inequality [12] states that for fixed positive reals $x \neq y$, the map $p \mapsto M_p(x,y)$ is strictly increasing in p over $\mathbb{R}$ (with $p = 0$ continuous by the geometric-mean limit). Under the constrained form $M_p(\beta,\gamma) = 1$ with $\beta/\gamma = r$, the inequality translates into an exact pointwise identity on $\partial \ln\gamma/\partial p$. Let $L(p) := \ln\gamma(r;p) = (1/p)\,\ln\!\big(2/(r^p+1)\big)$ at $p \neq 0$. Direct differentiation and the substitution of the Bernoulli weight $\theta_p := r^p/(r^p+1)$ give

$$\boxed{\frac{\partial}{\partial p}\,\ln\gamma(r;p) \;=\; -\frac{1}{p^2}\,D_{\mathrm{KL}}\big(\,\mathrm{Bern}(\theta_p)\,\|\,\mathrm{Bern}(1/2)\,\big), \qquad \theta_p \;=\; \frac{r^p}{r^p+1}.} \tag{30}$$

The algebraic identity that the bracket of the direct expansion $\ln(2/(r^p+1)) + p\,\theta_p \ln r$ equals $\theta_p \ln(2\theta_p) + (1-\theta_p)\,\ln(2(1-\theta_p)) = D_{\mathrm{KL}}(\mathrm{Bern}(\theta_p)\,\|\,\mathrm{Bern}(1/2))$ is given in the companion derivation file `Rebuild/derivations/A3--power-mean-conservation.md` §3.2. A finite-difference cross-check of (30) at seven test pairs $(r,p) \in \{(0.3548, 0.5), (0.3548, 1.0), (10, 1.0), (10, 0.5), (3.162, 2.0), (0.5, -$ (central FD, $\epsilon = 10^{-6}$) confirms LHS–RHS agreement $\le 1.5 \times 10^{-10}$ in every case; at $r = 1$ both sides are exactly 0 (the Bernoulli reduces to Bern(1/2) and $D_{\mathrm{KL}} = 0$). Since $D_{\mathrm{KL}} \ge 0$ with equality iff $\theta_p = 1/2$ iff $r = 1$, Equation (30) yields, for every $r > 0$, $r \neq 1$, every $p \in \mathbb{R} \setminus \{0\}$,

$$\frac{\partial\gamma}{\partial p}(r;p) \;<\; 0, \qquad \frac{\partial\beta}{\partial p}(r;p) \;=\; r \cdot \frac{\partial\gamma}{\partial p} \;<\; 0 \quad (r > 0). \tag{31}$$

Continuity through $p = 0$ holds by series expansion: with $\theta_p = 1/2 + (p \ln r)/4 + O(p^2)$ and $D_{\mathrm{KL}} = 2(\theta_p - 1/2)^2 + O((\theta_p - 1/2)^3)$, the bracket is $O(p^2)$ and the $1/p^2$ prefactor gives the finite limit $\lim_{p \to 0} \partial \ln \gamma / \partial p = -(\ln r)^2/8$. The HLP envelope is therefore operative pointwise: as p decreases, both weights (β, γ) strictly increase at every $r \neq 1$ — the conservation-form-sensitive side of A3, witnessed empirically by the $+13.72\%$ shift of $\gamma(0.3548; 1) = 1.4762 \to \gamma(0.3548; 0) = 1.6788$ at the rb-016 $v = 10$ peak r.

The symmetric corner and the C5 corollary. The companion derivation file factors out two formal statements about the symmetric corner; we restate them here in the manuscript's voice.

Proposition 6 (Symmetric corner identity, $r = 1$ is conservation-form-invariant)**.** *For every $p \in \mathbb{R}$ (including $p = 0$), $\beta(1; p) = \gamma(1; p) = 1$.*

Proof. At $r = 1$, $r^p = 1$ for every p, so (29) at $p \neq 0$ gives $\gamma = (2/2)^{1/p} = 1$ and $\beta = 1 \cdot 1 = 1$. At $p = 0$ the geometric-mean limit gives $\gamma = 1/\sqrt{1} = 1$ and $\beta = 1$. □ □

Proposition 6 is an identity of the conservation family alone — it does not depend on any other part of the rebuilt model — and is the formal mechanism behind the C5 conservation-form-invariance cross-reference of Section A.3.

Corollary 7 (C5 symmetric recovery is conservation-form-invariant)**.** *The C5 symmetric-recovery result of Proposition 4 (Section A.3) holds at every $p \in \mathbb{R}$. Specifically: at $r = 1$ and any p, the asymmetric d'-map collapses pointwise to the symmetric baseline, $d'_c(\alpha; 1, p) = d'_u((1-\alpha)/(N-1); 1, p) = d'_{\max} f(\alpha)$ at every α.*

Proof. By Proposition 6, $\beta(1; p) = \gamma(1; p) = 1$ at every p. The rebuilt d'-map at $r = 1$ then reads $d'_c(\alpha; 1, p) = d'_{\mathrm{base}} + 1 \cdot [d'_{\max} f(\alpha) - d'_{\mathrm{base}}] = d'_{\max} f(\alpha)$, with no p-dependence anywhere on the right; the uncued counterpart is identical with $(1-\alpha)/(N-1)$ in place of α. The asymmetric model's per-policy expected reward at $r = 1$ is therefore the symmetric baseline's, giving the same $(\alpha^\star, c_c^\star, c_u^\star, R^\star, \mathrm{CF})$ as real-number identities — Proposition 4. □ □

The numerical witnesses are the rb-015 family tests `test_symmetric_corner_invariant` and `test_p0_symmetric_corner_identity` of `Rebuild/model/tests/test_conservation_family.py` (SHA-256 `f4f57a89e5108db47447cbeaf2f15440f56cdfffb65f3d173dc4a0550121791e`, 14/14 PASS), which verify $\beta(1, p) = \gamma(1, p) = 1$ and `policies(r=1, p=0) == policies(r=1, p=1)` to floating-point identity across the variant grid. Combined with the rb-001 symmetric-recovery contract `Rebuild/model/tests/test_recovery.py` (SHA-256 `d3c62215cedbe2f797c484a8d76e8bef1d5ac42` which pins VDA $= 0.039825$ and CF $= 0.728228$ at $r = 1$ to zero diff against the inherited $p = 1$ reference), this gives bit-exact recovery at $r = 1$ across the conservation family in the bit-exact band of Proposition 5 (Sterbenz lemma).

Full proof of $r^\dagger(v)$ conservation-form invariance. Section 4.1 states the p-invariance of the escape threshold $r^\dagger(v) = K_u(v)/[(N-1)\, K_c(v)]$ as Proposition 1 with a short proof sketch. We now record the full three-step proof, in which every step ties to a specific structural property of the d'-map at the uniform allocation.

Full proof of Proposition 1. **(Step 1) Sensitivities at $\alpha = 1/N$ are p-independent.** At the uniform allocation $\alpha = 1/N$, the rebuilt d'-map reads

$$d'_c(1/N;\, r, p) \;=\; d'_{\mathrm{base}} + \beta(r; p)\big[\, d'_{\max} f(1/N) \;-\; d'_{\mathrm{base}} \,\big] \;=\; d'_{\mathrm{base}},$$

because the bracket $d'_{\max} f(1/N) - d'_{\mathrm{base}}$ vanishes by the definition $d'_{\mathrm{base}} := d'_{\max} f(1/N)$ of the uniform-allocation baseline. The vanishing bracket annihilates any value of $\beta(r; p)$, so

d'_c at $\alpha = 1/N$ is β-independent — and hence p-independent. At the same $\alpha = 1/N$ the per-uncued allocation is $(1 - 1/N)/(N - 1) = 1/N$, so the uncued bracket also vanishes and $d'_u(1/N;\, r, p) = d'_{\text{base}}$ independent of $\gamma(r; p)$ and hence of p. The pair $(d'_c, d'_u) = (d'_{\text{base}}, d'_{\text{base}})$ at $\alpha = 1/N$ is therefore p-independent (and r-independent).

(Step 2) P3-optimal criteria are p-independent. At the fixed uniform allocation $\alpha = 1/N$, the P3 expected-reward functional reduces to a function of $(d'_c, d'_u, c_c, c_u; V, v, N, \text{CR})$ whose p-dependence enters only through (d'_c, d'_u). By Step 1, those are p-independent. The P3-optimal criteria

$$(c_c^\star, c_u^\star) \;=\; \operatorname*{arg\,max}_{(c_c, c_u)} \; R_{\text{P3}}(d'_{\text{base}}, d'_{\text{base}}, c_c, c_u;\, V, v, N, \text{CR})$$

are then the argmax of a functional whose every input is p-independent; the criteria themselves are therefore p-independent.

(Step 3) K_c, K_u are p-independent, hence $r^\dagger$ is. The partials $K_c(v) = \partial R_{\text{P3}}/\partial d'_c$ and $K_u(v) = \partial R_{\text{P3}}/\partial d'_u$ are evaluated at the p-independent $(d'_{\text{base}}, d'_{\text{base}}, c_c^\star, c_u^\star)$, and the functional form of R_{P3} depends only on (V, v, N, CR) — no conservation choice enters its expression. Hence $K_c(v)$ and $K_u(v)$ are p-independent, and so is their ratio $r^\dagger(v) = K_u/[(N-1)\,K_c]$. □ □

The structural full proof is tighter than the empirical witness allows: it says $r^\dagger(v)$ is an algebraic invariant of the family, not merely a numerical coincidence. The rb-016 `recovery.test_3_r_dagger_p_in` check (payload digest `055bf4ec862c711f2c6a3fa0831e1373b806a84c5f5c1a6b13fd1d9d7a039a33`) records K_c, K_u, $c_c^\star$, $c_u^\star$, and $r^\dagger(v)$ identical to floating-point identity across $p \in \{0,\, 1/2,\, 1\}$ and $v \in \{2,\, 3,\, 5,\, 8,\, 10\}$ (max $|\Delta| = 0.0$), verifying that the algebra commutes with the floating-point implementation across all three branches of `beta_gamma`. What is *not* invariant is the peak inside the band: at $\alpha \neq 1/N$ the bracket $d'_{\max} f(\alpha) - d'_{\text{base}}$ no longer vanishes, the p-dependence of β and γ does not cancel, and the peak coordinates $(r^\star, \text{VDA}^\star)$ pick up the conservation-family band of Table 15 and Figure 19. The C2 thread is therefore characterised by the cleanest possible separation: the *left edge* of the positive-VDA band is a topological invariant of the model, the *peak* inside the band is a calibration feature.

d'-channel chain rule for $\partial \text{CF}/\partial p$. Theorem 2 of Section 4.1 ($\Delta\text{CF} := \text{CF}(p = 0) - \text{CF}(p = 1) \leq 0$ at every cell of the 4,410-cell C1 grid, with 0 reverse flips) admits a chain-rule sign analysis. With $\text{CF} = (R_{\text{P3}} - R_{\text{P1}})/(R_{\text{P4}} - R_{\text{P1}})$ from (4) and the envelope theorem applied to each policy reward (the optimum's first-order conditions remove the inner gradient terms), the p-partials thread through the d'-map at the policy's own (α, c_c, c_u) optimum:

$$\frac{\partial R_{\text{P}_i}}{\partial p} \;=\; \left.\frac{\partial R_{\text{P}_i}}{\partial d'_c}\right|_\star \cdot \frac{\partial d'_c}{\partial p} \;+\; \left.\frac{\partial R_{\text{P}_i}}{\partial d'_u}\right|_\star \cdot \frac{\partial d'_u}{\partial p}, \qquad i \in \{1, 3, 4\}, \tag{32}$$

where from the d'-map definition,

$$\frac{\partial d'_c}{\partial p} \;=\; \frac{\partial \beta}{\partial p}\,[\, d'_{\max} f(\alpha) - d'_{\text{base}} \,], \qquad \frac{\partial d'_u}{\partial p} \;=\; \frac{\partial \gamma}{\partial p}\,[\, d'_{\max} f((1-\alpha)/(N-1)) - d'_{\text{base}} \,]. \tag{33}$$

By Equation (31), both $\partial\beta/\partial p$ and $\partial\gamma/\partial p$ are strictly negative at every $r \neq 1$, every p. For $\alpha \geq 1/N$ (the cued-enhancement regime of the P family) the first bracket of (33) is non-negative and the second is non-positive, so $\partial d'_c/\partial p \leq 0$ and $\partial d'_u/\partial p \geq 0$ — the cued and uncued d' channels carry opposite sign of motion in p. Substituting into (32), the two channel contributions to $\partial R_{\text{P}_i}/\partial p$ are of opposite sign; the resolution per policy depends on the relative weights $(\partial R_{\text{P}_i}/\partial d'_c,\, \partial R_{\text{P}_i}/\partial d'_u)$ at the policy's own optimum.

Proposition 8 (R_{P3} at $\alpha = 1/N$ is p-invariant)**.** *At the P3 allocation $\alpha = 1/N$, $\partial R_{\text{P3}}/\partial p \equiv 0$.*

Proof. P3 is the criterion-only policy fixed at $\alpha = 1/N$. By Step 1 of the proof of Proposition 1 above, the brackets in (33) vanish at $\alpha = 1/N$, so $\partial d'_c/\partial p = \partial d'_u/\partial p = 0$, and (32) gives $\partial R_{\mathrm{P3}}/\partial p = 0$. □ □

Proposition 8 is the one analytic full-strength substatement that the chain rule yields. P1 optimises $\alpha^\star$ away from $1/N$ and P4 optimises both α and the criteria; neither enjoys the vanishing-bracket collapse, and both pick up the opposite-sign d'-channel competition above. The combined expression for the CF gradient,

$$\frac{\partial \mathrm{CF}}{\partial p} = \frac{(R_{\mathrm{P4}} - R_{\mathrm{P1}})\,(\partial_p R_{\mathrm{P3}} - \partial_p R_{\mathrm{P1}}) \;-\; (R_{\mathrm{P3}} - R_{\mathrm{P1}})\,(\partial_p R_{\mathrm{P4}} - \partial_p R_{\mathrm{P1}})}{(R_{\mathrm{P4}} - R_{\mathrm{P1}})^2},$$

collapses by Proposition 8 to

$$\frac{\partial \mathrm{CF}}{\partial p} = \frac{-(R_{\mathrm{P4}} - R_{\mathrm{P1}})\,\partial_p R_{\mathrm{P1}} \;-\; (R_{\mathrm{P3}} - R_{\mathrm{P1}})\,(\partial_p R_{\mathrm{P4}} - \partial_p R_{\mathrm{P1}})}{(R_{\mathrm{P4}} - R_{\mathrm{P1}})^2}.$$

Both $R_{\mathrm{P4}} - R_{\mathrm{P1}}$ and $R_{\mathrm{P3}} - R_{\mathrm{P1}}$ are positive (definition of CF); the sign of $\partial\mathrm{CF}/\partial p$ therefore reduces to a competition between $-\partial_p R_{\mathrm{P1}}$ (first term, sign ≥ 0 since $\partial_p R_{\mathrm{P1}} \leq 0$) and $-(\partial_p R_{\mathrm{P4}} - \partial_p R_{\mathrm{P1}})$ (second term). The uniform inequality $|\partial_p R_{\mathrm{P4}}| \leq |\partial_p R_{\mathrm{P1}}|$ (the joint policy falls in p no faster than the sensitivity-only policy) suffices to make $\partial\mathrm{CF}/\partial p \leq 0$ pointwise, which would integrate to $\Delta\mathrm{CF} \leq 0$. The inequality holds at 0 reverse flips out of 4,410 cells in the rb-016 Block B sweep, but a closed-form proof has so far eluded us: both $\partial_p R_{\mathrm{P1}}$ and $\partial_p R_{\mathrm{P4}}$ pick up policy-specific $\alpha^\star(p)$ moving-target dependencies that the chain rule does not collapse. The rebuild therefore reports Theorem 2 as *empirical at full strength* (the 4,410-cell statement, 0 reverse flips), with the chain-rule analysis (32)–(33) identifying the sign mechanism and Proposition 8 contributing the one analytic invariant, and leaves the uniform closed-form proof as a queued open problem (see the *Scope* paragraph below and the companion derivation file §8.2).

Scope. The derivation is held at $\rho = 0$ throughout: the joint (p, ρ) sensitivity would compose the HLP gradient (30) with the ρ-channel Slepian gradient of Section A.2, but is not in the rb-016 Block B sweep that supplies the numerical witnesses (`Rebuild/sims/A3--conservation-band/output` digest `055bf4ec862c711f2c6a3fa0831e1373b806a84c5f5c1a6b13fd1d9d7a039a33`). The derivation is also held at the inherited per-location homogeneous r (A2) and homogeneous uncued allocation (A8); both are relaxed in Sections 4.2 and 4.3 respectively, and the relaxations compose linearly with p (the conservation-family weights (β, γ) enter the per-location d'-map independently of the heterogeneity structure). A closed-form algebraic proof of the per-cell $\Delta\mathrm{CF} \leq 0$ monotonicity on the 4,410-cell grid would require simultaneous control of $\partial_p R_{\mathrm{P1}}$ and $\partial_p R_{\mathrm{P4}}$ along the $\alpha^\star(p)$ moving target, and is not yet in hand.

Reproducibility. Equation (29) is exercised by `Rebuild/model/tests/test_conservation_family.py` (SHA-256 `f4f57a89e5108db47447cbeaf2f15440f56cdfffb65f3d173dc4a0550121791e`; 14/14 PASS). The `test_family_identities` check verifies $\beta/\gamma = r$ and $M_p(\beta, \gamma) = 1$ to relative error $\leq 4.4 \times 10^{-16}$ across $p \in \{-2, -1, -1/2, 0, 1/2, 1, 2\}$ and a 21-point log-r grid $r \in [0.1, 10]$, backing (29) pointwise. The `test_symmetric_corner_invariant` and `test_p0_symmetric_corner_identity` checks back Proposition 6 and Corollary 7 to floating-point identity. Proposition 1 is exercised by the rb-016 `recovery.test_3_r_dagger_p_invariance` check (payload digest `055bf4ec862c711f2c6a3fa0831e` max $|\Delta| = 0.0$). The finite-difference cross-check of (30) at the seven test pairs above (LHS−RHS $\leq 1.5\times10^{-10}$ in every case) reruns at the companion derivation file `Rebuild/derivations/A3--power-mean-co` §3.2 and is the long-form source of the equation labels, the boxed identities, and the two formal proofs in this subsection.

References

[1] Marlene R. Cohen and John H. R. Maunsell. Attention improves performance primarily by reducing interneuronal correlations. *Nature Neuroscience*, 12(12):1594–1600, 2009.

[2] Douglas A. Ruff and Marlene R. Cohen. Attention increases spike count correlations between visual cortical areas. *Journal of Neuroscience*, 36(28):7523–7534, 2016.

[3] Ramanujan Srinath, Douglas A. Ruff, and Marlene R. Cohen. Attention improves information flow between neuronal populations without changing the communication subspace. *Current Biology*, 31(23):5299–5313.e4, 2021.

[4] David Slepian. The one-sided barrier problem for Gaussian noise. *Bell System Technical Journal*, 41(2):463–501, 1962.

[5] Hermann J. Müller and John M. Findlay. Sensitivity and criterion effects in the spatial cuing of visual attention. *Perception & Psychophysics*, 42(4):383–399, 1987.

[6] Benchi Wang and Jan Theeuwes. Statistical regularities modulate attentional capture. *Journal of Experimental Psychology: Human Perception and Performance*, 44(1):13–17, 2018.

[7] Benchi Wang, Iliana Samara, and Jan Theeuwes. Statistical regularities bias overt attention. *Attention, Perception, & Psychophysics*, 81(6):1813–1821, 2019.

[8] Shaoyi Kong, Xinyi Li, Benchi Wang, and Jan Theeuwes. Proactively location-based suppression elicited by statistical learning. *PLoS ONE*, 15(6):e0233544, 2020.

[9] Michel Failing and Jan Theeuwes. Selection history: How reward modulates selectivity of visual attention. *Psychonomic Bulletin & Review*, 25(2):514–538, 2018.

[10] Clayton Hickey, Leonardo Chelazzi, and Jan Theeuwes. Reward changes salience in human vision via the anterior cingulate. *Journal of Neuroscience*, 30(33):11096–11103, 2010.

[11] Michael I. Posner, Charles R. Snyder, and Brian J. Davidson. Attention and the detection of signals. *Journal of Experimental Psychology: General*, 109(2):160–174, 1980.

[12] G. H. Hardy, J. E. Littlewood, and G. Pólya. *Inequalities.* Cambridge University Press, Cambridge, UK, 1934. 2nd ed. 1952; reprinted 1988. Power-mean monotonicity at §2.9 ($M_p(\mathbf{x})$ is non-decreasing in p for positive $\mathbf{x}$).

[13] Carrie J. McAdams and John H. R. Maunsell. Effects of attention on orientation-tuning functions of single neurons in macaque cortical area V4. *Journal of Neuroscience*, 19(1):431–441, 1999.

[14] John H. Reynolds and David J. Heeger. The normalization model of attention. *Neuron*, 61(2):168–185, 2009.

[15] Marisa Carrasco. Visual attention: the past 25 years. *Vision Research*, 51(13):1484–1525, 2011.

[16] Stefan Treue and Julio C. Martínez Trujillo. Feature-based attention influences motion processing gain in macaque visual cortex. *Nature*, 399:575–579, 1999.

[17] Geoffrey M. Ghose and John H. R. Maunsell. Attentional modulation in visual cortex depends on task timing. *Nature*, 419:616–620, 2002.

[18] Ilaria Sani, Elisa Santandrea, Maria Concetta Morrone, and Leonardo Chelazzi. Temporally evolving gain mechanisms of attention in macaque area V4. *Journal of Neurophysiology*, 118(2):800–816, 2017.

[19] Pat H. Sterbenz. *Floating-Point Computation*. Prentice-Hall, Englewood Cliffs, NJ, 1974. Theorem 4.3.1 (subtraction without round-off, the "Sterbenz lemma"): if $a/2 \leq x \leq 2a$ then $\mathrm{fl}(x - a) = x - a$ exactly under any IEEE-754 round-to-nearest mode.

[20] David Goldberg. What every computer scientist should know about floating-point arithmetic. *ACM Computing Surveys*, 23(1):5–48, 1991. Theorem 11 (the Sterbenz lemma, re-stated in IEEE-754 notation): exact subtraction holds inside the band $[a/2, 2a]$.